LogRhythm Schema Dictionary and Guide

October 24, 2019

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Table of Contents

This document is a complete dictionary of the LogRhythm SIEM schema. This guide contains descriptions of every field, including the intent for the field, guidance for how to parse data into the field, use cases for each field, and sample logs showing correct, incorrect, and ambiguous examples.

The fields in this guide are organized according to the tabs in the Analyzer grid in the LogRhythm Web Console. To access the Analyzer grid on the Dashboards page or Analyze page, at the lower-right side of the page, click the **Logs** tab.

> Fields that are listed with [7.2] after the field name are not available in LogRhythm versions earlier than 7.2.1.

Origin vs. Impacted

Definition and Guidelines

LogRhythm presents log sources from the perspective of the impacted system, the origin system, or both. Although origin and impacted align with the network-centric view of "source" and "destination," origin and impacted are meant to represent a security-centric view, in which:

In a security-centric view:

- Origin represents:
    - The client in “client server.”
    - The attacker in a security context.
    - The cause of an observation.
    - The user account who performed an action.
- Impacted represents:
    - The server in "client server."
    - The target in a security context, or the device impacted by a security event.
    - The effect of an observation.
    - The user affected by an action.

Use in Schema

The use of origin and impacted is particularly important for understanding the schema. Origin and impacted apply to IP addresses, hosts, users, and other fields that describe the object in the log. These fields include:

- Hostname
- MAC address
- Interface
- IP address
- User

> For an IP address, the schema parses into fields called SIP and DIP, where SIP represents origin and DIP represents impacted.

What Determines Origin/Impacted?

The origin/impacted context can be defined and changed in multiple places:

- *Selecting correct parsing fields*. This is important when converting the network view of source and destination to the security view of origin and impacted. Origin is not always source and impacted is not always destination.
- *Rule definition with explicit options*. The rule can explicitly force a conversion of direction.
- *Automatic Host Contextualization (AHC)*. The AHC feature can change direction based on tables of well-known ports and protocols.

Examples

- O365 SharePoint. SIP is explicitly called out, but because O365 is the cloud, there is no discernable impacted hostname.

```
TS=2016-10-20T20:22:23
SESSID=8b157afd-eb80-45e4-926f-08d3f926cd63 COMMAND=AnonymousLinkUsed
USERTYPE=Regular USERKEY=anonymous WORKLOAD=SharePoint RESULTCODE=
OBJECT=https://lrhackathon.sharepoint.com/LogRhythm/Shared
Documents/abuse_ch_copy.txt USER=anonymous SIP=1.1.1.1 ITEMTYPE=File
EVENTSOURCE=SharePoint
USERAGENT=Mozilla/5.0 (Windows NT 6.3; Win64; x64) AppleWebKit/537.36 (KHTML,
like Gecko) Chrome/54.0.2840.59 Safari/537.36 DOMAIN= FILENAME= DESTINATION=
DESTINATIONFILENAME= USERSHAREDWITH= SHARINGTYPE= MODIFIEDPROPERTIES=
```

> This is not a security event, so apply the network-centric view of client vs. server. The client is referenced in the SIP, and therefore SIP (origin) is the IP Origin. The IP Impacted is undefined, but the Impacted Host can be inferred from the log source. It is ambiguous whether the log source is the agent calling the API or refers to O365.

- Oracle 10g Audit. Client is the source of the session, but also impacted by the logoff.

```
20101115202959.307904 AUDIT_TYPE=Standard Audit
STATEMENT_TYPE=LOGOFF BY CLEANUP RETURNCODE=0 AUDIT_OPTION= PRIV _USED=CREATE
SESSION OS_USER=shenja DB_USER=SYSTEM UHOST=WKST0005 TERM=UNKNOWN
OBJECT_SCHEMA= OBJECT_NAME= POLICY_NAME= NEW_OWNER= NEW_ NAME= EXT_NAME=
SQL_TEXT= COMMENT_ TEXT=Authenticated by: DATABASE; Client address:
(ADDRESS=(PROTOCOL=tcp)(HOST=1.1.1.1)(PORT=4888))
SES_ACTIONS= GLOBAL_UID= SESSION_ID=325213 PROXY_SESSIONID= STATEMENTID=1
ENTRYID=1 CLIENT_ID= ECONTEXT_ID= TRANSACTIONID= OS_PROCESS=610338
INSTANCE_NUMBER=0 ACTION=102 SQL_BIND= OBJ_PRIVILEGE= SYS_PRIVILEGE=
OS_PRIVILEGE=NONE SCN= GRANTEE= LOGOFF_TIME=11/15/2010 3:32:42 PM
LOGOFF_LREAD=1386 LOGOFF_PREAD=80 LOGOFF_LWRITE=36 LOGOFF_DLOCK=0
SESSION_CPU=10
```

> Because this is not a security log, the host is likely the client (in client server). The host becomes the Origin Host. The Impacted Host is the Oracle server (automatically resolved by the log source).

- Windows Application. <computer> is where the event log was written.

```
<Event
xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='MSSQLSERVER'/><EventID Qualifiers='49152'>32040</EventID><Level>Information</
Level><Task>Server</Task><Keywords>Classic</Keywords><TimeCreated
SystemTime='2013-12-18T20:19:48.000000000Z'/><EventRecordID>5652</
EventRecordID><Channel>Application</Channel><Computer>ACMEPREM01</Computer><Security/></
System><EventData>The
alert for 'oldest unsent transaction' has been raised. The current value of '3'
surpasses the threshold '1'.</EventData></Event>
```

> The computer is the Impacted Host because there is no other context. Because the log came from this computer, it is the source of the log message.

- Cb Response. The endpoint is where the file originated in the scan, but is also likely impacted.

```
02 07 2017 17:30:21 1.1.1.1 <USER:NOTE>
LEEF:1.0|CB|CB|5.1|watchlist.storage.hit.binary|cb_server=cbserver    cb_version=525
copied_mod_len=8704 digsig_result=Unsigned     digsig_result_code=2148204800
endpoint=PIA-EX2010-01|2018       file_desc=
  file_version=1.1.1.1   group=Default Servers      host_count=1
internal_name=rwl_hdls.dll is_64bit=false   is_executable_image=false
last_seen=2017-02-07T23:26:29.825Z      legal_copyright=    link_md5=https://pia-
carbla-01.smchcn.net/#/binary/5F897E95044D43F58E30806857092186
md5=5F897E95044D43F58E30806857092186   observed_filename=c:\\windows\\temp\\rwl_hdls.dll
orig_mod_len=8704   original_filename=rwl_hdls.dll   os_type=Windows
product_version=1.1.1.1   server_added_timestamp=2017-02-07T23:26:29.825Z
server_name=localhost   timestamp=1486510220.266   type=watchlist.storage.hit.binary
watchlist_2=2017-02-07T23:30:03.972203Z    watchlist_id=2      watchlist_name=Default:
Newly Loaded Modules
```

> Because this is a security event that occurred on the endpoint, the endpoint is the Impacted Host. The other hosts involved (for example, CB server or agent reading syslog) are not relevant to the security context.

- CylancePROTECT. The threat originated from the device and IP, but is also impacted by the threat and the quarantine.

```
05 09 2016 01:33:03 1.1.1.1 <SLOG:WARN> 1
2016-05-09T06:32:55.1224002Z sysloghost CylancePROTECT - - - Event Type:
Threat, Event Name: threat_quarantined, Device Name: GQ-6FPLVZ1, IP Address: (1.1.1.1),
File Name: SOP.EXE, Path: E:\HESS\Corrosion\HESS Okume Lab C drive
```

```
Backup\NALCO\Okume CD
training\programme\OkumeBandC\ProdWellManifolds\fscommand\, Drive Type:
Internal Hard Drive, SHA256:
8050FE3DCA43D594611492AA149AF09FC9669149602BB2945AFEA4148A24B175
, MD5:
59E0D058686BD35B0D5C02A4FD8BD0E0
, Status: Quarantined, Cylance Score: 97, Found
Date: 1/7/2016 5:03:51 PM, File Type: Executable, Is Running: False, Auto Run:
False, Detected By: BackgroundThreatDetection
```

> Because this is a security event, the Device Name is the Impacted Host.

Polyfields and Parsing Field Aggregation

Not all fields that are parsed in a rule are stored in the Data Indexer as parsed or displayed in the console as parsed. For example, the Web Console Duration field is a calculation based on one or more time-based parsing fields. Similarly, there are more than a dozen fields for bytes as a size, but only one value is stored and only one value is displayed.

Polyfields are a special type of display field used for aggregating across similar source data. For example, the Impacted Host polyfield could contain a hostname, an IP address, or a well-known entity. The hostname and IP address may also be stored separately. The polyfield generally has preference logic at the code level to determine which source field to display.

When reading this document, pay particular attention to fields that are called out as source data for a polyfield, or parsing fields that are transformed into final data fields.

Application Tab

The fields in the Application tab describe the *Impacted Object* referenced by the log. The Application tab contains the most fields.

The following fields are on the Application tab:

- Action [7.2]
- Amount
- Command
- Hash [7.2]
- IANA Protocol Name
- IANA Protocol Number
- Object
- Object Name
- Object Type [7.2]
- Parent Process ID [7.2]
- Parent Process Name [7.2]
- Parent Process Path [7.2]
- Policy [7.2]
- Process ID
- Process Name
- Quantity
- Rate
- Reason [7.2]
- Response Code [7.2]
- Result [7.2]
- Session
- Session Type [7.2]
- Size
- Status [7.2]
- Subject
- URL
- User Agent [7.2]
- Version

Action [7.2]

Action is a broad field for what was done as described in the log. Action is usually a secondary function of a command or process.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Action
Client Console Short Name	Action
Web Console Tab/Name	Action
Elasticsearch Field Name	action
Rule Builder Column Name	Action
Regex Pattern	<action>
NetMon Name	Not applicable

Field Relationships

- Command
- Status
- Result
- Response Code
- Process

Common Applications

- Firewall
- Proxy
- Antivirus
- IDS/IPS
- Vulnerability scanner
- RIM/FIM

Use Case

- Recording network traffic accepts, drops, or blocks.
- Secondary function of a command—for example, PowerShell (process), might issue "AD commandlet" (command), which might have an action of lock out user.
- Action describes a mechanism. The result describes a state outcome. A firewall action can "pass" traffic. The result might be "success."

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Capture more simplistic actions than command might.
- An Action is what you are trying to initiate via a command.
- Action, Process, and Command separation:
  - A process is something "running."
  - A command is an operating system command (for example, batch) or a user originated command to a system.
  - The Action is often the "result" of a process or command. The A/V process (Symantec) might have a command of "Run Scan", which could have an Action of Quarantine.
- In RIM/FIM, the Action would be "read, write, add, delete" or any other common action verb applied to the file or registry key.

Examples

- FortiGate

```
02 18 2015 16:13:49 1.1.1.1 <LOC7:INFO> date=2015-02-18 time=16:13:51
devname=FG22222222222217 devid=FGdfsdfds1111111 logid=1059028704 type=utm subtype=app-
ctrl eventtype=app-ctrl-all level=information vd="root" appid=16637 user="pete.store"
srcip=1.1.1.1 srcport=57227 dstip=1.1.1.1 dstport=53 proto=17 service="DNS"
sessionid=391322221 applist="APPC Monitor All" appcat="Update" app="Sophos.Update"
action=pass msg="Update: Sophos.Update," apprisk=low
```

> In this case, the firewall action is to "pass" the traffic because it is on an approved list.

Amount

The qualitative description of quantity (percentage or relative numbers).

Data Type

Double

Aliases

Use	Alias
Client Console Full Name	Amount
Client Console Short Name	Amount
Web Console Tab/Name	Amount
Elasticsearch Field Name	amount
Rule Builder Column Name	Amount
Regex Pattern	<amount>
NetMon Name	Not applicable

Field Relationships

- Quantity
- Rate
- Size

Common Applications

- Point of Sale
- Hardware Monitoring

Use Case

- Capturing price into amount and quantity of items purchased to quantity for fraud analytics.
- Monitoring disk or CPU use and thresholds.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Currency amounts can be captured here.
- Percentages can be captured here.

Examples

- Huawei Access Router

```
04 07 2014 15:43:50 1.1.1.1 <LOC7:WARN> Apr  7 2014 13:43:49 USABLDRRECFLOW01 %%01CPUP/
4/CPU_USAGE_HIGH(l)[1237]:The CPU is overloaded, and the tasks with top three CPU
occupancy are HardIrq(80.8%), TICK(6.8%), ROUT(2.2%) . (CpuUsage=83%, Threshold=80%)
```

Indicates the percent amount of CPU usage on router.

Command

The specific command executed that has been recorded in the log message.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Command
Client Console Short Name	Command
Web Console Tab/Name	Command
Elasticsearch Field Name	command
Rule Builder Column Name	Command
Regex Pattern	<command>
NetMon Name	Not applicable

Field Relationships

- Result
- Status
- Process
- Action

Common Applications

- PowerShell
- Windows Command Shell
- SSH
- Telnet
- Bash

Use Case

- Cron
- Sudo
- Auditing

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Vendor Message ID is a unique event type identifier.
- Command identifies an executable or script with arguments.
- May contain an executable, but is distinct from Process.
- Can describe the execution of a process.
- Command within a process.
- Often specifically called out as CMD or Command.
- Not Action (for example, Firewall Block/Allow).
- Not Result (Command can have a Result).
- Command may describe Action.

Examples

Correct Examples

- CrowdStrike FalconHost

```
12 14 2016 18:53:39 1.1.1.1 <USER:NOTE> CEF:0|CrowdStrike|FalconHost|1.0|ScanResults|AV
Scan Results In A Detection Summary Event|4| externalID=2222222222222222eee799
cn2Label=ProcessId cn2=148181079514282 shost=WIN-HPBKBMLLSST suser=pete.store
fname=GoogleUpdate.exe filePath=\\Device\\HarddiskVolume1\\Users\\pete.store\\AppData\
\Local fileHash=e361a8c5da2e3d1a0ed3be85ed906dad cs1Label=CommandLine cs1="C:\\Users\
\pete.store\\AppData\\Local\\GoogleUpdate.exe" sntdom=safaware cs2Label=ScanResultEngine
cs2=AVware cs3Label=ScanResultName cs3=Trojan-Downloader.Win32.Fraudload
cn4Label=ScanResultVersion cs4=1.1.1.1 cs6Label=FalconHostLink cs6=https://
falcon.crowdstrike.com/activity/detections/detail/
ec3f4ca727a04f025f2ea97647a61799/222222222 cn3Label=Offset cn3=1066242
```

Specifically called out Command Line, even though it is an executable.

- CrowdStrike FalconHost

```
12 15 2016 00:19:05 1.1.1.1 <USER:NOTE> CEF:0|CrowdStrike|FalconHost|1.0|ScanResults|AV
Scan Results In A Detection Summary Event|3|
externalID=022222222222222222ea584f3783f5b1eee9 cn2Label=ProcessId cn2=1482087830222222
shost= USABLDRRECFLOW01suser=Pete.Store fname=upnp.exe filePath=\\Device\
```

```
\HarddiskVolume1\\Users\\pete.store\\AppData\\Local\\Temp
fileHash=13804f8dc4e72ba103d5e34de895c9db cs1Label=CommandLine cs1="C:\\Users\\ALVINF~1\
\AppData\\Local\\Temp\\upnp.exe" -a 1.1.1.1 1604 1604 TCP sntdom=safaware
cs2Label=ScanResultEngine cs2=TrendMicro cs3Label=ScanResultName
cs3=TROJ_GEN.R0FBC0CI116 cn4Label=ScanResultVersion cs4=1.1.1.12 cs6Label=FalconHostLink
cs6=https://falcon.crowdstrike.com/activity/detections/detail/
02c60e7a579b4fea584f3783f5b1eee9/222222222 cn3Label=Offset cn3=1066392
```

Executable with arguments.

- AIX

```
02 20 2013 09:16:33 1.1.1.1 <SAU1:NOTE> Feb 20 09:16:33 Message forwarded from
USABLDRRECFLOW01: sudo:  dt14437 : TTY=pts/0 ; PWD=/dst/home/omg37 ; USER=root ;
COMMAND=/usr/bin/crontab -l
```

Command called out explicitly.

- Proofpoint Spam Firewall

```
12 07 2011 14:19:10 1.1.1.1 <USER:NOTE> Dec  7 14:19:10 filter_instance1 rprt
s=11huq2222 m=1 x=pB7JJAlE02222 mod=access cmd=run rule=spamsafe duration=0.000
```

Run is the Command, not the Process.

Incorrect Examples

- Check Point Firewall

```
26Feb2013 14:59:21 Product=VPN-1 & FireWall-1 OriginIP=1.1.1.1 Origin=
USABLDRRECFLOW01Action=encrypt SIP=1.1.1.1 Source= USABLDRRECFLOW01SPort=0 DIP=1.1.1.1
Destination= USABLDRRECFLOW01DPort=0 Protocol=icmp ICMPType=8 ICMPCode=0 IFName=eth1
IFDirection=inbound Reason=- Rule=32 Info=- XlateSIP=1.1.1.1 XlateSPort=- XlateDIP=-
XlateDPort=-
```

Encrypt is not a command. Encrypt is better parsed into Action.

- Juniper Firewall

```
04 22 2012 17:28:13 1.1.1.1 <USER:INFO> 1 2012-04-23T08:27:25.564  RT_FLOW -
RT_FLOW_SESSION_CLOSE [junos@21.1.1.1.2.41 reason="unset" source-address="1.1.1.1"
source-port="138" destination-address="1.1.1.1" destination-port="138" service-
name="junos-nbds" nat-source-address="1.1.1.1" nat-source-port="138" nat-destination-
address="1.1.1.1" nat-destination-port="138" src-nat-rule-name="None" dst-nat-rule-
```

```
name="None" protocol-id="17" policy-name="allowAll" source-zone-name="trust"
destination-zone-name="trust" session-id-32="21434" packets-from-client="1" bytes-from-
client="229" packets-from-server="0" bytes-from-server="0" elapsed-time="59"
application="UNKNOWN" nested-application="UNKNOWN" username="N/A" roles="N/A" packet-
incoming-interface="fe-0/0/7.0"]
```

RT_FLOW_SESSION_CLOSE is not a command. RT_FLOW_SESSION_CLOSE is VMID.

- Palo Alto Firewall

```
02 24 2015 15:21:01 1.1.1.1 <USER:INFO> Feb 24 15:21:01 1,2015/02/24
15:21:01,0011C100222,TRAFFIC,drop,0,2015/02/24 15:21:01,1.1.1.1,1.1.1.1,1.1.1.1,1.1.1.1,d
enyall,,,not-applicable,vsys1,dmz,inet,ethernet1/9,,LogRhythm-Receiver,2015/02/24
15:21:00,0,1,64812,443,0,0,0x0,tcp,deny,66,66,0,1,2015/02/24 15:21:02,0,any,
0,27629666933,0x0,United States,United States,0,1,0
```

Drop is not the Command. Drop is the Action. Denyall is not Command either. Denyall is closer to Result (could also be the name of a Policy).

Hash [7.2]

The hash value (for example, MD5 or SHA256) of a file, process, or object. The value is independent of the algorithm. Only the resulting hash is stored in this field.

Only three hash types are in common usage: MD5, SHA1, and SHA256.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

Alphanumeric string (0-512 characters, 64 average characters)

Aliases

Use	Alias
Client Console Full Name	Hash
Client Console Short Name	Hash
Web Console Tab/Name	Hash
Elasticsearch Field Name	hash
Rule Builder Column Name	Hash
Regex Pattern	<hash>
NetMon Name	Not applicable

Field Relationships

Object, Process, and Object Name fields. This is the hash for the process identified in process.

Common Applications

- IDS/IPS

- Vulnerability scanners
- Endpoint monitoring (for example, Cbresponse)
- Threat Intelligence feeds
- Antivirus

Use Case

Mapping hash value to threat feeds and known Indictators of Compromise (IOCs).

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Priority if there are multiple hashes is MD5 > SHA1 > SHA256, until strongly typed fields available.
- Make it as easy as possible to match to most common threat feeds.
- Do not include the hash type in the field (for example, remove MD5:).

Examples

- Cylance log sample

```
Sample - 05 09 2016 21:40:29 1.1.1.1 <SLOG:WARN> 1 2016-05-10T02:40:19.2905167Z
syslogghost CylancePROTECT - - - Event Type: AppControl, Event Name: pechange, Device
Name: US-JNTJKV1, IP Address: (1.1.1.1, 1.1.1.1,), Action: Deny, Action Type: PE File
Change, File Path: C:\Users\Public\TechTools\Host65, SHA256:
 8050FE3DCA43D594611492AA149AF09FC9669149602BB2945AFEA4148A24B175
```

> Parse the hash removing the algorithm header SHA256.

- Cb Response log sample

```
Sample - 05 13 2016 20:56:15 1.1.1.1 <USER:NOTE> LEEF:1.0|CB|CB|5.1|
watchlist.hit.binary|cb_server=cbserver    cb_version=511    company_name=Microsoft
Corporation    copied_mod_len=11616    digsig_issuer=Microsoft Windows Production PCA
2011    digsig_prog_name=Microsoft Windows    digsig_publisher=Microsoft Corporation
digsig_result=Signed    digsig_result_code=0    digsig_sign_time=2015-10-30T12:32:00Z
digsig_subject=Microsoft Windows    endpoint=[" USABLDRRECFLOW01"]
file_desc=recordflow console    file_version=10.0.10.0 (th2_release.151029-1700)
group=["Testing"]    host_count=1    internal_name=recflowcon    is_64bit=true
is_executable_image=false    last_seen=2016-05-14T03:42:10.709Z    legal_copyright=©
Record Flow LLC. All rights reserved.    md5=59E0D058686BD35B0D5C02A4FD8BD0E0observed_fi
lename=["c:\\windows\\system32\\downlevel\\api-ms-win-core-stringansi-l1-1-0.dll"]
orig_mod_len=11616    original_filename=apisetstub    os_type=Windows
product_name=Microsoft® Windows® Operating System    product_version=10.0.10586.0
server_added_timestamp=2016-05-14T03:42:10.709Z    server_name=USABLDRRECFLOW01
```

```
signed=Signed    timestamp=2016-05-14T03:42:10.709Z    type=watchlist.hit.binary
watchlist_id=4    watchlist_name=Newly Loaded Modules
```

> Parse the hash removing the type md5=.

IANA Protocol Name

The IANA Protocol Name representing the official registered name for well-known network protocols. For more information, see RFC 5237 and RFC 7045.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Known Application
Client Console Short Name	Not applicable
Web Console Tab/Name	Application
Elasticsearch Field Name	application/protocolName/serviceName
Rule Builder Column Name	<protname>
Regex Pattern	<protname>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain

- DIPv6E
- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- Origin Login
- Impacted Account
- IANA Protocol Number

Common Applications

- Firewalls
- IDS/IPS
- NetMon

Use Case

Classifying network traffic.

MPE/Data Masking Manipulations

Compares to list of IANA Protocol Names and is shown in Known Application in the Client Console or Application in the Web Console.

Usage Standards

- Only parse IANA Protocol Names in this field.
- If both Protocol Number and Protocol Name are present in a log, parse Protocol Number.
- For Protocol Names and Numbers, see https://www.iana.org/assignments/protocol-numbers/protocol-numbers.xhtml

Examples

- FortiGate

```
12 12 2016 12:18:55 1.1.1.1 <LOC7:ALRT> date=2016-12-12 time=12:18:55 devname=ABC-DEF-
FORTIGATE-02 devid=FG80050000000 logid=0419016385 type=utm subtype=ips
eventtype=signature level=alert vd=root severity=low srcip=1.1.1.1 srccountry="Reserved"
dstip=1.1.1.1 srcintf="WIFI_NETWORK" dstintf="VLAN" policyid=380 sessionid=24634444
action=dropped proto=1 service="PING" attack="Traceroute" icmpid=0x6425 icmptype=0x08
icmpcode=0x00 attackid=12466 profile="IPS_WEB_OUT" ref="http://Host1/ids/VID12345"
incidentserialno=123456789 msg="icmp: Traceroute," crscore=5 crlevel=low
```

> Service corresponds with proto=1 which is ICMP (Ping). Service can sometimes indicate an IANA Protocol Name instead of a process. For more information, see http://www.iana.org/assignments/protocol-numbers/protocol-numbers.xhtml.

- Juniper Firewall

```
11 06 2009 12:09:51 1.1.1.1 <SAU1:CRIT> dc-dp-1: NetScreen device_id=dc-dp-1
[Root]system-critical-00033: Src IP session limit! From 1.1.1.1:11698 to 1.1.1.1:49156,
proto UDP (zone DAVE-PK1 int  ethernet0/0.3). Occurred 16 times. (2010-11-06 12:09:50)
```

Proto shows the Protocol Name UDP instead of a number. Corresponds to protocol number 17. For more information, see http://www.iana.org/assignments/protocol-numbers/protocol-numbers.xhtml.

IANA Protocol Number

The Internet Assigned Numbers Authority (IANA) Protocol Number represents the official registered ID for well-known network protocols. For more information, see RFC 5237 and RFC 7045.

Data Type

Integer (0 to 255)

Aliases

Use	Alias
Client Console Full Name	Known Application
Client Console Short Name	Not applicable
Web Console Tab/Name	Application
Elasticsearch Field Name	application/protocolId/serviceName
Rule Builder Column Name	Protnum
Regex Pattern	<protnum>
NetMon Name	Application (remapped by syslog parser)

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain

- DIPv6E
- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- Origin Login
- Impacted Account
- IANA Protocol Name

Common Applications

- Firewalls
- IDS/IPS

Use Case

Classifying network traffic.

MPE/Data Masking Manipulations

Compares to a list of IANA Protocol Numbers and is shown in Known Application in the Client Console or Application in the Web Console.

Usage Standards

- Do not overload this field. It maps to a table in the SIEM (protocol).
- Only parse IANA Protocol Numbers in this field.
- If both the Protocol Number and Protocol Name are present in a log, parse the Protocol Number.
- For Protocol Names and Numbers, see https://www.iana.org/assignments/protocol-numbers/protocol-numbers.xhtml.

Examples

- FortiGate

```
12 12 2016 12:18:55 1.1.1.1 <LOC7:ALRT> date=2016-12-12 time=12:18:55 devname=ABC-DEF-
FORTIGATE-02 devid=FG000000000000 logid=042006385 type=utm subtype=ips
eventtype=signature level=alert vd=root severity=low srcip=1.1.1.1 srccountry="Reserved"
dstip=1.1.1.1 srcintf="WIFI_NETWORK" dstintf="VLAN" policyid=4 sessionid=5156446
action=dropped proto=1 service="PING" attack="Traceroute" icmpid=0x6425 icmptype=0x08
icmpcode=0x00 attackid=12466 profile="IPS_WEB_OUT" ref="http://Host1/ids/VID5555"
incidentserialno=5000000000 msg="icmp: Traceroute," crscore=5 crlevel=low
```

> Proto (short for protocol) typically indicates IANA Protocol Numbers or Protocol Names. In this case, proto represents a number. Proto=1 corresponds to ICMP (Ping). For more information, see http://www.iana.org/assignments/protocol-numbers/protocol-numbers.xhtml.

- Cisco Netflow

```
02 19 2014 06:41:03 NetFlow V9 CONN_ID=- Src=1.1.1.1 SPort=57534 InIfc=4 Dst=1.1.1.1
DPort=8612 OutIfc=9 Prot=17 ICMP_IPV4_TYPE=- ICMP_IPV4_CODE=- XLATE_SRC_ADDR_IPV4=-
XLATE_DST_ADDR_IPV4=- XLATE_SRC_PORT=- XLATE_DST_PORT=- FW_EVENT=- FW_EXT_EVENT=-
EVENT_TIME_MSEC=- IN_PERMANENT_BYTES=- DETAILS=CONN_ID=1632425523 ICMP_IPV4_TYPE=0
ICMP_IPV4_CODE=0 XLATE_SRC_ADDR_IPV4=1.1.1.1 XLATE_DST_ADDR_IPV4=1.1.1.1
XLATE_SRC_PORT=57534 XLATE_DST_PORT=8612 FW_EVENT=2 FW_EXT_EVENT=2013
EVENT_TIME_MSEC=1392835263526 IN_PERMANENT_BYTES=16 DefaultDevice TemplateID=263
```

Prot indicates an IANA Protocol Number, corresponding to UDP. For more information, see http://www.iana.org/assignments/protocol-numbers/protocol-numbers.xhtml.

Object

The resource (file) referenced or impacted by activity reported in the log, except when another schema field is more precisely relevant.

The following fields should be used if they are more relevant:

- *Process*. For anything clearly executable or running as a process.
- *Action*. Data explicitly classified as an action (for example, block traffic).
- *Result*. Result of a process (for example, HTTP result codes).
- *Status*. Explicit status as presented by log source.
- *Reason*. Explicit reason as presented by log source.
- *Policy*. Explicit policy.
- *Command*. Command executed by log source.
- *Threat Name*. Explicit threat name (for example, APT1).
- *CVE*. Explicit CVE in standard CVE format.
- *Hash*. Explicitly generated Hash field. For more information, see Hash.
- *Vendor Information*. Additional information from vendor (beyond the Vendor Message ID or VMID).
- *UserAgent*. User agent string for web traffic.
- Anything that can be inferred into the LogRhythm Entity, Location or Network.

Data Type

String (1000 characters maximum)

Aliases

Use	Alias
Client Console Full Name	Object
Client Console Short Name	Object
Web Console Tab/Name	Object
Elasticsearch Field Name	object
Rule Builder Column Name	Object
Regex Pattern	<object>

Use	Alias
NetMon Name	Not applicable

Field Relationships

- Object Name
- Object Type
- Hash

Common Applications

- Stores a resource being mentioned in the log message.
- Can be used in almost every log source type.

Use Case

Finding a specific known resource for log source type (for example, searching for a specific database name).

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

Do not use in the following cases:

- When another schema field is more appropriate to describe the resource (Process, Dname, Hash, Sender, Command, Recipient, Subject, etc.).
- When describing a LogRhythm-defined entity.
- To describe an event. Object describes an event's target.

Examples

Correct Examples

- Windows System Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-DHCP-Server' Guid='{6d44402c-a145-4dac-9a01-f0555b41ca84}'
EventSourceName='DhcpServer'/><EventID Qualifiers='0'>1020</EventID><Version>0</
Version><Level>Warning</Level><Task>None</Task><Opcode>Info</Opcode><Keywords>Classic</
Keywords><TimeCreated SystemTime='2016-08-02T13:14:16.000000000Z'/
><EventRecordID>1340877</EventRecordID><Correlation/><Execution ProcessID='0'
ThreadID='0'/><Channel>System</Channel><Computer>IVNMKDP.DKVM.IVAN.log.cz.ru.biz</
```

```
Computer><Security/></System><EventData><Data>1.1.1.1</Data><Data>100</Data><Data>0</
Data></EventData></Event>
```

> The IP represents the IP Scope of this DHCP log, so it is the referenced object in this context. It is not appropriate to use SIP/DIP/SNATIP/DNATIP because the data field does not represent a host.

- Windows System Log

```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
 Name='Microsoft-Windows-Kernel-PnP' Guid='{9c343432439-12340-48324d-abhh7-
e831c6sdf4539}'/><EventID>219</EventID><Version>0</Version><Level>Warning</
Level><Task></Task><Opcode>Info</Opcode><Keywords></Keywords><TimeCreated
SystemTime='2016-08-03T01:44:55.547851500Z'/><EventRecordID>5823877</
EventRecordID><Correlation/><Execution ProcessID='4' ThreadID='88'/><Channel>System</
Channel><Computer>IVNMKDP.DKVM.IVAN.log.cz.ru.biz</Computer><Security UserID='NT
AUTHORITY\SYSTEM'/></System><EventData><Data Name='DriverNameLength'>60</Data><Data
Name='DriverName'>PCI\VEN_8086&DEV_7020&SUBSYS_110434AF4&REV_01\3&13c0b0c5&0&0A</
Data><Data Name='Status'>3221226382</Data><Data Name='FailureNameLength'>15</Data><Data
Name='FailureName'>\Driver\usbuhci</Data><Data Name='Version'>0</Data></EventData></
Event>
```

> Object is the specific driver component that failed to load (\driver\usbuhci). ObjectName is the DriverName. Both are correct as the referenced object is the driver component, and ObjectName expands on this with the full driver name.

- Office 365 Exchange Logs

```
TS=2016-03-03T01:17:28 SESSID=50e4435a-45e6-42de-7ae3-08d13419636 COMMAND=Set-
TransportConfig USERTYPE=DcAdmin USERKEY=NT AUTHORITY\SYSTEM
(Microsoft.Exchange.ServiceHost) WORKLOAD=Exchange RESULTCODE=True OBJECT=ivantesto365.o
nmicrosoft.com\Transport Settings USER=NT AUTHORITY\SYSTEM
(Microsoft.Exchange.ServiceHost) SIP= OBJECTNAME=
PARAMETERS=[{"Name":"DomainController","Value":""},
{"Name":"Identity","Value":"lrtesto365.onmicrosoft.com"},
{"Name":"HygieneSuite","Value":"Premium"}] MODIFIEDPROPERTIES= EXTERNALACCESS=True
ORIGINATINGSERVER=ivandave0298 (15.31.05654.011) ORGANIZATIONNAME=ivantesto365.onmicroso
ft.com LOGONTYPE= MAILBOXOWNER= MAILBOXMASTER= LOGONUSERSID= LOGONUSERDISPLAYNAME=
USERAGENT= CLIENTIPADDRESS= CLIENTPROCESSNAME= CLIENTVERSION= FOLDER=
CROSSMAILBOXOPERATIONS= DESTMAILBOX= DESTMAILBOXOWNER= DESTMAILBOXMASTER= DESTFOLDER=
FOLDERS= AFFECTEDITEMS= ITEM= SENDASUSER= SENDONBEHALFOFUSER=
```

> ivantesto365.onmicrosoft.com\Transport Settings parses into Object because this setting is recorded as modified in the log.

Incorrect Examples

- Sensitive Data

```
04/10-07:08:54.002765  [**] [139:1:1] SDF_COMBO_ALERT [**] [Classification: Sensitive
Data was Transmitted Across the Network] [Priority: 2] {PROTO:254} 1.1.1.1 -> 1.1.1.1
```

> SDF_COMBO_ALERT parses into Object. This is incorrect because SDF_COMBO_ALERT indicates the type of log message, rather than what object is impacted or referenced in the log. In this example, the Object field should not be used.

- Cisco ACS

```
06 07 2013 09:13:19 1.1.1.1 <LOC6:NOTE> Jun  7 09:13:19 mrk-prd-acs
CSCOacs_TACACS_Accounting 0000819174 2 1  NetworkDeviceGroups=Location:All
Locations:DPDC, AuditSessionId=davemon:1.1.1.1:tty1:1.1.1.1, Response={Type=Accounting;
AcctReply-Status=Success; }
```

> The Success value is parsed incorrectly from the key status into Object. It should parse into Status instead. In this example, the Object field should not be used.

- Symantec Endpoint Server

```
01 28 2015 16:15:37 1.1.1.1 <LPTR:INFO> Jan 28 16:01:42 SymantecServer MVK-GDF-01:
hostname,Local: 1.1.1.1,Local: 0,Local: 010000000001,Remote: 1.1.1.1,Remote: ,Remote:
0,Remote: 5156165156RS,7,Inbound,Begin: 2015-01-28 15:54:39,End: 2015-01-28
15:54:39,Occurrences: 1,Application: ,Rule: Block all other traffic,Location: Corporate
Network,User: Dave_Store,Domain: DP,Action: Blocked
```

> The Location value should not parse into Object, as this can be inferred, and entities can be used to gather this type of data. Location should be tied to the entity structure. In this case, the Object field should not be used. Application in a log could be Process or Object, depending on the analysis of additional samples.

- Snare 2008 Event Log

```
08 28 2016 23:03:14 1.1.1.1 <USER:NOTE> Aug 28 23:03:14 DAVEWINDOW.loc.gregsports.com
 MSWinEventLog     1     Application 15450631    Sun Aug 28 23:03:14 2016    1026   .NET
Runtime       N/A    N/A    Error       DAVEWINDOW.loc.gregsports.com None
Application: pptviewerbackendwatchdog.exe Framework Version: v4.0.30319 Description: The
process was terminated due to an unhandled exception. Exception Info:
System.TypeInitializationException…
```

> Application should be parsed into Process because it is an executable. In this example, the Object field should not be used.

Ambiguous Examples

- Riverbed

```
01 24 2014 02:57:25 1.1.1.1 <LOC0:NOTE> Jan 24 02:57:25 IVNDPMVK01 rbmd[10763]:
[rbmd.NOTICE]: Connecting to appliance IVAN48564546TV
```

> The log notice may be a hostname or a device name (such as an AP). It is ambiguous whether this strictly meets the definition of object impacted, object referenced, or something else. In this case, the field could be a device, serial number, or other identifier. Object is not incorrect, but this log source should be researched further.

- Microsoft Application Log

```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
 Name='MsiInstaller'/><EventID Qualifiers='0'>1022</EventID><Level>Information</
Level><Task>None</Task><Keywords>Classic</Keywords><TimeCreated
SystemTime='2014-11-05T08:50:01.000000000Z'/><EventRecordID>9442</
EventRecordID><Channel>Application</Channel><Computer>IVNMKDP.DKVM.IVAN.log.cz.ru.biz</
Computer><Security UserID='NT AUTHORITY\SYSTEM'/></System><EventData>Product:
Microsoft .NET Framework 4.5 - Update 'KB2979578v2' installed successfully.</
EventData></Event>
```

> Product could parse into Process instead of Object. Object is not incorrect, but may be confusing. In this case, the product does not define a runnable process on the system, so Object is a better choice than Process.

- Windows Application Log

```
<Event xmlns='http://Host1/win/2004/08/events/event'><System><Provider
 Name='SQLSERVERAGENT'/><EventID Qualifiers='16384'>208</EventID><Level>Warning</
Level><Task>Job Engine</Task><Keywords>Classic</Keywords><TimeCreated
SystemTime='2015-07-23T18:20:39.000000000Z'/><EventRecordID>2042567</
EventRecordID><Channel>Application</Channel><Computer>IVNMKDP.DKVM.IVAN.log.cz.ru.biz </
Computer><Security/></System><EventData>SQL Server Scheduled Job 'LogRhythm Sunday
Maintenance' (0x7BN5C000E7A34C90000000D2F3) - Status: Failed - Invoked on: 2015-07-23
12:20:38 - Message: The job failed.  The Job was invoked by User sa.  The last step to
run was step 29 (LogRhythm Job Step Validation).  The job was requested to start at step
29 (LogRhythm Job Step Validation).</EventData></Event>
```

> Job name parses into Object. However, it is ambiguous whether job is an object, an action, or a process.

- Windows Security Log

```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
 Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-
a5ba-3e3b0328c30d}'/><EventID>4907</EventID><Version>0</Version><Level>Information</
```

```
Level><Task>Audit Policy Change</Task><Opcode>Info</Opcode><Keywords>Audit Success</
Keywords><TimeCreated SystemTime='2016-02-26T06:56:10.852896900Z'/
><EventRecordID>228903233</EventRecordID><Correlation/><Execution ProcessID='752'
ThreadID='768'/><Channel>Security</Channel><Computer>IVNMKDP.DKVM.IVAN.log.cz.ru.biz</
Computer><Security/></System><EventData><Data Name='SubjectUserSid'>IVN\dave.crowley</
Data><Data Name='SubjectUserName'>dave.crowley</Data><Data
Name='SubjectDomainName'>IVN</Data><Data Name='SubjectLogonId'>0x10be65</Data><Data
Name='ObjectServer'>Security</Data><Data Name='ObjectType'>File</Data><Data
Name='ObjectName'>C:\Windows\System32\autochk.exe</Data><Data Name='HandleId'>0xb0</
Data><Data Name='OldSd'>S:AI</Data><Data Name='NewSd'>S:(AU;SAFA;DCLCSDSDWDWO;;;WD)</
Data><Data Name='ProcessId'>0x3298</Data><Data Name='ProcessName'>C:\Program Files
(x86)\Google\Chrome\Application\chrome.exe</Data></EventData></Event>
```


File parses into Object, and the XML field is ObjectType, so it is a good candidate for ObjectType. Autochk and path parses into ObjectName, and XML calls this ObjectName as well.

Object Name

The resource name (filename) referenced or impacted by activity reported in the log, specifically related to what is parsed into Object.

Object Name is a friendly name or expanded information about the Object. Do not use Object Name if Object is not also parsed.

Object Name is normalized into the star schema of the Events database (LogRhythm_Events.dbo.Object).

Data Type

String (1000 characters maximum)

Aliases

Use	Alias
Client Console Full Name	Object Name
Client Console Short Name	Object Name
Web Console Tab/Name	Object Name
Elasticsearch Fieldname	objectName
Rule Builder Column Name	ObjectName
Regex Pattern	<objectname>
NetMon Name	Not applicable

Field Relationships

- Object is described by Object Name
- Object Type

Common Applications

Everywhere that Object is used and a friendly name exists.

Use Case

- Getting context about an Object.
- Not likely to be a primary search field.
- Not likely to be a major field in AIE rules.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Object and Object Name are context-sensitive to the log itself. They must be defined for each device and device family across multiple samples.
    - Object is primary and required to be filled first. Object Name is secondary and optional.
    - Object Name is an expanded or friendly name of the object, not necessarily the file or process name (Object).
- For any database log:
    - Object is the name of the database.
    - Object Name should only be used if there is a human readable name in addition.
- Do not use Object Name with any other speciality field, such as session, process, URL, and so on.

Examples

Correct Examples

- Windows Security Event Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Security-Auditing' Guid='{54559625-5478-4994-
a5ba-3e3b0328c30d}'/><EventID>4907</EventID><Version>0</Version><Level>Information</
Level><Task>Audit Policy Change</Task><Opcode>Info</Opcode><Keywords>Audit Success</
Keywords><TimeCreated SystemTime='2016-02-26T06:56:10.852896900Z'/
><EventRecordID>228903233</EventRecordID><Correlation/><Execution ProcessID='752'
ThreadID='768'/><Channel>Security</Channel><Computer>log.log.log</Computer><Security/></
System><EventData><Data Name='SubjectUserSid'>log\dave.crowley</Data><Data
Name='SubjectUserName'>dave.crowley</Data><Data Name='SubjectDomainName'>log</Data><Data
Name='SubjectLogonId'>0x10be65</Data><Data Name='ObjectServer'>Security</Data><Data
Name='ObjectType'>File</Data><Data Name='ObjectName'>C:\Windows\System32\autochk.exe</
Data><Data Name='HandleId'>0xb0</Data><Data Name='OldSd'>S:AI</Data><Data
Name='NewSd'>S:(AU;SAFA;DCL545RSDWDWO;;;WD)</Data><Data Name='ProcessId'>0x3298</
Data><Data Name='ProcessName'>C:\Program Files
(x86)\Google\Chrome\Application\chrome.exe</Data></EventData></Event>
```

> File parses into Object—though Object Type would be better. Autochk.exe parses into Object Name appropriately.

- Windows Security Event Log

```
<Event xmlns='http://Host3/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{548465416845-5478-4994-a5ba-3e3b0328c30d}'/><EventID>6144</EventID><Version>0</Version><Level>Informationen</Level><Task>Andere Richtlinienänderungsereignisse</Task><Opcode>Info</Opcode><Keywords>Überwachung erfolgreich</Keywords><TimeCreated SystemTime='2016-03-15T17:52:23.176154700Z'/><EventRecordID>57042720</EventRecordID><Correlation/><Execution ProcessID='524' ThreadID='2808'/><Channel>Security</Channel><Computer>Host2l</Computer><Security/></System><EventData><Data Name='ErrorCode'>0</Data><Data Name='GPOList'>{31b2f340-016d-11d2-945f-00c04fb984f9}      Default Domain Policy </Data></EventData></Event>
```

> The string for GPOList parses into Object. The Default Domain Policy parses into Object Name.

- Cisco Unified Communication Mgr

```
11 09 2009 00:22:45 1.1.1.1 <LOC7:ERRR> 157: : : 125: Nov 09 05:22:03.34 UTC : %CCM_CALLMANAGER-CALLMANAGER-3-DeviceTypeMismatch: Device type mismatch. Name of device.:DAV002454654BA Device type.:436 Database device type:435 App ID:Cisco CallManager Cluster ID:CORP-DP001 Node ID:CORP0004-D31005
```

> Cluster ID parses into Object. Node ID parses into Object Name.

- Voltage Securemail

```
01 29 2015 01:02:20 1.1.1.1 <USER:DBUG> voltage: LogMsgID="3", ServerNode="MVBK1", TenantID="LOG.BIZ.RU", SubTenant="<default>", Created="2015-01-29 01:02:20.673", Status="0", Summary="Authentication being handled for pete.store@recordflow.biz", EventLevel="Verbose", SessionID="1odz45646546dfdf3gscuijtpiv8", RequestID="1191", SourceName="IDAdapterEvents", EventName="Auth", Service="VSIBE", ClusterName="GH Data Center", ClusterUID="1", IPAddress="1.1.1.1", TenantUID="36", UserAgentType="2", Identity=" pete.store@recordflow.bizrecordflow.biz", AdapterType="vs.enrollment", AdapterID="2435885555110902908 8", Result="4", Duration="9", Details="null"
```

> Vs.enrollment parses into Object. The numeric string for AdapterID parses into Object Name.

Ambiguous Examples

- NAC System – FortiGate

```
07 23 2016 20:00:12 1.1.1.1 <LOC7:NOTE> date=2016-07-23 time=23:00:11 devname=logfw
devid=FG5555555RecFlw1600315 logid=0100043777 type=event subtype=system level=notice
vd="Transparent" logdesc="NAC anomaly quarantine" srcip=1.1.1.1 dstip=2.2.22
src_int="port1" dst_int="N/A" srcport=0 dstport=0 proto=0 service="ip" action=ban-ip
user="N/A" group="N/A" policyid=0 banned_src=dos banned_rule="tcp_dst_session"
sensor="DoS-policy1"
```

Banned_src and banned_rule parse into Object and Object Name, respectively. These are ambiguous because the source and rule are related to one another, but source refers to a denial of service attack, which is more of an action than a resource.

In this case, banned_rule could be parsed into Policy and banned_src could parse into Object (because the rule acted on the "dos" src).

- Postgres

```
07 15 2015 14:59:42 1.1.1.1 <LOC4:INFO> Jul 15 14:59:43 src@Host70lt0 postgres[26940]:
[708937-1] user=hasselhoff,db=recordflow_dev LOG:  duration: 929.018 ms  execute
<unnamed>: UPDATE jobs.TRIGGERS SET TRIGGER_STATE = $1 WHERE SCHED_NAME =
'schedulerFactoryBean' AND JOB_NAME = $2 AND JOB_GROUP = $3 AND TRIGGER_STATE = $4
```

Database and Log parse into Object Name and Object, respectively. A database meets the criteria of a resource referenced or impacted in this log. However, the log seems closer to a command, action, or result (log parses into Command).

The database value should parse into Object, and the log should parse into Command. Object Name should not be used.

- Two logs from FortiGate with URLs

```
08 21 2016 02:16:52 1.1.1.1 <LOC1:ALRT>
date=2016-08-21,time=02:17:46,devname=FG123456456,devid=FG5445645641,logid=0419016384,ty
pe=utm,subtype=ips,eventtype=signature,level=alert,vd="root",severity=low,srcip=1.1.1.1,
dstip=1.1.1.1,srcintf="port16",dstintf="port16",policyid=1,sessionid=22078931,action=det
ected,proto=6,service=tcp/
20480,attack="MS.IIS.Web.Server.Folder.Traversal.Evasion",srcport=53355,dstport=80,hostn
ame="1.1.1.1",direction=outgoing,attackid=15152,profile="all_default",ref="http://
www.fortinet.com/ids/VID5555",incidentserialno=1981412111,msg="web_server:


MS.IIS.Web.Server.Folder.Traversal.Evasion,",crscore=10,crlevel=medium
```

```
07 23 2016 20:00:12 1.1.1.1 <LOC7:ALRT> date=2016-07-23 time=23:00:11
devname=zackasdsd3343434 devid=FG5555121321 logid=0720018432 type=anomaly subtype=anomal
y level=alert vd="Transparent" severity=critical srcip=1.1.1.1 dstip=1.1.1.1
srcintf="port1" sessionid=0 action=detected proto=6 service=SNMP count=802
attack="tcp_src_session" srcport=36078 dstport=162 attackid=4544654 policyid=1 ref="http
```

```
://www.fortinet.com/ids/VID1511112" msg="anomaly: tcp_src_session, 1251 > threshold
1250, repeats 802 times" crscore=50 crlevel=critical
```

The domain of the URL parses into Object Name in the referrer field in both logs. Strictly speaking, this is a referenced object, but Object is not used in the first log, so there is no relation. In the second log, Subtype parses into Object and the domain of the URL parses into Object Name. There is no relation between these fields in the second instance, as subtype describes the event rather than a resource.

In these logs, the ref field defines an outside URL to additional information. It is not the object of the log or the name of the object. The ref field should parse into the Vendor Information field. There is no need to have an Object or Object Name for this log source.

- Entrust entillgence messaging server - User Credentials

```
06 07 2013 09:29:36 1.1.1.1 <LOC3:WARN> ECD[12901]: b7fd WARN ECD: (31516556428) Warning
of credential expiry.  Details [[friendlyName=Onboard SSL credential for www.recordflow.b
iz][days since expiry:161]]
```

Friendly name parses into Object Name and the subsequent hostname parses into Object. Object should parse into impacted host (dname) in this log. Object Name is strictly correct with the usage of object for the hostname, but would probably be better for Object after that is changed to dname.

If onboard SSL Credential parses into Object, then Object Name is empty. Also, the rule name and common event probably captures it already "credential expiry." Look at other samples to see if there are other types of credential besides the one shown here.

- Microsoft Antimalware

```
4/24/2013 4:03 PM TYPE=Warning USER= COMP=Host1 SORC=Microsoft Antimalware CATG=(0)
EVID=1116 MESG=Microsoft Antimalware has detected malware or other potentially unwanted
software.  For more information please see the following: http://Host3/fwlink/?
linkid=37020&name=Worm:Win32/Vobfus.PQ&threatid=2147680921        Name: Worm:Win32/
Vobfus.PQ    ID: 214764421     Severity: Severe        Category: Worm    Path: file:_C:
\Documents and Settings\All Users\Application Data\Symantec\SRTSP\Quarantine\APQ7.tmp
Detection Origin: Local machine     Detection Type: Concrete      Detection Source:
Real-Time Protection        User: NT AUTHORITY\SYSTEM     Process Name: C:\Program Files
(x86)\Symantec AntiVirus\RHost2     Signature Version: AV: 1.1.1.1, AS: 1.1.1.1, NIS:
1.1.1.1      Engine Version: AM: 1.1.9402.0, NIS: 1.1.1.1
```

The object is the target file (apq7.tmp), as it is being acted on. The name is a friendly descriptor and thus is the Object Name.

Object Type [7.2]

The resource type (file type) referenced or impacted by activity reported in the log, specifically related to what is parsed into Object. Object Type is a categorization field in comparison to Object Name, which is a specific description of the value in Object.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String (0-512 characters, 64 average characters)

Aliases

Use	Alias
Client Console Full Name	Object Type
Client Console Short Name	Object Type
Web Console Tab/Name	Application/Object Type
Elasticsearch Field Name	objectType
Rule Builder Column Name	ObjectType
Regex Pattern	<objecttype>
NetMon Name	Not applicable

Field Relationships

- Object Type is a categorization of the resource described in Object.
- Object Type is a broader classification whereas Object Name is a specific name or description.

Common Applications

- AV software
- HTTP access logs

Use Case

Sub-classification when the event type is not enough.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Object Type does not require an Object. For example, a file scanner might create a log looking for .gif and not find any. The Object Type would be GIF, but there is no Object because no files were found.
- Do not use Object Type with any other specialty field, such as Hash, Process, Subject, and so on. Object Type only applies to Object.

Examples

- HTTP access log. Object Type could contain the MIME type of file(s)
- Windows Security Event Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-
a5ba-3e3b0328c30d}'/><EventID>4907</EventID><Version>0</Version><Level>Information</
Level><Task>Audit Policy Change</Task><Opcode>Info</Opcode><Keywords>Audit Success</
Keywords><TimeCreated SystemTime='2016-02-26T06:56:10.852896900Z'/
><EventRecordID>228903233</EventRecordID><Correlation/><Execution ProcessID='752'
ThreadID='768'/><Channel>Security</Channel><Computer>USLT0775JCROW.schq.safaware.com</
Computer><Security/></System><EventData><Data
Name='SubjectUserSid'>SAFAWARErecordflow\julian.crowley</Data><Data
Name='SubjectUserName'>julian.crowley</Data><Data Name='SubjectDomainName'>SAFAWARE</
Data><Data Name='SubjectLogonId'>0x10be75</Data><Data Name='ObjectServer'>Security</
Data><Data Name='ObjectType'>File</Data><Data Name='ObjectName'>C:
\Windows\System32\autochk.exe</Data><Data Name='HandleId'>0xb0</Data><Data
Name='OldSd'>S:AI</Data><Data Name='NewSd'>S:(AU;SAFA;DCLCRPCRSDWDWO;;;WD)</Data><Data
Name='ProcessId'>0x3298</Data><Data Name='ProcessName'>C:\Program Files
(x86)\Google\Chrome\Application\chrome.exe</Data></EventData></Event>
```


> In this case, Object is authchk.exe. Object Name is blank even though the source log explicitly calls it out. If the log had a field that called Auto check process or some other expanded description or friendly name of the object, then that value would parse into Object Name. Object Type would parse into File.

- MS Forefront TMG Web Proxy

```
1.1.1.1 anonymous Windows-Update-Agent Y 2014-12-22 17:45:02 w3proxy APPGATEDR - -
1.1.1.1 80 31 221 359 http TCP HEAD http://ds.download.windowsupdate.com/v11/2/
microsoftupdate/redir/v6-muauth.cab?14546421745 application/octet-stream Inet 200
0x40800000 [System] Allow all HTTP traffic from Forefront TMG to all networks (for CRL
downloads) Req ID: 11c05fb1; Compression: client=No, server=No, compress rate=0%
decompress rate=0% Local Host External 0x180 Allowed 2014-12-22 17:45:02 - - - - Allowed
Malware Inspection Disabled for the Matching Policy Rule Unknown - - 0 - 0 - - - - - - 0
0 - 0 - - Feature disabled None ds.download.windowsupdate.com 50937 -
```

> Application/octet-stream parses into Object Type, and v6-muauth.cab parses into Object (if possible). No Object Name is parsed.

- Trend Micro Deep Discovery Inspector

```
06 05 2016 01:04:09 1.1.1.1 <LOC3:INFO> CEF:0|Trend Micro|Deep Discovery Inspector|
3.82.1133|200127|Notable Characteristics of the analyzed sample|6|rt=Jun 05 2016
03:03:49 GMT+04:00 dvc=1.1.1.1 dvchost=uascdiscover.merto.uasc.corp
dvcmac=00:00:00:00:00:00 deviceExternalId=4449875B3A-46561482-3301-FCA4-11156
fname=recordflow.exe fileHash=

9B822B964971D32EC4C97920CDD0D4620F767BC8107D2F

fileType=WIN32 EXE fsize=905216 cs1Label=PolicyCategory cs1=Autostart or other system
reconfiguration msg=Key: HKEY_CURRENT_USER\\Software\\Microsoft\\Windows NT\
\CurrentVersion\\Windows Messaging Subsystem\\Profiles\\Outlook\
\ve9375CFF0413d11d3B88A00104B2A6676\\\nValue: \nType: REG_NONE cs3Label=SandboxImageType
cs3=UASC2 cs2Label=PolicyName cs2=Modifies important registry entries to perform rogue
functions
```

> Win32 EXE parses into Object Type, recordflow.exe parses into Object, and the registry name parses into Object Name.

- Cylance Protect

```
08 23 2016 08:39:29 1.1.1.1 <SLOG:WARN> 1 2016-08-23T13:39:12.2911991Z sysloghost
CylancePROTECT - - - Event Type: Threat, Event Name: threat_changed, Device Name:
USABLDRRECFLOW01, IP Address: (1.1.1.1), File Name: creative Host77, Path: c:\program
files (x86)\adobe\adobe creative cloud\acc\, Drive Type: Internal Hard Drive, SHA256:
8050FE3DCA43D594611492AA149AF09FC9669149602BB2945AFEA4148A24B175, MD5:
59E0D058686BD35B0D5C02A4FD8BD0E0, Status: Abnormal, Cylance Score: 100, Found Date:
8/3/2016 4:22:21 PM, File Type: Executable, Is Running: True, Auto Run: False, Detected
By: FileWatcher
```

> Executable parses into Object Type, and creative Host77 parses into Object.

Parent Process ID [7.2]

The Parent Process ID of a system or application process that is of interest.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String (16 characters)

Aliases

Use	Alias
Client Console Full Name	Parent Process ID
Client Console Short Name	Parent Process ID
Web Console Tab/Name	Application/Parent Process ID
Elasticsearch Field Name	parentProcessId
Rule Builder Column Name	ParentProcessID
Regex Pattern	<parentprocessid>
NetMon Name	Not applicable

Field Relationships

- Parent Process Name
- Parent Process Path
- Process Name
- Process ID
- Object
- Object Name
- Object Type

- Session
- Session Type

Common Applications

- Endpoint devices (for example, Carbon Black)
- Windows logs

Use Case

Identifying that Office is the source for a PowerShell process that is malicious.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

Parse the most obvious meaningful parent ID, which is typically a top-level root.

Examples

- Windows Event Log - Sysmon


```
<Event xmlns='http://Host3/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{22222222-C22A-43E0-BF4C-06F5698FFBD9}'/><EventID>1</EventID><Version>5</Version><Level>Information</Level><Task>Process Create (rule: ProcessCreate)</Task><Opcode>Info</Opcode><Keywords></Keywords><TimeCreated SystemTime='2016-09-19T17:50:20.719863700Z'/><EventRecordID>230097</EventRecordID><Correlation/><Execution ProcessID='3716' ThreadID='3740'/><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer> USABLDRRECFLOW01</Computer><Security UserID='NT AUTHORITY\SYSTEM'/></System><EventData>Process Create:
 UtcTime: 2016-09-19 17:50:20.718
 ProcessGuid: {FCC7BD93-255C-57E0-0000-00109BAC260D}
 ProcessId: 127228
 Image: C:\Windows\System32\Host2
 CommandLine: Host2 /c echo ffsuli > \\.\pipe\ffsuli
 CurrentDirectory: C:\Windows\system32\
 User: NT AUTHORITY\SYSTEM
 LogonGuid: {FCC7BD93-8F2C-57DC-0000-22222222222}
 LogonId: 0x3E7
 TerminalSessionId: 0
 IntegrityLevel: System
 Hashes: SHA1=

811627E612944FE5DADF2A14763A08111143C27E

ParentProcessGuid: {FCC7BD93-8F2B-57DC-0000-222222222222}
 ParentProcessId: 504
```

```
ParentImage: C:\Windows\System32\Host1
ParentCommandLine: C:\Windows\system32\Host1</EventData></Event>
```

Parent Process ID is specifically called out in this log.

- Cb Response

```
08 30 2016 02:20:42 1.1.1.1 <USER:NOTE> LEEF:1.0|CB|CB|1.1.1.1623.1033|
watchlist.storage.hit.process|cb_server=cbserver       cb_version=1.1.1.1623.1033
childproc_count=1   cmdline=C:\\Windows\\system32\\cmd.exe /c ping provisionserver >nul
2>nul      crossproc_count=1   filemod_count=0       host_type=workstation
last_update=2016-08-30T08:02:01.670Z    modload_count=11       netconn_count=0
os_type=windows     parent_guid=222222222-0000-2010-01d2-0294ad4c889c
parent_id=2222222222       parent_name=scsdiscovery.exe     parent_pid=8208
parent_unique_id=222222-0000-2010-01d2-0294ad4c889c-2222222222       path=c:\\windows\
\syswow64\\cmd.exe     process_guid=000001c3-0000-097c-01d2-222222222
process_id=000001c3-0000-097c-01d2-22222222222 process_name=cmd.exe
process_pid=2428    regmod_count=0      server_name=localhost.localdomain
start=2016-08-30T08:01:24.874Z timestamp=1472548449.903
type=watchlist.storage.hit.process
unique_id=000001c3-0000-097c-01d2-22222222222-00000001      username=SYSTEM
watchlist_155=2016-08-30T09:10:02.525745Z      watchlist_id=155
watchlist_name=Command Line
```

Parent_pid (Process ID) called out specifically.

Parent Process Name [7.2]

The parent process name of a system or application process.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String (255 characters maximum)

Aliases

Use	Alias
Client Console Full Name	Parent Process Name
Client Console Short Name	Parent Process Name
Web Console Tab/Name	Parent Process Name
Elasticsearch Field Name	parentProcessName
Rule Builder Column Name	ParentProcessName
Regex Pattern	<parentprocessname>
NetMon Name	Not applicable

Field Relationships

- Parent Process ID
- Parent Process Path
- Process Name
- Process ID
- Object
- Object Name
- Object Type

- Session
- Session Type

Common Applications

- Endpoint devices (for example, Carbon Black)
- Windows logs

Use Case

Identifying that Office is the source for a PowerShell process that is malicious.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Parse the most obvious meaningful parent process (typically top-level root).
- Parent Process Name must match the Parent Process ID.
- Do not capture the process path in the name. That goes in Parent Process Path.

Examples

- Cb Response

```
08 30 2016 02:20:42 1.1.1.1 <USER:NOTE> LEEF:1.0|CB|CB|1.1.1.1623.1033|
watchlist.storage.hit.process|cb_server=cbserver       cb_version=1.1.1.1623.1033
childproc_count=1    cmdline=C:\\Windows\\system32\\cmd.exe /c ping provisionserver >nul
2>nul       crossproc_count=1    filemod_count=0       host_type=workstation
last_update=2016-08-30T08:02:01.670Z     modload_count=11       netconn_count=0
os_type=windows      parent_guid=000001c3-0000-2010-01d2-0294ad4c889c
parent_id=7575139489275778785     parent_name=scsdiscovery.exe       parent_pid=8208
parent_unique_id=000001c3-0000-2010-01d2-0294ad4c889c-22222222222       path=c:\
\windows\\syswow64\\cmd.exe      process_guid=000001c3-0000-097c-01d2-2222222222
process_id=000001c3-0000-097c-01d2-22222222222 process_name=cmd.exe
process_pid=2428     regmod_count=0      server_name=localhost.localdomain
start=2016-08-30T08:01:24.874Z timestamp=1472548449.903
type=watchlist.storage.hit.process
unique_id=000001c3-0000-097c-01d2-222222222222-00000001     username=SYSTEM
watchlist_155=2016-08-30T09:10:02.525745Z      watchlist_id=155
watchlist_name=Command Line
```

> Parent_Name is the parent process name in this instance.

- Windows Event Log - Sysmon

```
<Event xmlns='http://Host3/win/2004/08/events/event'><System><Provider Name='Microsoft-
Windows-Sysmon' Guid='{5770385F-C22A-43E0-BF4C-06F5698FFBD9}'/><EventID>1</
EventID><Version>5</Version><Level>Information</Level><Task>Process Create (rule:
ProcessCreate)</Task><Opcode>Info</Opcode><Keywords></Keywords><TimeCreated
SystemTime='2016-09-19T17:50:20.719863700Z'/><EventRecordID>230097</
EventRecordID><Correlation/><Execution ProcessID='3716' ThreadID='3740'/
><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>LRXM</
Computer><Security UserID='NT AUTHORITY\SYSTEM'/></System><EventData>Process Create:
 UtcTime: 2016-09-19 17:50:20.718
 ProcessGuid: {FCC7BD93-255C-57E0-0000-222222222222}
 ProcessId: 127228
 Image: C:\Windows\System32\Host2
 CommandLine: Host2 /c echo ffsuli > \\.\pipe\ffsuli
 CurrentDirectory: C:\Windows\system32\
 User: NT AUTHORITY\SYSTEM
 LogonGuid: {FCC7BD93-8F2C-57DC-0000-2222222222}
 LogonId: 0x3E7
 TerminalSessionId: 0
 IntegrityLevel: System
 Hashes: SHA1=

811627E612944FE5DADF2A14763A08111143C27E

 ParentProcessGuid: {FCC7BD93-8F2B-57DC-0000-22222222222}
 ParentProcessId: 504
 ParentImage: C:\Windows\System32\Host1
 ParentCommandLine: C:\Windows\system32\Host1</EventData></Event>
```


Obfuscated process name, but this would be appropriate for Parent Process Name.

Parent Process Path [7.2]

The full path of a parent process of a system or application process.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String (892 characters maximum)

Aliases

Use	Alias
Client Console Full Name	Parent Process Path
Client Console Short Name	Parent Process Path
Web Console Tab/Name	Parent Process Path
Elasticsearch Field Name	parentProcessPath
Rule Builder Column Name	ParentProcessPath
Regex Pattern	<parentprocesspath>
NetMon Name	Not applicable

Field Relationships

- Parent Process ID
- Parent Process Name
- Process Name
- Process ID
- Object
- Object Name
- Object Type

- Session
- Session Type

Common Applications

- Endpoint devices (for example, Carbon Black)
- Windows logs

Use Case

- Identifying where parent executing process resides on target device.
- Tracking malware installation locations.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Parent process path must match the parent process ID/name.
- Do not capture the process path in this field, only the parent process path.
- Parse out the OS-dependent path using whichever separators are native to that OS.

Examples

- Windows Event Log - Sysmon


```
<Event xmlns='http://Host3/win/2004/08/events/event'><System><Provider Name='Microsoft-
Windows-Sysmon' Guid='{2222222222-C22A-43E0-BF4C-06F5698FFBD9}'/><EventID>1</
EventID><Version>5</Version><Level>Information</Level><Task>Process Create (rule:
ProcessCreate)</Task><Opcode>Info</Opcode><Keywords></Keywords><TimeCreated
SystemTime='2016-09-19T17:50:20.719863700Z'/><EventRecordID>230097</
EventRecordID><Correlation/><Execution ProcessID='3716' ThreadID='3740'/
><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>LRXM</
Computer><Security UserID='NT AUTHORITY\SYSTEM'/></System><EventData>Process Create:
 UtcTime: 2016-09-19 17:50:20.718
 ProcessGuid: {FCC7BD93-255C-57E0-0000-22222222222}
 ProcessId: 127228
 Image: C:\Windows\System32\Host2
 CommandLine: Host2 /c echo ffsuli > \\.\pipe\ffsuli
 CurrentDirectory: C:\Windows\system32\
 User: NT AUTHORITY\SYSTEM
 LogonGuid: {2222222222-8F2C-57DC-0000-2222222}
 LogonId: 0x3E7
 TerminalSessionId: 0
 IntegrityLevel: System
 Hashes: SHA1=

811627E612944FE5DADF2A14763A08111143C27E
```

```
ParentProcessGuid: {22222222222-8F2B-57DC-0000-2222222222222}
 ParentProcessId: 504
 ParentImage: C:\Windows\System32\Host1
 ParentCommandLine: C:\Windows\system32\Host1</EventData></Event>
```

ParentImage contains a path to the parent process.

Policy [7.2]

The specific policy referenced (for example, Firewall or Proxy) in a log message.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Policy
Client Console Short Name	Policy
Web Console Tab/Name	Policy
Elasticsearch Field Name	policy
Rule Builder Column Name	Policy
Regex Pattern	<policy>
NetMon Name	Not applicable

Field Relationships

- Group
- Login
- Account
- Domain
- Object (disambiguation—policy was historically stored as object in some cases)

Common Applications

- Firewall
- Antivirus
- IDS/IPS
- Directory
- Vulnerability scanners
- Audit tools
- Proxies
- IT management

Use Case

- Tracking group policy
- Correlating AV and vulnerability scanners
- Compliance
- Policy violations

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Only store explicitly called out Policy values from log.
- You can store policy synonyms (for example, Standard).
- Capture the broadest policy if multiple different policy types are defined in the log.

Examples

- SourceFire IDS

```
10 02 2016 20:30:22 1.1.1.1 <LOC6:WARN> Oct  2 23:27:07 mtl-corp-sen-01 CORPvDC:
Protocol: TCP, SrcIP: 1.1.1.1, DstIP: 1.1.1.1, SrcPort: 54217, DstPort: 443, TCPFlags:
0x0, IngressInterface: s1p6, EgressInterface: s1p5, IngressZone:
Ingress_CORP_recflow_FROM_NX, EgressZone: Egress_CORP_recflow_TO_ASA, DE: Primary
Detection Engine (f20ae1fc-2be2-22e3-9bcc-222222222222), Policy: RECFLOW_CORP_Sensor,
ConnectType: End, AccessControlRuleName: Rules_Inspection_CORP_RF_Log,
AccessControlRuleAction: Allow, UserName: No Authentication Required, Client: SSL
client, ApplicationProtocol: HTTPS, WebApplication: recflow, InitiatorPackets: 9,
ResponderPackets: 9, InitiatorBytes: 1017, ResponderBytes: 4258, NAPPolicy:
RF_CORP_PREPROCESSORS, DNSResponseType: No Error, Sinkhole: Unknown, URLCategory:
Unknown, URLReputation: Risk unknown, URL: https://www.recordflow.biz
```

> Policy is parsed here as it is explicitly called out. NAPPolicy can go unparsed as there is a broader policy name field.

- Sourcefire IDS

```
05 22 2014 06:12:49 1.1.1.1 <SLOG:ERRR> 2014-05-22 10:12:47.494
recmetric:SOURCE[recflow.Host4]:REC2604E:[ALARM] Policy[ForTheRecord-Media]
User[recflow\Domain Usersrecflowusers,Medium Mandatory Level@Mandatory Label...
\ercflow,Host4] Process[\SystemRoot\System32\Host2] Action[read_dir] Res[M:
\Media\QueryBuilder]  Effect[DENIED Code (1U,2U,3U,4U,5U,6U,7U,8U,9U,10U,11U,12P,13P,
14U,15U,16U,17A,18U,19M)]
```

ForTheRecord-Media parses into Policy as it is explicitly called out.

Process ID

System or application process ID.

Data Type

Integer

Aliases

Use	Alias
Client Console Full Name	Process ID
Client Console Short Name	Process ID
Web Console Tab/Name	Process ID
Elasticsearch Field Name	processId
Rule Builder Column Name	ProcessID
Regex Pattern	<processid>
NetMon Name	Not applicable

Field Relationships

- Process Name
- Parent Process ID
- Parent Process Name
- Parent Process Path

Common Applications

Anything that tracks applications/processes.

Use Case

Identifying what is running on a system.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Process ID should be the unique identifier (typically a PID).
- Store HEX representation by preference, but allow decimal if that's what log source provides.

Examples

- *nix

```
03 21 2014 10:13:00 1.1.1.1 <CLK1:INFO> crond[2596]: (root) CMD (/usr/lib64/sa/sa1 1 1)
```

> In *nix logs, the Process and ProcessID frequently follow the syslog facility and severity. In this case, crond is followed by the ProcessID 2596 in square braces.

- Cb Response

```
08 30 2016 02:20:42 1.1.1.1 <USER:NOTE> LEEF:1.0|CB|CB|1.1.1.1623.1033|
watchlist.storage.hit.process|cb_server=cbserver       cb_version=1.1.1.1623.1033
childproc_count=1    cmdline=C:\\Windows\\system32\\cmd.exe /c ping provisionserver >nul
2>nul       crossproc_count=1    filemod_count=0       host_type=workstation
last_update=2016-08-30T08:02:01.670Z     modload_count=11       netconn_count=0
os_type=windows     parent_guid=11111111-0000-2010-01d2-0294ad4c889c
parent_id=7575139489111111 parent_name=scsdiscovery.exe     parent_pid=8208
parent_unique_id=222222222-0000-2010-01d2-0294ad4c889c-00000001       path=c:\\windows\
\syswow64\\cmd.exe     process_guid=222222-0000-097c-01d2-0294b431d3b1
process_id=2222222222222222       process_name=cmd.exe       process_pid=2428
regmod_count=0      server_name=localhost.localdomain
start=2016-08-30T08:01:24.874Z       timestamp=1472548449.903
type=watchlist.storage.hit.process
unique_id=000001c3-0000-097c-01d2-0294b431d3b1-00000001     username=SYSTEM
watchlist_155=2016-08-30T09:10:02.525745Z      watchlist_id=155
watchlist_name=Command Line
```

> Process_pid called out specifically.

Process Name

System or application process described by log message.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Process
Client Console Short Name	Process
Web Console Tab/Name	Process Name
Elasticsearch Field Name	process
Rule Builder Column Name	Process
Regex Pattern	<process>
NetMon Name	Varies by protocol

Field Relationships

- Parent Process ID
- Parent Process Name
- Parent Process Path
- Process
- Process ID
- Object
- Object Name
- Object Type
- Session
- Session Type

Common Applications

Any application.

Use Case

Monitoring timer jobs (for example, cron, or Windows scheduler).

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

Process Name should contain the identified process (for example, PowerShell.exe).

Examples

- Cb Response

```
08 30 2016 02:20:42 1.1.1.1 <USER:NOTE> LEEF:1.0|CB|CB|1.1.1.1623.1033|
watchlist.storage.hit.process|cb_server=cbserver       cb_version=1.1.1.1623.1033
childproc_count=1   cmdline=C:\\Windows\\system32\\cmd.exe /c ping provisionserver >nul
2>nul      crossproc_count=1   filemod_count=0       host_type=workstation
last_update=2016-08-30T08:02:01.670Z    modload_count=11       netconn_count=0
os_type=windows     parent_guid=22221c3-0000-2010-01d2-0294ad4c889c
parent_id=75751394892752222       parent_name=scsdiscovery.exe       parent_pid=8208
parent_unique_id=2222222-0000-2010-01d2-0294ad4c889c-00002222       path=c:\\windows\
\syswow64\\cmd.exe     process_guid=000001c9-2222-097c-01d2-0294b431d3b1
process_id=000001c3-0000-097c-01d2-222222222   process_name=cmd.exe
process_pid=2428    regmod_count=0      server_name=localhost.localdomain
start=2016-08-30T08:01:24.874Z timestamp=1472548449.903
type=watchlist.storage.hit.process
unique_id=000001c3-0000-097c-01d2-0294b431d3b1-00000001     username=SYSTEM
watchlist_155=2016-08-30T09:10:02.525745Z      watchlist_id=155
watchlist_name=Command Line
```

> Process_name called out specifically.

- Windows Event Log – System


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Service Control Manager' Guid='{222222-a6d7-4695-8e1e-26931d2012f4}'
EventSourceName='Service Control Manager'/><EventID Qualifiers='16384'>7036</
EventID><Version>0</Version><Level>Information</Level><Task>None</Task><Opcode></
Opcode><Keywords>Classic</Keywords><TimeCreated
SystemTime='2016-08-01T08:58:46.675586600Z'/><EventRecordID>823261</
EventRecordID><Correlation/><Execution ProcessID='512' ThreadID='8508'/
```

```
><Channel>System</Channel><Computer> USABLDRRECFLOW01</Computer><Security/></
System><EventData><Data Name='param1'>Windows Error Reporting Service</Data><Data
Name='param2'>stopped</Data><Binary>570065007200530076222222222031000000</Binary></
EventData></Event>
```

Param1 in the 7036 event indicates the service (process) status.

- *nix

```
03 21 2014 10:13:00 1.1.1.1 <CLK1:INFO> crond[2596]: (root) CMD (/usr/lib64/sa/sa1 1 1)
```

In *nix logs, the process frequently follows the syslog facility and severity, in this case Cron Daemon.

Quantity

Quantity is a numeric integer count of something.

Data Type

Integer

Aliases

Use	Alias
Client Console Full Name	Quantity
Client Console Short Name	Quantity
Web Console Tab/Name	Quantity
Elasticsearch Field Name	quantity
Rule Builder Column Name	Quantity
Regex Pattern	<quantity>
NetMon Name	Not applicable

Field Relationships

- Amount
- Rate
- Size

Common Applications

Not heavily used.

Use Case

Aggregated logs and UDLA queries for fraud detection.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Not used for percentages.
- Not used for currency.
- Used to capture specific integer numbers.
- Use Quantity to represent numbers, and Amount to represent percentages.

Examples

- Unisys Stealth


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='StealthUI'/><EventID Qualifiers='0'>109</EventID><Level>Information</
Level><Task></Task><Keywords>Classic</Keywords><TimeCreated
SystemTime='2016-12-09T05:26:38.000000000Z'/><EventRecordID>8449</
EventRecordID><Channel>Application</Channel><Computer>psb-stl-em.LRPMBD.local</
Computer><Security/></System><EventData><Data>User query for logs retrieved 19 records
out of 1631 total records </Data></EventData></Event>
```


Quantity indicates the number of logs retrieved.

Rate

Defines a number per unit of time. Always expressed as a fraction.

Data Type

Double

Aliases

Use	Alias
Client Console Full Name	Rate
Client Console Short Name	Rate
Web Console Tab/Name	Rate
Elasticsearch Field Name	rate
Rule Builder Column Name	Rate
Regex Pattern	<rate>
NetMon Name	Not applicable

Field Relationships

- Size
- Quantity
- Amount

Common Applications

Flow rate

Use Case

Determining frequency.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Rarely used except where specifically called out as a rate.
- There is no quantifier of what the time is (second, minute, year, fortnight).

Examples

- SFlow Log

```
sFlow v5 AGENTIP=1.1.1.1 OPAQUE=flow_sample ENTERPRISE=0 FORMAT=2202
SAMPLENAME=AppOperation INPUTINTERFACE=21 OUTPUTINTERFACE=2 SAMPLEDATA=96 APPLICATION=_
OPERATION=_4 ATTRIBUTES=_ STATUS_DESC=TS REQ_BYTES=2323 RESP_BYTES=34343 USEC=12
STATUS=1 DETAILS=SubAgentId=0 AgentUpTime=142864 DatagramSequence=47998 SampleRate=16384
SamplePool=18939904
```

> SampleRate could be a rate. SamplePool could be a size because it refers to the capacity of the pool.

Reason [7.2]

The justification for an action or result.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Reason
Client Console Short Name	Reason
Web Console Tab/Name	Reason
Elasticsearch Field Name	reason
Rule Builder Column Name	Reason
Regex Pattern	<reason>
NetMon Name	Not applicable

Field Relationships

- Action
- Command
- Policy
- Result
- ResponseCode

Common Applications

Understanding why an action or command was executed, or why a result or ResponseCode was generated.

Use Case

- IDS/IPS
- Email filtering
- Firewall blocking
- Antivirus
- Vulnerability scanning

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- If the log explicitly calls out a policy, use policy instead.
- Reason should be free text. If it is an industry standard code use ResponseCode.
- Result should be used for what and Reason should be used for why.

Examples

- eSafe Email Security

```
05 01 2012 16:21:21 1.1.1.1 <LOC5:ERRR> eSafeCR: Alert from eSafe    Scan result: SMTP
error  Protocol: SMTP  File Name\Mail Subject:  Business Plan & Financials  Source:
1.1.1.1  Destination: 1.1.1.1  Mail Sender: Peter.Store@recordflow.biz  Mail Recipients:
pete.store@recordflow.biz  Details: Delivery Msg #911 - Email b0eeb3e8 NOT sent after
multiple retries, likely reason: 554 delivery error: dd This user doesn't have a
recordflow.biz account (pete.store@recordflow.biz) [0] - recordflow.biz.
```

> The Reason field (554) parses into ResponseCode because 554 is an SMTP response. The text after could be parsed into Reason. Obtain other samples to determine whether there is a legitimate pattern in the log.

- Alcatel-Lucent Wireless Controller

```
12 10 2012 09:08:56 1.1.1.1 <LOC1:DBUG> Dec 10 09:09:03 DAVE authmgr[1600]: <124004>
<DBUG> <DAVE-03 1.1.1.1>  Setting user 00:00:00:00:00:00 aaa profile to default-dot1x,
reason: bbq_set_aaa_profile_defaults
```

> This is an assumed Policy, but additional logs and product knowledge is needed to confirm. There would not be a Reason in this log because the reason is that it is policy.

- NetApp CIFS Security Audit Event Log

```
04/11/2016 16:55 TYPE=FailureAudit USER= COMP=Computer SORC=Security CATG=Logon/Logoff
EVID=537 MESG=Logon Failure:        Reason:           An unexpected error occurred
during logon    User Name:  -     Domain:           -        Logon Type: 3     Logon
Process:    Data ONTAP        Authentication Package:    Extended Security
Workstation Name: -     Status code:      -        Substatus code:    -     Caller User
Name: -     Caller Domain:    -        Caller Logon ID:  -     Caller Process ID:
3170862     Transited Services:   -     Source Network Address: 1.1.1.1     Source
Port:      0        Caller Process Name:
```

Logon failure is the event, and unexpected error parses into Reason.

Response Code [7.2]

The explicit and well-defined response code for an action or command in a log. Response Code differs from Result in that response code should be well structured and easily identifiable as a code.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Response Code
Client Console Short Name	Response Code
Web Console Tab/Name	Response Code
Elasticsearch Field Name	responseCode
Rule Builder Column Name	ResponseCode
Regex Pattern	<responsecode>
NetMon Name	Not applicable

Field Relationships

- Status
- Result
- Action
- Command
- VMID

Common Applications

- Web server
- Proxy
- Mail server

Use Case

Anything that captures HTTP or SMTP traffic.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Response Code should be industry standard. If it is a vendor standard, use VMID.
- If the value is unstructured text, use Result instead.
- This supplants VMID completely for parsing HTTP and SMTP response codes. In other words, VMID should be tied to a vendor while HTTP codes are an independent standard.
- This field can be extended to non-IT industry response codes. For example, credit card response codes if ATM or POS logs are parsed, and ICS/SCADA-specific protocols.

Examples

- IBM WebSphere DataPower Integration

```
03 23 2014 13:14:32 1.1.1.1 <USER:INFO> Mar 23 13:14:26USABLDRRECFLOW01 [Service_Router]
[mpgw][info] mpgw(Routing_Int_MPG): trans(1954389697)[1.1.1.1]: HTTP response code 200
 for 'https://1.1.1.1:54010/legacy/eg/aggregate'
```

> 200 parsed into Response code.

- Microsoft IIS

```
::1, Host1st@Host2, 8/25/2015, 15:25:43, W3SVC2, USABLDRRECFLOW01, ::1, 171, 327, 512,
500, 0, GET, /, |88|800a0009|Subscript_out_of_range:_'[number:_1]',
```

> HTTP response code.

- Microsoft ActiveSync 2010

```
2012-08-26 00:07:52 1.1.1.1 GET /owa/1.1.1.1/scripts/premium/flogon.js - 443 - 1.1.1.1
Mozilla/5.0+(Windows+NT+6.1;+WOW64)+AppleWebKit/537.1+(KHTML,+like+Gecko)+Chrome/
21.0.1180.83+Safari/537.1 304 0 0 281
```

> HTTP response code from ActiveSync.

- Microsoft IIS SMTP

```
2012-03-29 07:30:50 1.1.1.1 USABLDRRECFLOW01SMTPSVC1 CDESMTP 1.1.1.1 0 HELO - +CDENETMON
250 0 55 14 0 SMTP - - - -
```

> SMTP response code.

- Bluecoat Proxy

```
06 29 2015 14:26:18 1.1.1.1 <USER:NOTE> date=2015-06-29 time=19:25:57 time-taken=65 c-
ip=1.1.1.1 cs-username=- cs-auth-group=- x-exception-id=- sc-filter-result=OBSERVED cs-
categories="Technology/Internet" cs(Referer)=http://www.amazon.com/Travel-Mattress-
Healing-Magnetic-Cover/dp/B0029OMC6A cs-status=500 s-action=TCP_NC_MISS cs-method=GET
rs(Content-Type)=text/xml cs-uri-scheme=http cs-host=fls-na.amazon.com cs-uri-port=80
cs-uri-path=/1/amazon-clicks/1/OP cs-uri-query=?
requestId=1J6GGDGMDB10asdvasehQ2&childRequestId=152CJ96fgnfhjkjTW28Z5AP&widgetName=varia
nt_ads_below_fold&searchResultNumber=1&impressionRankOnAsin=3 cs-uri-extension=-
cs(User-Agent)=Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko s-
ip=1.1.1.1 cs-bytes=1217 rs-bytes=293
```

> Despite Status being the key, the value is an HTTP response code.

Result [7.2]

Result is for the outcome of a command operation or action. For example, the result of “quarantine" might be "success."

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Result
Client Console Short Name	Result
Web Console Tab/Name	Result
Elasticsearch Field Name	result
Rule Builder Column Name	Result
Regex Pattern	<result>
NetMon Name	Not applicable

Field Relationships

- *Action*. The Action should be what generated the result.
- *Command*. A Command could also be a generator of a result.
- *Status*. Status is similar to Result, but reserved for explicitly defined result values. Result is an outcome, whereas a Status can be independent of the action.

Common Applications

- Endpoint protection such as CarbonBlack or Cylance
- IDS/IPS

Use Case

- Determining whether an action or command succeeded or failed. Validating normal operational process.
- Monitoring backup processes to see if they were successful.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Result is the outcome of an occurence and should be tied to a command, action, or policy.
- Result should not contain industry standard response codes such as HTTP response codes.
- If given a choice, use VMID/Vendor Info if the log is just a message and not tied to an action/command. Use Result if the log contains a clear action/command. For example, VMID/Vendor Info might be tied to "Attempted quarantine" and the result might be "success.”
- Do not take result in the log literally. It could be a result, a VMID, or a status.

Examples

- F5 BIG-IP ASM

```
03 22 2012 14:19:54 a4eg01-1-admi <LOC1:NOTE> Mar 22 14:19:54 USABLDRRECFLOW01 local/
USABLDRRECFLOW01-1 notice apd[4096]: 01490102:5: de71deef: Access policy result:
Network_Access
```

Access policy result shows Network Access as the result of a policy being applied. Network Access parses into Result.

- Vamsoft ORF

```
01 27 2013 18:54:25 1.1.1.1 <MAIL:INFO> Jan 27 18:52:57 fe80::1111:11e1:31111:dsfsd%13
ORFEE: SRC:SMTPSVC-1,CLASS:Blacklist,ACT:Reject,FP: OnArrival,IP:1.1.1.1,SND:no-
reply@Host34,RCPT: pstore@Host2;agent414@Host2,TEXT:Email blacklisted by the SPF test
(sender forged per policy of "Host34", SPF result: Fail).
```

Fail or SPF Fail parses into Result, reject from the ACT field parses into Action, and Blacklist or Sender Forged parses into Policy.

- Windows Event Log – Trend Micro AV

```
<Event xmlns='http://Host3/win/2004/08/events/event'><System><Provider Name='Trend Micro
OfficeScan Server'/><EventID Qualifiers='32773'>600</EventID><Level>Warning</
Level><Task>System</Task><Keywords>Classic</Keywords><TimeCreated
SystemTime='2016-07-26T22:37:03.000000000Z'/><EventRecordID>152848</
EventRecordID><Channel>Application</Channel><Computer>Host2</Computer><Security
UserID='NT AUTHORITY\SYSTEM'/></System><EventData>Virus/Malware: Unauthorized File
Encryption

Endpoint: USABLDRRECFLOW01

Domain: safaware\

File: \\safaware\thinnerapp\lotusnotes\bin\lotus Host1

Date/Time: 7/26/2016 18:35:25

Result: Virus successfully detected, cannot perform the Quarantine action

</EventData></Event>
```


Showing result of AV scan and attempted remediation action to Quarantine.

- Cisco IDS/IPS


```
<evStatus eventId="1332222222222228024371874" vendor="Cisco" xmlns="http://
www.cisco.com/cids/2006/08/cidee">> USABLDRRECFLOW01</hostId><appName>mainApp</
appName><appInstanceId>1260</appInstanceId></originator><time offset="-300"
timeZone="GMT-06:00">1345793398595703000</time><autoUpgradeServerCheck><uri>http://
breon.moore@1.1.1.1//swc/esd/06/273556262/contract/recordflowconsole.pkg</
packageFileName><result status="true"></result></autoUpgradeServerCheck></evStatus>
```


This is a Result instead of a Status because it represents an outcome of a task or operation. Status represents a state independent of an operation being performed. AutoUpgradeServerCheck may parse into Action.

Session

Unique user or system session identifier.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Session
Client Console Short Name	Session
Web Console Tab/Name	Session
Elasticsearch Field Name	session
Rule Builder Column Name	Session
Regex Pattern	<session>
NetMon Name	SessionID

Field Relationships

- Account
- Login
- SessionType
- Protname
- Protnum
- IP Address Fields
- Process
- ProcessID

Common Applications

- SSH
- Remote Desktop
- Telnet
- FTP
- Web Application
- Shell
- Web Browser

Use Case

- NetMon session identifier.
- User session for a web session or computer session.
- Session ID for a VoIP call.
- Session record for a vulnerability scan.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Unique non-permanent identifier for a user/system session.
- Session Token identifier/number.
- Used for tracking activity associated with a session.
- Not ProcessID.

Examples

- Linux Host

```
10 15 2010 10:50:31 1.1.1.1 <SAU1:INFO> Oct 15 10:50:30 USABLDRRECFLOW01: [ID 702911
Host7] 700 Auth_method_success, Username: pete.store, Auth method: keyboard-interactive,
Session-Id: 10707
```

> Session-ID parses into Session.

- Windows Event Log

```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-22222222222}'/
><EventID>4742</EventID><Version>0</Version><Level>Information</Level><Task>Computer
Account Management</Task><Opcode>Info</Opcode><Keywords>Audit Success</
Keywords><TimeCreated SystemTime='2016-02-24T19:46:19.175040100Z'/
><EventRecordID>4814831973</EventRecordID><Correlation/><Execution ProcessID='560'
ThreadID='8892'/><Channel>Security</Channel><Computer> USABLDRRECFLOW01</
Computer><Security/></System><EventData><Data Name='ComputerAccountChange'>-</Data><Data
```

```
Name='TargetUserName'> USABLDRRECFLOW01$</Data><Data Name='TargetDomainName'>SAFAWARE</
Data><Data Name='TargetSid'>SAFAWARE\ USABLDRRECFLOW01$</Data><Data
Name='SubjectUserSid'>NT AUTHORITY\ANONYMOUS LOGON</Data><Data
Name='SubjectUserName'>ANONYMOUS LOGON</Data><Data Name='SubjectDomainName'>NT
AUTHORITY</Data><Data Name='SubjectLogonId'>0x3e6</Data><Data Name='PrivilegeList'>-</
Data><Data Name='SamAccountName'>-</Data><Data Name='DisplayName'>-</Data><Data
Name='UserPrincipalName'>-</Data><Data Name='HomeDirectory'>-</Data><Data
Name='HomePath'>-</Data><Data Name='ScriptPath'>-</Data><Data Name='ProfilePath'>-</
Data><Data Name='UserWorkstations'>-</Data><Data Name='PasswordLastSet'>2/24/2016
12:46:19 PM</Data><Data Name='AccountExpires'>-</Data><Data Name='PrimaryGroupId'>-</
Data><Data Name='AllowedToDelegateTo'>-</Data><Data Name='OldUacValue'>-</Data><Data
Name='NewUacValue'>-</Data><Data Name='UserAccountControl'>-</Data><Data
Name='UserParameters'>-</Data><Data Name='SidHistory'>-</Data><Data Name='LogonHours'>-
</Data><Data Name='DnsHostName'>-</Data><Data Name='ServicePrincipalNames'>-</Data></
EventData></Event>
```


SubjectLogonID parses into Session. Used to track user activity from login to logout.

- Windows Event Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Security-Auditing' Guid='{222222-5478-4994-a5ba-3e3b0328c30d}'/
><EventID>4624</EventID><Version>0</Version><Level>Information</Level><Task>Logon</
Task><Opcode>Info</Opcode><Keywords>Audit Success</Keywords><TimeCreated
SystemTime='2016-02-09T00:45:00.703363000Z'/><EventRecordID>2269912024</
EventRecordID><Correlation/><Execution ProcessID='520' ThreadID='12080'/
><Channel>Security</Channel><Computer> USABLDRRECFLOW01</Computer><Security/></
System><EventData><Data Name='SubjectUserSid'>NT AUTHORITY\SYSTEM</Data><Data
Name='SubjectUserName'> USABLDRRECFLOW01$</Data><Data
Name='SubjectDomainName'>SAFAWARE</Data><Data Name='SubjectLogonId'>0x3e7</Data><Data
Name='TargetUserSid'>NT AUTHORITY\SYSTEM</Data><Data Name='TargetUserName'>SYSTEM</
Data><Data Name='TargetDomainName'>NT AUTHORITY</Data><Data Name='TargetLogonId'>0x3e7</
Data><Data Name='LogonType'>5</Data><Data Name='LogonProcessName'>Advapi  </Data><Data
Name='AuthenticationPackageName'>Negotiate</Data><Data Name='WorkstationName'></
Data><Data Name='LogonGuid'>{00000000-0000-0000-0000-000000000000}</Data><Data
Name='TransmittedServices'>-</Data><Data Name='LmPackageName'>-</Data><Data
Name='KeyLength'>0</Data><Data Name='ProcessId'>0x200</Data><Data Name='ProcessName'>C:
\Windows\System32\services.exe</Data><Data Name='IpAddress'>-</Data><Data
Name='IpPort'>-</Data></EventData></Event>
```


TargetLogonID is parsed instead of SubjectLogonID. Using Target because it is the initiation of a new session that can be tracked separate from the initiator session. For example, Process Run As a different user in Windows.

Session Type [7.2]

The type of session described in the log (for example, console, CLI, or web). This field is free text.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String (128 characters)

Aliases

Use	Alias
Client Console Full Name	Session Type
Client Console Short Name	Session Type
Web Console Tab/Name	Session Type
Elasticsearch Field Name	sessionType
Rule Builder Column Name	SessionType
Regex Pattern	<sessiontype>
NetMon Name	Not applicable

Field Relationships

- See IANA Protocol Number and IANA Protocol Name
- Session
- Login
- Account
- Domain
- Process
- ProcessID

- Protname
- Protnum

Common Applications

- Windows security log lists all types of sessions (logon type)
- Linux authentication methods

Use Case

Tracking how users are interacting with a system.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- SessionType can exist without Session.
- Session can exist without a defined Session Type.

Examples

- Linux Host

```
10 15 2010 10:50:31 1.1.1.1 <SAU1:INFO> Oct 15 10:50:30 USABLDRRECFLOW01: [ID 702911
Host7] 700 Auth_method_success, Username: pete.store, Auth method: keyboard-interactive,
Session-Id: 10707
```

> Keyboard-Interactive parses into Session Type.

- Windows Event Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Security-Auditing' Guid='{2222222-5478-4994-a5ba-3e3b0328c30d}'/
><EventID>4624</EventID><Version>0</Version><Level>Information</Level><Task>Logon</
Task><Opcode>Info</Opcode><Keywords>Audit Success</Keywords><TimeCreated
SystemTime='2016-02-09T00:45:00.703363000Z'/><EventRecordID>2269912024</
EventRecordID><Correlation/><Execution ProcessID='520' ThreadID='12080'/
><Channel>Security</Channel><Computer> USABLDRRECFLOW01</Computer><Security/></
System><EventData><Data Name='SubjectUserSid'>NT AUTHORITY\SYSTEM</Data><Data
Name='SubjectUserName'>USBO1PDC02$</Data><Data Name='SubjectDomainName'>SAFAWARE</
Data><Data Name='SubjectLogonId'>0x3e7</Data><Data Name='TargetUserSid'>NT
AUTHORITY\SYSTEM</Data><Data Name='TargetUserName'>SYSTEM</Data><Data
Name='TargetDomainName'>NT AUTHORITY</Data><Data Name='TargetLogonId'>0x3e7</Data><Data
Name='LogonType'>5</Data><Data Name='LogonProcessName'>Advapi  </Data><Data
Name='AuthenticationPackageName'>Negotiate</Data><Data Name='WorkstationName'></
```

```
Data><Data Name='LogonGuid'>{00000000-0000-0000-0000-000000000000}</Data><Data
Name='TransmittedServices'>-</Data><Data Name='LmPackageName'>-</Data><Data
Name='KeyLength'>0</Data><Data Name='ProcessId'>0x200</Data><Data Name='ProcessName'>C:
\Windows\System32\services.exe</Data><Data Name='IpAddress'>-</Data><Data
Name='IpPort'>-</Data></EventData></Event>
```


LogonType parses into Session Type. Establishes the LogonID as a Service. Service session can be tracked with Session 0x3e7.

Size

Numeric description of capacity (for example, disk size). Size is best thought of as a limit rather than a current measurement. Use Amount for non-specific measurements.

Data Type

Double

Aliases

Use	Alias
Client Console Full Name	Size
Client Console Short Name	Size
Web Console Tab/Name	Size
Elasticsearch Field Name	size
Rule Builder Column Name	Size
Regex Pattern	<size>
NetMon Name	Not applicable

Field Relationships

- Amount
- Quantity
- Rate
- [prefix]Bytes

Common Applications

- IT Operations (drive size)
- CPU usage (for example, threshold limit on a CPU alert)

Use Case

Used in conjunction with other numeric tags such as bytes or megabytes, can show a disk capacity (<size>) and the usage in <megabytes>.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Use size for capacity measures, use quantity, amount, or [prefix]bytes for measuring current value.
- If there is no label to an explicit size in the log, use <size> when the value is an integer.

Examples

- Threat Defense

```
07 18 2015 23:30:02 1.1.1.1 <LOC6:INFO> Jul 18 23:30:02 ATD-3000 ATD2ESM[26906]: {"CPU Alert": {"CPU Usage":83.7, "CPU Threshold":75.0}}
```

> Size could be based on the CPU Threshold. Amount could be used for the CPU Usage.

Status [7.2]

The vendor's perspective on the state of a system, process, or entity. Status should not be used as the result of an action.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Status
Client Console Short Name	Status
Web Console Tab/Name	Status
Elasticsearch Field Name	status
Rule Builder Column Name	Status
Regex Pattern	<status>
NetMon Name	Not applicable

Field Relationships

- ResponseCode
- Action
- Command
- Process
- Result
- Policy

Common Applications

- Inventory trackers
- SNMP analysis
- Heartbeat detection

Use Case

- IT operations
- Deployment monitors

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

Status should refer to the state, not the result of an action.

Examples

Correct Examples

- Elastic search – red/yellow/green
- Raid array – drive up/down
- Service monitoring – email server up/down

Incorrect Examples

- Cisco Secure ACS

```
06 06 2013 09:12:45 1.1.1.1 <LOC6:NOTE> Jun  6 09:12:45 USABLDRRECFLOW01
CSCOacs_TACACS_Accounting 0000817989 2 1  AuditSessionId=firemon:1.1.1.1:tty1:1.1.1.1,
Response={Type=Accounting; AcctReply-Status=Success; }
```

Accounting Status was Success, but this is a Result, not a Status.

- Tectia SSH Server

```
84479804 | 8/7/2013 4:00:23 AM | None | N/A | USABLDRRECFLOW01 | Information | 0 | SSH
Tectia Server | 709 Publickey_auth_warning, Username: MET_INTNET\SSHVRZCOMM, Algorithm:
publickey, "Unknown key type for `d:
\transops\crit\sshusers\SSHBBQCOM\.ssh2\id_rsa_pub' (status: Key type given not
recognized).", Session-Id: 28172
```

Key value pair showing the status of the public key, but this should be a Reason not a Status.

- Windows Event Log


```
<Event xmlns='http://Host1/win/2004/08/events/event'><System><Provider
Name='SQLSERVERAGENT'/><EventID Qualifiers='16384'>208</EventID><Level>Warning</
Level><Task>Job Engine</Task><Keywords>Classic</Keywords><TimeCreated
SystemTime='2015-07-23T18:20:39.000000000Z'/><EventRecordID>2042567</
EventRecordID><Channel>Application</Channel><Computer> USABLDRRECFLOW01</
Computer><Security/></System><EventData>SQL Server Scheduled Job 'LogRhythm Sunday
Maintenance' (0x7A222222222E72222F538A9DE038D2F3) - Status: Failed - Invoked on:
2015-07-23 12:20:38 - Message: The job failed.  The Job was invoked by User sa.  The
last step to run was step 29 (LogRhythm Job Step Validation).  The job was requested to
start at step 29 (LogRhythm Job Step Validation).</EventData></Event>
```


Showing a failed status for maintenance job. That is a Result, not a Status.

Subject

Originally meant to be the subject of an email. In 7.2 schema, this field becomes a secondary "category" field that can be used in several ways.

Data Type

String (255 characters maximum)

Aliases

Use	Alias
Client Console Full Name	Subject
Client Console Short Name	Subject
Web Console Tab/Name	Subject
Elasticsearch Field Name	subject
Rule Builder Column Name	Subject
Regex Pattern	<subject>
NetMon Name	Not applicable

Field Relationships

- Email fields (if email) for context
- Look at VMID, Vendor Info, and other category fields before using Subject

Common Applications

- Proxies
- NGFW
- NetMon

Use Case

- Classifying traffic (for example, secondary family of http traffic destinations).
- Categorizing data within the log, not the actual log message (use VMID, Vendor Info instead).
- UEBA—sub category of anomaly type.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

Use Subject as a category field only if another field is not more directly named (for example, Vendor Info).

Incorrect Examples

- Microsoft Event Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='MetaFrameEvents'/><EventID Qualifiers='49152'>10001</EventID><Level>Error</
Level><Task>None</Task><Keywords>Classic</Keywords><TimeCreated
SystemTime='2016-07-20T07:13:01.000000000Z'/><EventRecordID>5950393</
EventRecordID><Channel>Application</Channel><Computer> USABLDRRECFLOW01</
Computer><Security/></System><EventData>A usable server cannot be found on which to
launch the application. Application: Citrix AppCenter, Client: USABLDRRECFLOW01
(address: 1.1.1.1;;;), User pete.store. Check your worker group definitions and load
balancing policies to verify appropriate servers are assigned for Citrix AppCenter. </
EventData></Event>
```


> Based on the current standard this is incorrect; the above parses a description of the event into Subject. The Vendor Info tag can supplant this usage. This needs to parse into Vendor Info.

- Another Microsoft Event Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='MOVEit Central'/><EventID Qualifiers='32768'>3</EventID><Level>Warning</
Level><Task>None</Task><Keywords>Classic</Keywords><TimeCreated
SystemTime='2016-09-22T01:18:14.000000000Z'/><EventRecordID>1325287</
EventRecordID><Channel>Application</Channel><Computer> USABLDRRECFLOW01</
Computer><Security/></System><EventData>Task "Symitar Email Notifications": Could not
log task end: [Microsoft][SQL Server Native Client 10.0]Communication link failure</
EventData></Event>
```


> Subject is parsing the entire event data. This is too broad and makes any kind of normalization impossible. This should be parsed into multiple fields including Object, Action, and Vendor Info.

- Blue Coat Proxy Log

```
2016-07-21 20:42:18 3148 1.1.1.1 http://www.amazon.com/Travel-Mattress-Healing-Magnetic-
Cover/dp/B0029OMC6A RCF\Internet_users 1.1.1.1 1.1.1.1 Unavailable - Host3_exception
DENIED "Spam;Malicious Outbound Data/Botnets;Scam/Questionable/Illegal" -  200
TCP_DENIED GET text/html;%20charset=UTF-8 http Host2 80 /Host1 - ico "Mozilla/5.0
(Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko" 1.1.1.1 3323 260 - "none"
"none" unavailable
```

> Subject parsing out the web content category. This might be OK if Subject definition is broadened to something more akin to category.

- Windows Application Event Log

```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-
Windows-EventSystem' Guid='{899daace-4868-4295-afcd-9eb8fb497561}' EventSourceName='EventSystem'/
><EventID Qualifiers='32768'>4609</EventID><Version>0</Version><Level>Warning</Level><Task>Event
Service</Task><Opcode></Opcode><Keywords>Classic</Keywords><TimeCreated
SystemTime='2016-10-21T14:39:07.000000000Z'/><EventRecordID>1919714</EventRecordID><Correlation/
><Execution ProcessID='0' ThreadID='0'/><Channel>Application</Channel><Computer> USABLDRRECFLOW01</
Computer><Security/></System><EventData><Data Name='param1'>d:\recflow\com \security.cpp</Data><Data
Name='param2'>75</Data><Data Name='param3'>822706e5</Data></EventData></Event>
```

> Return Code parses into Subject for lack of a better field. Response Code should be used for this instead.

- Windows Application Event Log

```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='MsiInstaller'/><EventID Qualifiers='0'>11728</EventID><Level>Information</
Level><Task>None</Task><Keywords>Classic</Keywords><TimeCreated
SystemTime='2016-11-15T18:44:56.000000000Z'/><EventRecordID>38096</
EventRecordID><Channel>Application</Channel><Computer> USABLDRRECFLOW01</
Computer><Security UserID='SAFAWARE\pete.store/></System><EventData><Data>Product:
LogRhythm Console -- Configuration completed successfully.</Data><Data>(NULL)</
Data><Data>(NULL)</Data><Data>(NULL)</Data><Data>(NULL)</Data><Data>(NULL)</
Data><Data></
Data><Binary>7B38354632314132452D364144432D344638312D38454544211111111111130333237357D</
Binary></EventData></Event>
```

> Another example of an event description in Subject. This could be parsed into Vendor Information.

- Windows Security Event Log

```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-
a5ba-3e3b0328c30d}'/><EventID>4656</EventID><Version>1</Version><Level>Information</
```

```
Level><Task>Removable Storage</Task><Opcode>Info</Opcode><Keywords>Audit Success</
Keywords><TimeCreated SystemTime='2016-02-23T00:34:58.244632600Z'/
><EventRecordID>7148428</EventRecordID><Correlation/><Execution ProcessID='504'
ThreadID='512'/><Channel>Security</Channel><Computer> USABLDRRECFLOW01</
Computer><Security/></System><EventData><Data Name='SubjectUserSid'>NT
AUTHORITY\SYSTEM</Data><Data Name='SubjectUserName'> USABLDRRECFLOW01$</Data><Data
Name='SubjectDomainName'>SAFAWARE</Data><Data Name='SubjectLogonId'>0x3e7</Data><Data
Name='ObjectServer'>Security</Data><Data Name='ObjectType'>File</Data><Data
Name='ObjectName'>\Device\Floppy0</Data><Data Name='HandleId'>0x328</Data><Data
Name='TransactionId'>{00000000-0000-0000-0000-000000000000}</Data></EventData></Event>
```


Removeable Storage parses into Subject. Object and Object Name are in use already. Object Type could be used in this instance, possibly rearranging use of Object and Object Name, as they are File and \Device\Floppy0, respectively.

URL

The URL referenced or impacted by activity reported in the log.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	URL
Client Console Short Name	URL
Web Console Tab/Name	URL
Elasticsearch Field Name	url
Rule Builder Column Name	URL
Regex Pattern	<url>
NetMon Name	Not applicable

Field Relationships

- Domain Impacted (or Domain prior to 7.4)
- Domain Origin (or Domain in 7.4 and later)
- Session
- Response Code
- Protocol Number
- Protocol Name

Common Applications

- Proxy
- IDS/IPS
- Network monitoring

- Firewall
- Web servers/DNS

Use Case

- Tracking user web activity.
- Tracking and comparing hostile domains with lists of known bad web domains.

MPE/Data Masking Manipulations

Data Masking is used for QNAME format URL (14)DB001560E6EBC5(9)soasdfgtu(3)com(0.

Usage Standards

Do not use the vendor's link to details, which parses into Vendor Info.

Examples

- Blue Coat Proxy

```
08 27 2011 19:00:00 1.1.1.1 <USER:NOTE> 2011-08-27 02:05:36 151 3.1.4.2 - - - OBSERVED
"Email" http://Host10.com/neo/launch?.rand=6upoddav8e6  204 TCP_NC_MISS POST text/json
http Host10 80 /neo/stat - - "Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 5.2; WOW64;
Trident/4.0; .NET CLR 1.1.4322; .NET CLR 2.0.50727; .NET CLR 3.0.04506.30; .NET CLR
3.0.4506.2152; .NET CLR 3.5.30729; InfoPath.2)" 1.1.1.1 492 1434 -
```

> Highlighted URL from proxy log parses into URL.

- Windows DNS

```
11/21/2011 10:14:05 AM 0F8C PACKET  00000000089853C0 UDP Snd 1.1.1.1  fa93 R Q [8385 A
DR NXDOMAIN] A (14)HP001560E6EBC5(9)sonalysts(3)com(0)
```

> (14)DB001560E6EBC5(9)soasdfgtu(3)com(0(14)DB001560E6EBC5(9)soasdfgtu(3)com(0 with length octets. This is often a use case for data masking to replace the length octet with a period.

User Agent [7.2]

The User Agent string from web server logs (for example, Mozilla/5.0 (Windows NT 6.1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/41.0.2228.0 Safari/537.36).

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String (255 characters maximum)

Aliases

Use	Alias
Client Console Full Name	User Agent
Client Console Short Name	User Agent
Web Console Tab/Name	User Agent
Elasticsearch Field Name	userAgent
Rule Builder Column Name	UserAgent
Regex Pattern	<useragent>
NetMon Name	Not applicable

Field Relationships

- Full URL

Common Applications

- Web server logs
- Firewalls

Use Case

- Detecting malicious or malformed user agents.
- Searching for user agents as IOCs.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

Parse the full user agent string into the field.

Examples

- Juniper SSLVPN

```
07 31 2007 10:24:57 1.1.1.1 <LOC6:INFO> SSLVPN: id=sslvpn sn=0006222222B74
time="2007-07-31 10:24:57" vp_time="2007-07-31 15:24:57 UTC" fw=1.1.1.1 pri=6 m=18
src=1.1.1.1 dst=1.1.1.1 user="pete.store" usr="pete.store" msg="NetExtender"
rule=access-policy proto=NetExtender agent="Mozilla/4.0 (compatible; MSIE 6.0; Windows
NT 5.1; SV1)"
```

> Mozilla/4.0… parses into User Agent.

- MS IIS Web Log

```
10 30 2007 15:41:49 USABLDRRECFLOW01/1.1.1.1 <USER:NOTE> Oct 30 15:41:53 recflow/1.1.1.1
IISWebLog 3 2007-10-30 19:41:47 W3SVC414557987 recflow 1.1.1.1 POST /DataPHost2 - 443 -
1.1.1.1 HTTP/1.1 Mozilla/4.0+(compatible;
+MSIE+6.0;+Windows+5.2.3790.0;+MS+.NET+Remoting;+MS+.NET+CLR+1.1.4322.2407+) - - Host1
200 0 0 2277 1993 0Full UserAgent string capture
```

- Bluecoat Proxy

```
2010-03-01 20:23:45 1 1.1.1.1 pete.store safaware\Domain%20Users - OBSERVED "Sports/
Recreation" http://espn.go.com/free-online-games/  200 TCP_HIT GET image/jpeg http
a.espncdn.com 80 /i/espnarcade/GOM/116x67_gom_touch.jpg - jpg "Mozilla/4.0 (compatible;
MSIE 7.0; Windows NT 5.1; InfoPath.2; .NET CLR 2.0.50727; .NET CLR 3.0.04506.30; .NET
CLR 3.0.04506.648; .NET CLR 3.0.4506.2152; .NET CLR 3.5.30729)" 1.1.1.1 4318 443 -
```

Version

The software or hardware device version described in either the process, object, or entity.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Version
Client Console Short Name	Version
Web Console Tab/Name	Version
Elasticsearch Field Name	version
Rule Builder Column Name	Version
Regex Pattern	<version>
NetMon Name	Varies by protocol (most commonly ProtocolVersion)

Field Relationships

- Object (version describes object)
- Process (version describes process)
- Entity
- Host Fields
- User Agent (previously version was abused to contain user agent)

Common Applications

- Vulnerability scanners
- Virus scanners
- Asset inventory

Use Case

If multiple versions are contained in log, the priority is to capture the version of the object of the log, not the version of the product creating the log.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

Prioritize the version of an end object over the version of a product generating the log.

Examples

Correct Examples

- Cb Response

```
05 13 2016 19:56:26 1.1.1.1 <USER:NOTE> LEEF:1.0|CB|CB|5.1|watchlist.storage.hit.binary|
cb_server=cbserver cb_version=211 company_name=RecordFlow Technology Ltd.
copied_mod_len=1022272 digsig_result=Unsigned digsig_result_code=2148204222 endpoint=
USABLDRRECFLOW01|2 file_desc=SysAid  Agent file_version=1.1.1.1 group=RecordFlow HQ
host_count=1 internal_name=AgentStuffManager.dll is_64bit=true is_executable_image=false
last_seen=2016-05-14T02:49:18.142Z legal_copyright=© Copyright 2013 RecordFlow
Technologies Ltd. md5=59E0D058686BD35B0D5C02A4FD8BD0E0observed_filename=c:\\program
files\\sysaid\\agentstuffsmanager.dll orig_mod_len=1022976
original_filename=AgentstuffManager.dll os_type=Windows product_name=SysAid  Agent
product_version=1.1.1.1 server_added_timestamp=2016-05-14T02:49:18.142Z
server_name=localhost.localdomain timestamp=1463194218.586
type=watchlist.storage.hit.binary watchlist_4=2016-05-14T02:50:03.177584Z watchlist_id=4
watchlist_name=Newly Loaded Modules
```

> File version parses into Version. Cb_version is not parsed because the device sending the log is not very useful.

- Windows Event Log

```
10/23/2007 10:07 AM TYPE= USER= Safaware\pete.store COMP= USABLDRRECFLOW01
SORC=BPService CATG=Authentication\Interactive EVID=1000 MESG=Biometric authentication
was performed.    Username: pete.store Domain: Safaware Workstation: Safaware \
USABLDRRECFLOW01Security score: 75  Threshold: 30  Enrollment client: BPDave
Authentication client: BPDave  Client version: 3.0  AuthTag: 222222-dff3-4a70-
b1940157ab9d2d22  Effective settings from: pete.store  Keyboard:
```

> Client Version parses into Version. This could be useful for software auditing.

- CylanceProtect

```
Cylance08 24 2016 07:11:50 1.1.1.1 <SLOG:WARN> 1 2016-08-24T12:11:30.2394853Z sysloghost
CylancePROTECT - - - Event Type: Device, Event Name: SystemSecurity, Device Name:
USABLDRRECFLOW01, Agent Version: 1.2.1370.119, IP Address: (), MAC Address: (), Logged
On Users: (Safaware\pete.store), OS: Microsoft Windows 7 Enterprise Service Pack 1 x64
6.1.7601
```

> Cylance Agent version parses into Version. This could be used for ensuring all agents are up to date.

Incorrect Examples

- Windows Event Log

```
4/3/2007 10:50 AM TYPE=FailureAudit USER=User1 COMP=Host1 SORC=Security CATG=Detailed
Tracking EVID=861 MESG=The Windows Firewall has detected an application listening for
incoming traffic.    Name: -  Path: D:\stuff\jboss-3.2.3\bin\JavaSHost3  Process
identifier: 5668  User account: SYSTEM  User domain: NT AUTHORITY  Service: Yes  RPC
server: No  IP version: IPv4  IP protocol: TCP  Port number: 4087  Allowed: No  User
notified: No
```

> IP Version is not the kind of version needed.

- Windows Event Log

```
<Event xmlns='http://Host2/win/2004/08/events/event'><System><Provider Name='Microsoft-
Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0322g22d}'/
><EventID>6272</EventID><Version>1</Version><Level>Information</Level><Task>Network
Policy Server</Task><Opcode>Info</Opcode><Keywords>Audit Success</Keywords><TimeCreated
SystemTime='2010-06-01T21:40:38.228246300Z'/><EventRecordID>26101649</
EventRecordID><Correlation/><Execution ProcessID='452' ThreadID='1500'/
><Channel>Security</Channel><Computer>Host1</Computer><Security/></
System><EventData>Network Policy Server granted access to a user.    User:    Security
ID:    Safaware\pete.store    Account Name:    pete.store    Account Domain:    UNR    Fully
Qualified Account Name: UNR\rhickok    Client Machine:    Security ID:    NULL SID
Account Name:    -    Fully Qualified Account Name: -    OS-Version:    -    Called Station
Identifier:   000B8222222    Calling Station Identifier:   00000000000    NAS:    NAS IPv4
Address:   1.1.1.1    NAS IPv6 Address:   -    NAS Identifier:    -    NAS Port-Type:
Wireless - IEEE 802.11    NAS Port:    0    RADIUS Client:    Client Friendly Name:   Aruba
Controller 1    Client IP Address:    1.1.1.1    Authentication Details:    Connection
Request Policy Name: Use Windows authentication for all users    Network Policy Name:
RCF WPA    Authentication Provider:   Windows    Authentication Server:   Host1
Authentication Type:   MS-CHAPv2    EAP Type:    -    Account Session Identifier:   -
```

```
Logging Results:   Accounting information was written to the local log file.
Quarantine Information:   Result:    Full Access   Session Identifier:   -  </
EventData></Event>
```

OS-Version, if populated, would be more appropriate to parse.

Kbytes/Packets Tab

Schema fields that are displayed in the Kybtes/Packets tab of the Web Console. All fields in this section are polyfields:

- Host (Impacted) Kbytes Rcvd
- Host (Impacted) Kbytes Sent
- Host (Impacted) Kbytes Total
- Host (Impacted) Packets Rcvd
- Host (Impacted) Packets Sent
- Host (Impacted) Packets Total

These fields are calculated from source data that is parsed in fields with different names, different units, or both.

[prefix] [Bits/Bytes] [blank/In/Out]

Parsing fields for the size in bytes of the object described in the log.

- bitsin
- bitsout
- bytesin
- bytesout
- kilobitsin
- kilobitsout
- kilobytesin
- kilobytesout
- megabitsin
- megabitsout
- megabytein
- megabyteout
- gigabitsin
- gigabitsout
- gigabytein
- gigabyteout
- terabitsin
- terabitsout
- terabytesin
- terabytesout
- petabitsin
- petabitsout
- petabytesin
- petabytesout

Data Type

Double

Aliases

Use	Alias
Client Console Full Name	Various
Client Console Short Name	Various
Web Console Tab/Name	Various
Elasticsearch Field Name	Various
Rule Builder Column Name	Various
Regex Pattern	Various
NetMon Name	Various

Field Relationships

Kbytes tab in the Web Console.

Common Applications

- Network flows
- File sizes

Use Case

Anything measurable in terms of bytes/bits.

MPE/Data Masking Manipulations

Normalized to bytes.

Usage Standards

- Only use once per log (enforced by Super User console).
- Use whichever prefex is the best possible match and let the MPE do the conversion.

Examples

- SQLServer 2012 Error Log

```
2013-08-01 14:13:23.35 Server      Detected 3839 MB of RAM. This is an informational
message; no user action is required.
```

> Parse 3839 into MegaBytes.

- Adtran Switch

```
05 10 2014 22:23:57 1.1.1.1 <KERN:INFO> May 10 22:23:54 bbq22222 FIREWALL: id=firewall
time="2014-05-10 22:23:54" fw=BBQ2222 pri=6 rule=23 proto=53/udp src=1.1.1.1 dst=1.1.1.1
msg="Connection timed out.Bytes transferred : 228 Src 62725 Dst 53 from Private policy-
class on interface vlan 1" agent=AdFirewall
```

> Indicating 228 bytes transferred out.

- BlackBerry Enterprise Server

```
<2013-03-28 15:32:39.268 EDT>:[20945]:<BBQ-BES01_1>:<INFO >:<LAYER = IPPP, DEVICEPIN =
2ab20a5d, DOMAINNAME = USABLDRRECFLOW01, CONNECTION_TYPE = DEVICE_CONN, ConnectionId =
1374706373, DURATION(ms) = 7465, MFH_KBytes = 3.479, MTH_KBytes = 2.946,
MFH_PACKET_COUNT = 7, MTH_PACKET_COUNT = 5>
```

> MFH is Kbytesout MTH is KBytesIn.

Packets [Total/In/Out]

Number of packets received by Impacted Host (in) or sent by Impacted Host (out) or captured in either direction (total). Often stored in all three fields.

Data Type

Double

Aliases

Use	Alias
Client Console Full Name	Host (Impacted) Packets Rcvd Host (Impacted) Packets Sent Host (Impacted) Packets Total
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Impacted) Packets Rcvd Host (Impacted) Packets Sent Host (Impacted) Packets Total
Elasticsearch Field Name	itemsPacketsIn itemsPacketsOut impactedHostTotalPackets
Rule Builder Column Name	PacketsIn PacketsOut
Regex Pattern	<packetsin> <packetsout> <packets>
NetMon Name	TotalPackets

Field Relationships

- Packets In/Out
- Items In/Out

Common Applications

Network traffic analysis.

Use Case

- Evaluating how much network traffic a given application generates.
- Measuring average packet size as an indicator of protocol abuse.

MPE/Data Masking Manipulations

Conversion to In/Out.

Usage Standards

Capture total packets if possible.

Examples

- Tectica SSH server

```
84540711 | 8/8/2013 1:40:01 AM | None | N/A | USABLDRRECFLOW01| Information | 0 | SSH
Tectia Server | 1300 Channel inbound statistics, Username: uninitialized, Session-Id:
29936, Channel Id: 0, Packet count: 15, Packet size: 127
```

> Packet count should be Packets.

Classification Tab

This section contains the fields displayed in the Classification tab. These fields relate to metadata around the log, focusing on adding context to the information in other tabs that are more descriptive of the object.

The following fields are on the Classification tab:

- CVE [7.2]
- Severity
- Threat ID [7.2]
- Threat Name [7.2]
- Vendor Info [7.2]
- Vendor Message ID

There are several polyfields and injected/configuration data in this section, including:

- Classification
- Common Event
- Priority
- Direction

CVE [7.2]

CVE ID (for example, CVE-1999-0003) from vulnerability scan data.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String (64 characters maximum)

Aliases

Use	Alias
Client Console Full Name	Not applicable
Client Console Short Name	Not applicable
Web Console Tab/Name	Classification/CVE
Elasticsearch Field Name	cve
Rule Builder Column Name	CVE
Regex Pattern	<cve>
NetMon Name	Not applicable

Field Relationships

- Object (prior parsing for CVE)
- VMID (prior parsing for CVE)
- Threat Name
- VMID

Common Applications

- Vulnerability scanners
- F5
- Qualys
- IDS (Bro, Snort)
- NGFW (Palo Alto, Check Point)

Use Case

- Cross referencing threat feeds.
- Finding an entry point for an attack.
- Locating what is vulnerable to CVE and what is the impact if exposed.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Use most common format standard CVE-YYYY-#######.
- A malformed CVE can be represented as CVE-MAP-NOMATCH. Parse that as a valid CVE because that is what the log message says.

Examples

- Symantec Endpoint Protection

```
05 23 2014 20:21:58 1.1.1.1 <LPTR:CRIT> May 23 20:07:35 SymantecServer USABLDRRECFLOW01:
USABLDRRECFLOW01,[SID: 27517] Attack: OpenSSL Heartbleed CVE-2014-0160 3 attack blocked.
Traffic has been blocked for this application: SYSTEM,Local: 1.1.1.1,Local:
000000000000,Remote: ,Remote: 1.1.1.1,Remote: 000000000000,Inbound,TCP,Intrusion ID:
0,Begin: 2014-05-23 19:48:52,End: 2014-05-23 19:48:52,Occurrences: 1,Application:
SYSTEM,Location: Coprorate Network,User: pete.store,Domain: safaware,Local Port
443,Remote Port 52901,CIDS Signature ID: 27517,CIDS Signature string: Attack: OpenSSL
Heartbleed CVE-2014-0160 3,CIDS Signature SubID: 73036,Intrusion URL: ,Intrusion Payload
URL:
```

CVE-2014-0160 parsed into CVE.

- Cb Response

```
05 18 2016 09:51:39 1.1.1.1 <USER:NOTE> LEEF:1.0|CB|CB|511|feed.storage.hit.binary|
alliance_data_nvd=["10473","10472","10475","10470","10435"]       alliance_link_nvd=http
://web.nvd.nist.gov/view/vuln/detail?vulnId\=CVE-2013-3353
alliance_score_nvd=100     alliance_updated_nvd=2015-08-03T23:55:33.000Z
cb_server=cbserver  cb_version=511      company_name=Adobe Systems Incorporated
computer_name= USABLDRRECFLOW01  copied_mod_len=7790179     digsig_result=Unsigned
```

```
digsig_result_code=2148204800    endpoint=[" USABLDRRECFLOW01|26"," USABLDRRECFLOW01|
13"," USABLDRRECFLOW01|39"," USABLDRRECFLOW01|35"," USABLDRRECFLOW01|14"]
feed_id=13   feed_name=nvd file_desc=Adobe Acrobat Annot Plug-In
file_version=1.1.1.1       group=RecordFlow HQ host_count=5 hostname=
USABLDRRECFLOW01    ioc_attr={}  ioc_type=md5
ioc_value=4c6b53d9f75cb772e43f65960f905919       is_64bit=false
is_executable_image=false  last_seen=2016-05-18T00:01:11.682Z
legal_copyright=Copyright 1984-2012 Adobe Systems Incorporated and its licensors. All
rights reserved.    md5=59E0D058686BD35B0D5C02A4FD8BD0E0    observed_filename=["c:\
\program files (x86)\\adobe\\reader 11.0\\reader\\plug_ins\\annots.api"]
orig_mod_len=7790179       original_filename=Annot.api       os_type=Windows
product_name=Adobe Acrobat Annot  product_version=1.1.1.1    report_id=10435
report_score=100    sensor_id=14
server_added_timestamp=2016-05-17T15:26:48.469Z       server_name=localhost.localdomain
timestamp=1463589930.842       type=feed.storage.hit.binary
watchlist_4=2016-05-17T15:30:03.182Z
```

CVE parsed into CVE field from URL (may not be sustainable). Not predictable enough to parse.

- ForcePoint

```
10 28 2016 15:22:15 1.1.1.1 <KERN:INFO> CEF:0|FORCEPOINT|Alert|unknown|278069|HTTP_SHS-
Microsoft-Windows-MHTML-Information-Disclosure-CVE-2011-0096-3|7|spt=3811
destinationServiceName=HTTP deviceExternalId=Davestown node 2 dst=1.1.1.1
requestMethod=POST cat=Potential Compromise requestURL=Host2 app=tcp_service_5080 rt=Oct
28 2016 15:22:14 deviceFacility=Inspection destinationTranslatedPort=5080
sourceTranslatedPort=3811 destinationTranslatedAddress=1.1.1.1
sourceTranslatedAddress=1.1.1.1 act=Permit deviceOutboundInterface=2 proto=6 dpt=5080
src=1.1.1.1 dvc=1.1.1.1 dvchost=1.1.1.1 cs1Label=RuleId cs1=1073.1
```

CVE showing inline within CEF vendor info. Full header could be VMID or VendorInfo.

- McAfee Network Security Manager

```
03 27 2014 08:29:30 1.1.1.1 <SAU1:WARN> Mar 27 08:29:35 SyslogAlertForwarder: 2014-03-27
08:29:32 EDT!N/A!N/A!22222222222!0x4510fa00!Signature!Medium!Medium!Unknown!Exploit!
code-execution!Inbound!Inconclusive!1.1.1.1!1.1.1.1!80!24683!http!tcp!BBQ!BBQ!Proxy
Traffic (8A-8B)!signature!CVE-2013-3861!Not Forwarded!Unknown!No error!Unknown!HTTP:
JSON Parsing Vulnerability
```

CVE within exclamation delimiters.

Severity

The vendor's view of the severity or level of log message.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Severity
Client Console Short Name	Severity
Web Console Tab/Name	Severity
Elasticsearch Field Name	severity
Rule Builder Column Name	Severity
Regex Pattern	<severity>
NetMon Name	Severity for alarms only

Field Relationships

- Status
- VMID
- Vendor Info
- ThreatId
- ThreatName

Common Applications

- Syslog reports severity in the format <loc0:info>, with info being the severity level.
- Windows Event Log severity

Use Case

- Anything that generates alarms or analyzes risk.
- Almost every log format has a severity.

MPE/Data Masking Manipulations

Multilingual logs might have severity in native language. Use masking to convert to standard English. (See Windows logs, for example.)

Usage Standards

- Represent the severity the way the vendor/log source does in the clearest text way. Do not attempt to convert 0-5 to low/medium/high or red/yellow/green unless the vendor defines 0 = low.
- Do not misuse for level of confidence (for example, from an AV log).

Examples

- Windows Event Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Security-Auditing' Guid='{2222222-5478-4994-a5ba-3e3b0328c30d}'/
><EventID>5058</EventID><Version>0</Version><Level>Information</Level><Task>Other System
Events</Task><Opcode>Info</Opcode><Keywords>Audit Success</Keywords><TimeCreated
SystemTime='2016-02-02T00:24:23.559228400Z'/><EventRecordID>7670651176</
EventRecordID><Correlation/><Execution ProcessID='572' ThreadID='3136'/
><Channel>Security</Channel><Computer> USABLDRRECFLOW01</Computer><Security/></
System><EventData><Data Name='SubjectUserSid'>NT AUTHORITY\NETWORK SERVICE</Data><Data
Name='SubjectUserName'> USABLDRRECFLOW01$</Data><Data
Name='SubjectDomainName'>SAFAWARE</Data><Data Name='SubjectLogonId'>0x3e4</Data><Data
Name='ProviderName'>Microsoft Software Key Storage Provider</Data><Data
Name='AlgorithmName'>%%2432</Data><Data Name='KeyName'>le-a1f08494-0ec3-4902-9d6c-
caeeda9ce4f6</Data><Data Name='KeyType'>%%2499</Data><Data Name='KeyFilePath'>C:
\ProgramData\Microsoft\Crypto\RSA\MachineKeys\222222222229530509a71f1</Data><Data
Name='Operation'>%%2458</Data><Data Name='ReturnCode'>0x0</Data></EventData></Event>
```


<Level> tags in Windows indicate severity of the log message.

- Syslog - Apache Access Log


```
11 14 2013 17:19:04 1.1.1.1 <LOC5:INFO> Nov 14 22:19:04 USABLDRRECFLOW01access_http_log:
[14/Nov/2013:22:19:04 +0000] 1.1.1.1 1.1.1.1 HTTP/1.1 "POST /foundation/
getStandingsAjax.jsp HTTP/1.1" 2764 https://www.recordflow.biz
```


Any Syslog message contains a header that indicates severity level.

- Syslog – Crowdstrike Falconhost CEF

```
12 14 2016 11:39:44 1.1.1.1 <USER:NOTE> CEF:0|CrowdStrike|FalconHost|1.0|
DetectionSummaryEvent|Detection Summary Event|2| externalID=222222222222222222
cn2Label=ProcessId cn2=148191318711589 cn1Label=ParentProcessId cn1=148191316778231
shost=TheNarrowSea suser=IIS1$ msg=An administrative/reconnaissance tool (xcopy.exe,
ping.exe, tasklist.exe, ftp.exe, autoruns.exe) was spawned under an IIS worker process.
fname=systeminfo.exe filePath=\\Device\\HarddiskVolume1\\Windows\\System32
cs1Label=CommandLine cs1=systeminfo
fileHash=59E0D058686BD35B0D5C02A4FD8BD0E0sntdom=TARGETNET cs6Label=FalconHostLink cs6=ht
tps://falcon.crowdstrike.com/activity/detections/detail/2222222222/2222222222
 cn3Label=Offset cn3=1066147 deviceCustomDate1Label=ProcessStartTime
deviceCustomDate1=2016-12-14 18:39:42
```

> In this Syslog example, the Syslog severity is ignored in favor of the CEF format header which includes its own severity level.

Threat ID [7.2]

The ID number of a threat when available from an IDS/IPS signature, endpoint protection, or firewall log.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Not applicable
Client Console Short Name	Not applicable
Web Console Tab/Name	Threat ID
Elasticsearch Field Name	threatId
Rule Builder Column Name	ThreatID
Regex Pattern	<threatid>
NetMon Name	Not applicable

Field Relationships

- Threat Name
- VMID
- Vendor Message
- Object
- Object Name
- Object Type
- Process

- Process ID

Common Applications

- IDS/IPS
- Vulnerability scanners
- Proxy

Use Case

Correlating threats.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Signatures
- Numeric or string identifiers for threats under different names

Examples

- Cisco IDS/IPS

```
<sd:evIdsAlert eventId="222222222" vendor="Cisco" severity="high" xmlns:sd="http://
example.org/2003/08/sdee">bhiips xmlns:cid="http://www.cisco.com/cids/2006/08/
cidee">sensorApp xmlns:cid="http://www.cisco.com/cids/2006/08/cidee">9055 offset="-300"
timeZone="GMT-05:00">1232562570119108000</sd:time><sd:signature description="MSSQL
Resolution Service Stack Overflow" id="4703" cid:version="S367" cid:type="other"
cid:created="20000101" xmlns:cid="http://www.cisco.com/cids/2006/08/cidee">0 sample
truncated.
```

> Signature ID of a threat detectedeStreamer.

- eStreamer

```
LOGTYPE=INT_EVT_51_IPV4 R_ID=24105 R_REV=9 S_IP=1.1.1.1 S_PORT=58730 D_IP=1.1.1.1
D_PORT=8080 U_ID=0 U= R_NAME=MALWARE-OTHER HTTP POST request to a GIF file
CLASSIFICATION_ID=22 CLASSIFICATION=Detection of a Non-Standard Protocol or Event
PROT_NUM=6 PROT= ING_IF=s1p5 EG_IF=s1p1 BLOCKED=NotBlocked MANAGED_DEV_ID=9
EVT_ID=263305 EVT_T=01/15/2015 20:42:56 GEN_ID=1 PRI_ID=2 PRI=medium
IMPACT_FLAGS=MonitoredHost, MappedHost, ServerPortOrIp IMPACT=Orange BBQ_LABEL=0
VLAN_ID=0 POL=Intrusion Policy - Corporate AP_PROT=HTTP ACS_CTL_R=File Inspection Rule
ACS_CTL_POL=Access Control Policy - CORPORATE nnq_nnq_Z=Corporate EG_bbq_Z=OOB
```

R_ID=24105 is the Threat ID from this IDS signature log.

- Symantec Endpoint

```
05 22 2014 11:08:02 1.1.1.1 <LPTR:CRIT> May 22 10:55:13 SymantecServer USABLDRRECFLOW01:
USABLDRRECFLOW01,[SID: 25238] Fake App Attack: Misleading Application Website attack
blocked. Traffic has been blocked for this application: \DEVICE\HARDDISKVOLUME1\PROGRAM
FILES\GOOGLE\CHROME\APPLICATION\CHROME.EXE,Local: 1.1.1.1,Local:
000000000000,Remote: ,Remote: 1.1.1.1,Remote: 000000000000,Inbound,TCP,Intrusion ID:
0,Begin: 2014-05-22 10:53:42,End: 2014-05-22 10:53:42,Occurrences: 1,Application: /
DEVICE/HARDDISKVOLUME1/PROGRAM FILES/GOOGLE/CHROME/APPLICATION/CHROME.EXE,Location:
Coprorate Network,User: Christina_McCloud,Domain: INDY,Local Port 4295,Remote Port
80,CIDS Signature ID: 25238,CIDS Signature string: Fake App Attack: Misleading
Application Website,CIDS Signature SubID: 70185,Intrusion URL: pcfaster.info/usdown/?
sence=asdifas892nsndsafusaljnsxckad,Intrusion Payload URL:
```

SID is the signature ID of the detected threat.

Threat Name [7.2]

The name of a threat described in the log message (for example, malware, exploit name, or signature name). Do not overload with Policy.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Not applicable
Client Console Short Name	Not applicable
Web Console Tab/Name	Threat Name
Elasticsearch Field Name	threatName
Rule Builder Column Name	ThreatName
Regex Pattern	<threatname>
NetMon Name	Not applicable

Field Relationships

- Threat ID
- VMID
- Vendor Message
- Object
- Object Name
- Object Type

- Process
- ProcessID
- Policy
- Reason

Common Applications

- IDS/IPS
- Vulnerability scanners
- Proxy

Use Case

- Threat Name frequency for reporting.
- Identifying threats.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Signature names
- Malware names
- Vulnerability names
- Exploit names
- Can be used independently of Threat ID (for example, AV detections, or identifying malicious processes or objects)

Examples

- Cisco IDS/IPS

```
<sd:evIdsAlert eventId="2222222222222" vendor="Cisco" severity="high" xmlns:sd="http://
example.org/2003/08/sdee">bhiips xmlns:cid="http://www.cisco.com/cids/2006/08/
cidee">sensorApp xmlns:cid="http://www.cisco.com/cids/2006/08/cidee">9055 offset="-300"
timeZone="GMT-05:00">1232562570119108000</sd:time><sd:signature description="MSSQL
Resolution Service Stack Overflow" id="4703" cid:version="S367" cid:type="other"
cid:created="20000101" xmlns:cid="http://www.cisco.com/cids/2006/08/cidee">0:...log
sample truncated.
```

The description describes the threat indicated by signature ID 4703.

- Qualys Vulnerability Scanner

```
HOSTIP=1.1.1.1 HOSTNAME= USABLDRRECFLOW01HOSTOS=Linux 2.6 PORT= PROTOCOL= QID=115731
DETECTIONTYPE=Potential STATUS=New FIRSTFOUND=2010-10-05 01:20:11Z LASTFOUND=2010-10-05
```

```
01:20:11Z VULNERABILITY=Apache 1.3 and 2.0 Web Server Multiple Vulnerabilities
 VULNERABILITYTYPE=Vulnerability or Potential Vulnerability CATEGORY=Local
SEVERITYLEVEL=3 PATCHABLE=1 KBLASTUPDATE=2010-09-13 18:52:19Z CVE=CVE-2006-5752(http://
cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2006-5752),CVE-2007-3304(http://
cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-3304)
```

Name of vulnerability.

- eStreamer

```
LOGTYPE=INT_EVT_51_IPV4 R_ID=24105 R_REV=9 S_IP=1.1.1.1 S_PORT=58730 D_IP=1.1.1.1
D_PORT=8080 U_ID=0 U= R_NAME=MALWARE-OTHER HTTP POST request to a GIF file
 CLASSIFICATION_ID=22 CLASSIFICATION=Detection of a Non-Standard Protocol or Event
PROT_NUM=6 PROT= ING_IF=s1p5 EG_IF=s1p1 BLOCKED=NotBlocked MANAGED_DEV_ID=9
EVT_ID=263305 EVT_T=01/15/2015 20:42:56 GEN_ID=1 PRI_ID=2 PRI=medium
IMPACT_FLAGS=MonitoredHost, MappedHost, ServerPortOrIp IMPACT=Orange MPLS_LABEL=0
VLAN_ID=0 POL=Intrusion Policy - Corporate AP_PROT=HTTP ACS_CTL_R=File Inspection Rule
ACS_CTL_POL=Access Control Policy - CORPORATE ING_SEC_Z=Corporate BBQ_SEC_Z=OOB
```

R_NAME represents the signature ID (R_ID=24105) of the threat.

- Symantec Endpoint

```
05 22 2014 11:08:02 1.1.1.1 <LPTR:CRIT> May 22 10:55:13 SymantecServer
USABLDRRECFLOW01USABLDRRECFLOW01,[SID: 25238] Fake App Attack: Misleading Application
Website attack blocked. Traffic has been blocked for this application:
\DEVICE\HARDDISKVOLUME1\PROGRAM FILES\GOOGLE\CHROME\APPLICATION\CHROME.EXE,Local:
1.1.1.1,Local: 000000000000,Remote: ,Remote: 1.1.1.1,Remote:
000000000000,Inbound,TCP,Intrusion ID: 0,Begin: 2014-05-22 10:53:42,End: 2014-05-22
10:53:42,Occurrences: 1,Application: /DEVICE/HARDDISKVOLUME1/PROGRAM FILES/GOOGLE/
CHROME/APPLICATION/CHROME.EXE,Location: Coprorate Network,User: pete.store,Domain:
safaware,Local Port 4295,Remote Port 80,CIDS Signature ID: 25238,CIDS Signature string:
Fake App Attack: Misleading Application Website,CIDS Signature SubID: 70185,Intrusion
URL: recordflow.biz,Intrusion Payload URL:
```

“Fake App Attack: Misleading Application Website attack” is the name of the possible threat detected of signature ID 25238.

Vendor Info [7.2]

Description of specific vendor log or event identifier for the log. Human readable elaboration that directly correlates to the VMID.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Not applicable
Client Console Short Name	Not applicable
Web Console Tab/Name	Vendor Info
Elasticsearch Field Name	vendorInfo
Rule Builder Column Name	VendorInfo
Regex Pattern	<vendorinfo>
NetMon Name	Not applicable

Field Relationships

- VMID
- Subject

Common Applications

Any device that generates predetermined message types or categories that are differentiated by a brief description or identification number.

Use Case

Understanding VMID for correlating events without depending on the rule name, common event/classification.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- To be used when VMID is present.
- To be used rarely when VMID is not present.
    - Capturing long event descriptions such as a sentence.
- Not for subrules.

Examples

- Windows Event Log Security

```
<Event xmlns='http://Host2/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{2222222-5478-4994-a5ba-3e3b0328c30d}'/><EventID>4663</EventID><Version>0</Version><Level>Information</Level><Task>Kernel Object</Task><Opcode>Info</Opcode><Keywords>Audit Success</Keywords><TimeCreated SystemTime='2009-07-07T23:24:49.212Z'/><EventRecordID>451107</EventRecordID><Correlation/><Execution ProcessID='4' ThreadID='88'/><Channel>Security</Channel><Computer> USABLDRRECFLOW01</Computer><Security/></System><EventData>An attempt was made to access an object.

Subject:

   Security ID:         USABLDRRECFLOW01\Administrator

   Account Name:        Administrator

   Account Domain:             USABLDRRECFLOW01

   Logon ID:            0x2a9fe

Object:

   Object Server:       Security

   Object Type: SymbolicLink

   Object Name: \GLOBAL??\C:

   Handle ID:   0x3c0
```

Process Information:

```
Process ID:  0x8d0
Process Name: C:\Windows\Host10
```

Access Request Information:

```
Accesses:    Use symbolic link
Access Mask: 0x1</EventData></Event>
```

> Describes in human readable form what the event ID (VMID) translates to.

- CyberArk Privileged Threat Analytics

```
CEF:0|CyberArk|PTA|3.1|21|Suspected credentials theft|9|duser=pete.store dst=
USABLDRRECFLOW01cs2Label=eventID cs2=5b720c983420f5222222d
deviceCustomDate1Label=detectionDate deviceCustomDate1=1422836202000 cs3Label=link cs3=h
ttps://1.1.1.1/incidents/5b722222224979d
```

> Suspected Credentials Theft describes VMID 21.

Vendor Message ID

The specific vendor log or event identifier for the log used to describe a type of event.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Vendor Message ID
Client Console Short Name	Vendor Message ID
Web Console Tab/Name	Vendor Message ID
Elasticsearch Field Name	vendorMessageId
Rule Builder Column Name	VMID
Regex Pattern	<vmid>
NetMon Name	Not applicable

Field Relationships

- Vendor Information
- Threat Name
- Threat ID

Common Applications

Any device that generates predetermined message types or categories that are differentiated by a brief description or identification number.

Use Case

Correlating events.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Describes or identifies an event type
- Sometimes human readable
- Usually numeric
- Can be used for subrules
- Indexed field, do not use subrule tags when making subrules off VMID
- Not for Response Codes
- Not for Threat IDs (signatures)
- Not Event Record ID

Examples

- Windows Event Log Security


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Security-Auditing' Guid='{222222222-5478-4994-
a5ba-3e3b0328c30d}'/><EventID>4624</EventID><Version>0</Version><Level>Information</
Level><Task>Logon</Task><Opcode>Info</Opcode><Keywords>Audit Success</
Keywords><TimeCreated SystemTime='2016-02-09T00:45:00.703363000Z'/><EventRecordID>226991
2024</EventRecordID><Correlation/><Execution ProcessID='520' ThreadID='12080'/
><Channel>Security</Channel><Computer> USABLDRRECFLOW01</Computer><Security/></
System><EventData><Data Name='SubjectUserSid'>NT AUTHORITY\SYSTEM</Data><Data
Name='SubjectUserName'> USABLDRRECFLOW01$</Data><Data
Name='SubjectDomainName'>SAFAWARE</Data><Data Name='SubjectLogonId'>0x3e7</Data><Data
Name='TargetUserSid'>NT AUTHORITY\SYSTEM</Data><Data Name='TargetUserName'>SYSTEM</
Data><Data Name='TargetDomainName'>NT AUTHORITY</Data><Data Name='TargetLogonId'>0x3e7</
Data><Data Name='LogonType'>5</Data><Data Name='LogonProcessName'>Advapi  </Data><Data
Name='AuthenticationPackageName'>Negotiate</Data><Data Name='WorkstationName'></
Data><Data Name='LogonGuid'>{00000000-0000-0000-0000-000000000000}</Data><Data
Name='TransmittedServices'>-</Data><Data Name='LmPackageName'>-</Data><Data
Name='KeyLength'>0</Data><Data Name='ProcessId'>0x200</Data><Data Name='ProcessName'>C:
\Windows\System32\services.exe</Data><Data Name='IpAddress'>-</Data><Data
Name='IpPort'>-</Data></EventData></Event>
```


> The Event ID number is the Vendor Message ID. Event Record ID is not Vendor Message ID. This describes the individual instance of a log.

- Cisco ASA

```
02 03 2015 08:37:17 1.1.1.1 <LOC3:NOTE> :Feb 03 08:37:17 PST: %ASA-session-5-302013:
Built outbound TCP connection 1001222224 for outside:1.1.1.1/80 (1.1.1.1/80) to shr-web-
prod:1.1.1.1/58291 (1.1.1.1/58291)
```

> For Cisco ASA and Cisco products generally, this is where the identifier for the type of event is kept.

- FireEye Web MPS

```
02 01 2016 17:13:19 1.1.1.1 <LOC4:WARN> fenotify-609081.warning: CEF:0|FireEye|MPS|
1.1.1.1875|IM|infection-match|1|rt=Feb 01 2016 23:13:10 UTC src=1.1.1.1 cn3Label=cncPort
cn3=80 cn2Label=sid cn2=84575103 shost= USABLDRRECFLOW01proto=tcp spt=51997 dst=1.1.1.1
cs5Label=cncHost cs5=1.1.1.1 dvchost= USABLDRRECFLOW01dvc=1.1.1.1 smac=00:00:00:00:00:00
cn1Label=vlan cn1=0 dpt=80 externalId=609081 cs4Label=link cs4=https://
romaslcmp01.mayo.edu/event_stream/events_for_bot?ev_id\=609081 act=blocked
cs6Label=channel cs6=GET THINGS dmac=00:00:00:00:00:00 cs1Label=sname
cs1=Exploit.Kit.Angler
```

> For FireEye Web MPS, and CEF messages generally, the type of event is described here in a human readable form.

Host Tab

The Host Tab contains fields that help identify the impacted or origin host, such as IP, hostname, MAC address, and so on.

The following fields are on the Host tab:

- DIP/DestinationIP/Impacted IP
- DIPv4
- DIPv6
- DIPv6E
- Impacted Hostname
- Impacted Hostname or IP
- Impacted Interface
- Impacted MAC Address
- Impacted NAT IP
- IP Address (Origin)
- Origin Hostname
- Origin Hostname or IP
- Origin Interface
- Origin MAC Address
- Origin NAT IP
- Serial Number [7.2]
- SIPv4
- SIPv6
- SIPv6E

Many fields in this tab are polyfields including:

- Host (Origin)
- Host (Impacted)
- IP Address (Origin)
- IP Address (Impacted)
- Hostname (Origin)
- Hostname (Impacted)
- Known Host (Origin)
- Known Host (Impacted)

DIP/DestinationIP/Impacted IP

The host IP that was affected by the activity (for example, target or server). Destination IP in IPv4 or IPv6 format.

Data Type

IP

Aliases

Use	Alias
Client Console Full Name	Host (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Impacted)
Elasticsearch Field Name	impactedIp
Rule Builder Column Name	DIP
Regex Pattern	<dip>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIPv4
- DIPv6
- DIPv6E
- Impacted Hostname
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname or IP
- Impacted NAT IP
- Origin Port
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Networked equipment

Use Case

Host context

MPE/Data Masking Manipulations

Polyfield – Impacted Host

Usage Standards

- Do not override/overload, use <dip> not (?<dip>.*?).
- Impacted is server (In Client-Server Model).
- Impacted is Target (In Attacker-Target Model).
- Use when you see an Impacted IP address IPv4 or IPv6, unless it is an IPv4 address mapped to IPv6, in which case use <dipv6e>.

Examples

- FireEye Web MPS

```
02 01 2016 17:13:19 1.1.1.1 <LOC4:WARN> fenotify-609081.warning: CEF:0|FireEye|MPS|
1.1.1.1875|IM|infection-match|1|rt=Feb 01 2016 23:13:10 UTC src=1.1.1.1 cn3Label=cncPort
cn3=80 cn2Label=sid cn2=84575103 shost= USABLDRRECFLOW01proto=tcp spt=51997 dst=1.1.1.1
cs5Label=cncHost cs5=1.1.1.1 dvchost=romaslcmp01 dvc=1.1.1.1 smac=00:00:00:00:00:00
cn1Label=vlan cn1=0 dpt=80 externalId=609081 cs4Label=link cs4=THINGS
dmac=00:00:00:00:00:00 cs1Label=sname cs1=Exploit.Kit.AnglerDIPv4
```

> Src= in this instance is the host IP impacted by the infection match described in the log. (Attacker-Target). Dst= is the command and control server and therefore the closest Origin (attacker) to be inferred from the log.

- Brocade Switch

```
03 01 2017 02:08:41 1.1.1.1 <LOC6:NOTE> Mar  1 02:08:38 USABLDRRECFLOW01dataplane[2287]:
fw rule INTERNAL-IN:10000 block udp(17) src= USABLDRRECFLOW01/0:00:00:0a:ea:e8/
fe80::e0c0:f0f0:e00c:2029(546) dst=/22:22:2:1:0:2/ff22::2:2(547) len=159 hoplimit=1
len=119
```

Dst= IPv6 address following the MAC ID. Network context showing direction src->dst.

DIPv4

Constituent element of <dip> for only IPv4 parsing (not generally used).

Data Type

IP

Aliases

Use	Alias
Client Console Full Name	Host (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Impacted)
Elasticsearch Field Name	impactedIpV4
Rule Builder Column Name	DIP
Regex Pattern	<dipv4>
NetMon Name	Not applicable

Field Relationships

- Nested element of <dip> default regex
- Cannot be used with <dipv6>

Common Applications

IPv4 only network equipment

Use Case

Use when parsing a log that only contains IPv4 addresses where the very small performance gain over the standard DIP parsing field is necessary.

MPE/Data Masking Manipulations

Polyfield – Impacted Host

Usage Standards

- This field is rarely used.
- Is redundant to <dip>.
- If you are 100% certain an IPv4 address will always appear.
- Only use if you need an extremely minute performance improvement.

Examples

- Trend Micro Deep Security

```
11 19 2014 08:21:12 10.100.6.64 <LOC0:INFO> Nov 19 03:25:07 USABLDRRECFLOW01 dsa_mpnp:
REASON=IPv4_Packet HOSTID=230078 ACT=Deny IN=0C:0B:05:07:B0:05 OUT=
MAC=00:00:00:00:00:00:00:BE:00:00:00:0D:00:0d SRC=2.2.2.2 DST=1.1.1.1 LEN=86 PROTO=ICMP
SPT=0 DPT=0 CNT=1
```

DIPv6

Constituent element of <sip> for only IPv6 parsing (not generally used).

Data Type

IP

Aliases

Use	Alias
Client Console Full Name	Host (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Impacted)
Elasticsearch Field Name	impactedIpV6
Rule Builder Column Name	DIP
Regex Pattern	<dipv6>
NetMon Name	Not applicable

Field Relationships

<dipv6> is a nested element of <dip>

Common Applications

IPv6 only network equipment

Use Case

Use when parsing a log that only contains IPv6 addresses where the very small performance gain over the standard DIP parsing field is necessary.

MPE/Data Masking Manipulations

Polyfield – Impacted Host

Usage Standards

- This is rarely used.
- Is redundant to <dip>.
- If you are 100% certain an IPv4 address will always appear.
- Use if you need an extremely minute performance improvement.

Examples

- Trend Micro Deep Security

```
11 19 2014 08:21:12 10.100.6.64 <LOC0:INFO> Nov 19 03:25:07 USABLDRRECFLOW01 dsa_mpnp:
REASON=IPv6_Packet HOSTID=230078 ACT=Deny IN=0C:0B:05:07:B0:05 OUT=
MAC=00:00:00:00:00:00:00:BE:00:00:00:0D:00:0d SRC=fe80:0:0:0:0cd0:000f:bd2f:000b DST=ff0
1:0:0:0:0:0:0:1 LEN=86 PROTO=ICMPv6 SPT=0 DPT=0 CNT=1
```

DST= shows impacted IPv6 Address.

DIPv6E

The Impacted IPv4 IP address that was mapped to (for example, target or server).

Data Type

IP

Aliases

Use	Alias
Client Console Full Name	Host (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Impacted)
Elasticsearch Field Name	impactedIpV6
Rule Builder Column Name	DIP
Regex Pattern	<dipv6e>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- Impacted Hostname
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname or IP
- Impacted NAT IP
- Origin Port
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Networked equipment

Use Case

Host context

MPE/Data Masking Manipulations

Polyfield – Origin Host

Usage Standards

- Do not override/overload, use <dipv6e> not (?<dipv6e>.*?).
- Impacted is Server (In Client-Server Model).
- Impacted is Target (In Attacker-Target Model).
- Use when you see an Impacted IPv4 address mapped to IPv6.

Examples

- Windows Event Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Iphlpsvc' Guid='{66a5c15c-4f8e-4044-bf6e-71d896038977}'/
><EventID>4200</EventID><Version>0</Version><Level>Information</Level><Task>None</
Task><Opcode>Info</Opcode><Keywords></Keywords><TimeCreated
SystemTime='2016-08-02T19:20:14.492842100Z'/><EventRecordID>5823520</
EventRecordID><Correlation/><Execution ProcessID='920' ThreadID='3936'/
><Channel>System</Channel><Computer> USABLDRRECFLOW01</Computer><Security UserID='NT
AUTHORITY\SYSTEM'/></System><EventData><Data Name='ProtocolType'>1</Data><Data
Name='Interface'>isatap.{f7eec065-6118-437c-8414-eeeeeeeeeeee}</Data><Data
Name='Address'>fe80::5efe:1.1.1.1</Data></EventData></Event>
```


Impacted Address is IPv4 address mapped to IPV6. Traditional IP parsers do not work with this type of address.

Impacted Hostname

The host that was affected by the activity (for example, target or server).

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Host (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Impacted)
Elasticsearch Field Name	impactedName
Rule Builder Column Name	DName
Regex Pattern	<dname>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname or IP
- Impacted NAT IP
- Origin Port
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Networked equipment

Use Case

Host context

MPE/Data Masking Manipulations

Polyfield – Impacted Host

Usage Standards

- Impacted is Server (In Client-Server Model).
- Impacted is Target (In Attacker-Target Model).
- Can be used for parsing fully qualified domain names for non-world wide web context hostnames.

Examples

- Windows Event Log


```
<Event xmlns='http://Host2/win/2004/08/events/event'><System><Provider Name='NETLOGON'/
><EventID Qualifiers='0'>5805</EventID><Level></Level><Task>None</Task><Keywords></
Keywords><TimeCreated SystemTime='2014-02-06T06:03:06.000000000Z'/
><EventRecordID>156578</EventRecordID><Channel>System</Channel><Computer>
USABLDRRECFLOW01</Computer><Security/></System><EventData>The session setup from the
computer USABLDRRECFLOW02failed to authenticate. The following error occurred:

Access is denied.</EventData></Event>
```


<Computer> is the origin of the log message here, but also the domain controller which the origin is trying to authenticate against and is therefore impacted. Client-Server (origin-impacted) relationship applies here. Computer client trying to authenticate is the origin of the request to the server.

Impacted Hostname or IP

The host that was affected by the activity (for example, target or server).

Data Type

- String
- IP

Aliases

Use	Alias
Client Console Full Name	Host (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Impacted)
Elasticsearch Field Name	impactedName, impactedIp
Rule Builder Column Name	Not applicable
Regex Pattern	(<dipn>)
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login

- DIPv6E
- Impacted Hostname
- Impacted NAT IP
- Origin Port
- Impacted Account
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Not applicable

Use Case

See DIP/DestinationIP/Impacted IP and Impacted Hostname.

MPE/Data Masking Manipulations

See DIP/DestinationIP/Impacted IP and Impacted Hostname.

Usage Standards

- Use when a log can contain either an IP or a hostname in the same location.
- Must be wrapped in parenthesis to function (<dipn>).
- Do not overload/override.

Examples

- Aruba Clear Pass
  - `10 22 2015 16:23:22 1.1.1.1 <LOC1:INFO> 2015-10-22 16:23:22,956 [Th 12047 Req 8677508 SessId R0014aec9-06-5628c022] INFO  RadiusServer.Radius - rlm_ldap: found user host/ USABLDRRECFLOW01com in AD:dc-del4-1.synapse.com`
  - `10 22 2015 13:58:51 1.1.1.1 <LOC1:INFO> 2015-10-22 13:58:51,299 [Th 7649 Req 1708827 SessId R00060774-01-5628c16b] INFO  RadiusServer.Radius - rlm_ldap: searching for user 000000000 in AD:1.1.1.1`

> Server being queried (impacted) in log can be represented by an IP or a Hostname.

- Cisco Router
  - `03 02 2009 11:26:27 ATC-CW2K <LOC0:CRIT> Mar 2 11:26:54 USABLDRRECFLOW01ITMGSC: %local0-2-EVENT: 09$Partition=0]PartitionName=&)MODE=3;Alert ID=00061D0}Event ID=001KMPZ|Status=Active^Severity=Critical^Managed Object=1.1.1.1^Managed Object Type=Wireless^CUSTID=Security_Group^CUSTREV=*^Description=HighQueueDropRate::Component=IF-1.1.1.1/1 [Do0];Type=IEEE80211;OutputPacketNoErrorRate=0.11666667 PPS;DuplexMode=FULLDUPLEX;InputPacketQueueDropRate=0.0125 PPS;InputPacketQueueDropPct=48.07692 %;MaxSpeed=54000000;OutputPacketQueueDropPct=0.0`

- `03 02 2009 11:24:57 ATC-CW2K <LOC0:CRIT> Mar 2 11:25:24 USABLDRRECFLOW01 ITMGSC: %local0-2-EVENT: 09$Partition=0]PartitionName=&)MODE=3;Alert ID=0002O5E}Event ID=001KMPT|Status=Active^Severity=Critical^Managed Object=Host2^Managed Object Type=Routers^CUSTID=Security_Group^CUSTREV=*^Description=Unresponsive::Component=1.1.1.1 [Host2];IPStatus=OK;InterfaceName=IF-Host2/19 [Gi0/0.80] [1.1.1.1] [WAAS INTERFACE];InterfaceType=L2VLAN;InterfaceOperStatus=UP;NetworkNumber=1.1.1.1;Inter r`

> In the above two logs Managed Object= can contain either a hostname or an IP address. In both cases, the host/IP are impacted as the object being managed not the manager.

Impacted Interface

The network port or interface which was affected by the activity (for example, target or server).

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Interface (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	Interface (Impacted)
Elasticsearch Field Name	impactedInterface
Rule Builder Column Name	DInterface
Regex Pattern	<dinterface>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- IANA Protocol Number
- IANA Protocol Name

Common Applications

- Switches
- Firewalls
- Network equipment

Use Case

Troubleshooting connectivity.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Impacted is Server (In Client-Server Model).
- Impacted is Target (In Attacker-Target Model).
- If you have more than just a port number (for example, a switch ID), capture full interface name including switch ID.
- A Wireless Access Point can be an interface.

Examples

- Aerohive Access Point

```
05 28 2013 18:38:30 1.1.1.1 <LOC6:INFO> ah_auth: Notify driver to disassoc
2222:cccc:ffff from wifi1.3
```

> Disassociation of client from access point where the AP is impacted server. The client-server (origin-impacted) relationship applies.

- FortiGate

```
02 25 2010 13:56:25 1.1.1.1 <LOC5:ALRT> date=2010-02-25 time=13:56:25
devname=FG3222222222 device_id=FG22222222222 log_id=0419016384 type=ips
subtype=signature pri=alert fwver=040003 severity=critical carrier_ep="N/A"
profile="scan" src=1.1.1.1 dst=1.1.1.1 src_int="port1" dst_int="port2" policyid=48
serial=1514122225 status=detected proto=6 service=2612/tcp vd="root" count=1 src_port=80
dst_port=2612 attack_id=107347979 sensor="all_default" ref="http://Host1/ids/
VID107347979" user="N/A" group="N/A" incident_serialno=128862693 msg="http_decoder:
HTTP.Request.Smuggling"
```

Firewall log showing a signature detection with interface destination (impacted). In this case, the destination (impacted) is represented as destination from the Firewall perspective.

Impacted MAC Address

The MAC Address that was affected by the activity.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	MAC Address (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	MAC Address (Impacted)
Elasticsearch Field Name	impactedMac
Rule Builder Column Name	DMAC
Regex Pattern	<dmac>
NetMon Name	DestMAC

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- IANA Protocol Number
- IANA Protocol Name

Common Applications

- Firewall
- IDS/IPS
- Vulnerability scanners

Use Case

- Differentiating hosts and interfaces.
- Detecting MAC ID cloning.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Can be in any format of MAC address
  - MM:MM:MM:SS:SS:SS
  - MM-MM-MM-SS-SS-SS
  - MMM.MMM.SSS.SSS
  - MM MM MM SS SS SS
- Impacted is Server (In Client-Server Model)
- Impacted is Target (In Attacker-Target Model)

Examples

- FireEye Web MPS

```
02 01 2016 17:13:19 1.1.1.1 <LOC4:WARN> fenotify-609081.warning: CEF:0|FireEye|MPS|
1.1.1.1875|IM|infection-match|1|rt=Feb 01 2016 23:13:10 UTC src=1.1.1.1 cn3Label=cncPort
cn3=80 cn2Label=sid cn2=84575103 shost= USABLDRRECFLOW01proto=tcp spt=51997 dst=1.1.1.1
cs5Label=cncHost cs5=1.1.1.1 dvchost= USABLDRRECFLOW01dvc=1.1.1.1 smac=00:00:00:00:00:00
 cn1Label=vlan cn1=0 dpt=80 externalId=609081 cs4Label=link cs4=THINGS
dmac=00:00:00:00:00:00 cs1Label=sname cs1=Exploit.Kit.AnglerDIPv4
```

> smac= in this log is the target MAC Address (impacted).

- Brocade Switch

```
03 01 2017 02:08:41 1.1.1.1 <LOC6:NOTE> Mar  1 02:08:38 ch3p1gw4 dataplane[2287]: fw
rule INTERNAL-IN:10000 block udp(17) src=dp0p160p1/0:50:56:9a:ea:e8/
fe80::e9c4:f7f6:e72c:2029(546) dst=/33:33:0:1:0:2/ff02::1:2(547) len=159 hoplimit=1
len=119
```

dst= with a possible destination hostname followed by destination (impacted) MAC Address.

Impacted NAT IP

The Impacted Network Address Translated IP address (for example, target or server).

Data Type

IP

Aliases

Use	Alias
Client Console Full Name	NAT IP Address (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	NAT IP Address (Impacted)
Elasticsearch Field Name	impactedNatIp
Rule Builder Column Name	DNATIP
Regex Pattern	<dnatip>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname
- Impacted Hostname or IP
- Origin Port
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Network equipment

Use Case

Internal host context

MPE/Data Masking Manipulations

Polyfield – Impacted Host

Usage Standards

- Do not override/overload, use <dnatip> not (?<dnatip>.*?).
- NAT Impacted is Server (In Client-Server Model).
- NAT Impacted is Target (In Attacker-Target Model).
- Use when you see an Impacted IP address IPv4 or IPv6.

Examples

- Cisco Netflow

```
02 19 2014 06:40:29 NetFlow V9 CONN_ID=- Src=1.1.1.1 SPort=62173 InIfc=4 Dst=1.1.1.1
DPort=8080 OutIfc=3 Prot=6 ICMP_IPV4_TYPE=- ICMP_IPV4_CODE=- XLATE_SRC_ADDR_IPV4=-
XLATE_DST_ADDR_IPV4=- XLATE_SRC_PORT=- XLATE_DST_PORT=- FW_EVENT=- FW_EXT_EVENT=-
EVENT_TIME_MSEC=- IN_PERMANENT_BYTES=- DETAILS=CONN_ID=1632431052 ICMP_IPV4_TYPE=0
ICMP_IPV4_CODE=0 XLATE_SRC_ADDR_IPV4=1.1.1.1 XLATE_DST_ADDR_IPV4=1.1.1.1
XLATE_SRC_PORT=61695 XLATE_DST_PORT=8080 FW_EVENT=2 FW_EXT_EVENT=2015
EVENT_TIME_MSEC=1392835229440 IN_PERMANENT_BYTES=8807 DefaultDevice TemplateID=263
```

XLATE-DST-ADDR (Translated) indicates an impacted IP (destination in a network context) that utilizes Network Address Translation (NAT). SIP and DIP (Origin and Impacted) are indicated here with src= and dst=.

IP Address (Origin)

The IP address of the origin system. Often referred to as Source IP (in NetMon, Rule Builder and other parts of the system).

Data Type

- IP
- IPv4 in octets
- IPv6 (no support for CIDR or IPv6e)

Aliases

Use	Alias
Client Console Full Name	Host (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	IP Address (Origin)
Elasticsearch Field Name	originIp
Rule Builder Column Name	SIP
Regex Pattern	<sip>
NetMon Name	SrcIP

Field Relationships

- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain

- DIPv6
- DIPv6E
- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- Origin Port
- Impacted Domain
- Origin Login
- Impacted Account
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Everything that communicates through a network.

Use Case

Indicating the host relationship to the log message—for example, if it is an origin threat, impacted by a threat, the client, or the server.

MPE/Data Masking Manipulations

Polyfield – Origin Host

Usage Standards

- Do not override/overload, use <sip> not (?<sip>.*?).
- Origin is Client (In Client-Server Model).
- Origin is Attacker (In Attacker-Target Model).
- Use when you see an Origin IP address IPv4 or IPv6, unless it is an IPv4 address mapped to IPv6, in which case use <sipv6e>.

Examples

- Office 365

```
TS=2016-10-20T20:22:23 SESSID=8b157afd-eb80-45e4-926f-222222222
COMMAND=AnonymousLinkUsed USERTYPE=Regular USERKEY=anonymous WORKLOAD=SharePoint
RESULTCODE= OBJECT= https://www.recordflow.biz /Shared Documents/abuse_ch_copy.txt
USER=anonymous SIP=1.1.1.1 ITEMTYPE=File EVENTSOURCE=SharePoint USERAGENT=Mozilla/5.0
(Windows NT 6.3; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/54.0.2840.59
Safari/537.36 DOMAIN= FILENAME= DESTINATION= DESTINATIONFILENAME= USERSHAREDWITH=
SHARINGTYPE= MODIFIEDPROPERTIES=
```

> SIP (IPv4) in this case is Origin (source) connecting to O365 Cloud service. Client-Server are Origin-Impacted in this context.

- LogBinder

```
Jun 11 14:53:48 1.1.1.1 25000 LOGbinder EX|2.0|success|
2014-06-11T14:53:48.0000000-05:00|Undocumented Exchange mailbox operation|
name="occurred" label="Occurred" value="6/11/2014 2:53:48 PM"|name="operation"
label="Operation" value=""|name="result" label="Result" value="Succeeded"|
name="originatingserver" label="Originating Server" value=" USABLDRRECFLOW01
(14.02.0341.000)"|name="mailboxguid" label="Mailbox GUID" value="9db94f90-2222-2222-
b6c8-48200020026f"|name="mailboxowner" label="Mailbox Owner" value="n/a"|
name="mailboxownerupn" label="Mailbox Owner UPN" value="pete.store@recordflow.biz"|
name="mailboxownersid" label="Mailbox Owner SID"
value="S-1-5-21-2141518605-3280587107-2299868870-500"|name="folderid" label="Folder ID"
value="n/a"|name="foldername" label="Folder Name" value="\\Inbox"|
name="performedusername" label="Performed User Name" value="Administrator"|
name="performedusersid" label="Performed User SID"
value="S-1-5-21-222222222222-3280587107-2299868870-500"|name="performedlogontype"
label="Performed Logon Type" value="Owner"|name="clientinfo" label="Client Info"
value="Client\=OWA"|name="clientipaddress" label="Client IP Address"
value="fe80::b000:00c0:e000:f00e%00"|name="clientprocessname" label="Client Process
Name" value="n/a"|name="clientversion" label="Client Version" value="n/a"|
name="additionalinfo" label="Additional Information" value="Owner\= [Administrator];
LastAccessed\= [2013-03-06T04:41:48.0670508-05:00];"
```

IPv6 address for client. Client-Server are Origin-Impacted in this context.

Origin Hostname

The hostname from which activity originated (for example, attacker or client).

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Host (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Origin)
Elasticsearch Field Name	originHostName
Rule Builder Column Name	SName
Regex Pattern	<sname>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Impacted Hostname
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname or IP
- Impacted NAT IP
- Origin Port
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Networked equipment.

Use Case

Host context

MPE/Data Masking Manipulations

Polyfield – Origin Host

Usage Standards

- Origin is Client (In Client-Server Model).
- Origin is Attacker (In Attacker-Target Model).
- Can be used for parsing fully qualified domain names for non-world wide web context hostnames.

Examples

- Windows Event Log

```
<Event xmlns='http://Host2/win/2004/08/events/event'><System><Provider Name='NETLOGON'/
><EventID Qualifiers='0'>5805</EventID><Level></Level><Task>None</Task><Keywords></
Keywords><TimeCreated SystemTime='2014-02-06T06:03:06.000000000Z'/
><EventRecordID>156578</EventRecordID><Channel>System</Channel><Computer>
USABLDRRECFLOW01</Computer><Security/></System><EventData>The session setup from the
computer USABLDRRECFLOW02 failed to authenticate. The following error occurred:

Access is denied.</EventData></Event>
```

> Origin Host is the system trying to authenticate. <Computer> is the origin of the log message here, but also the domain controller which the origin is trying to authenticate against. Client-Server (origin-impacted) relationship applies here.

Origin Hostname or IP

The hostname or IP from which activity originated (for example, attacker or client).

Data Type

- String
- IP

Aliases

Use	Alias
Client Console Full Name	Host (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Origin)
Elasticsearch Field Name	originName, originIp
Rule Builder Column Name	SIP, SName
Regex Pattern	(<sipn>)
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- Origin Port
- Impacted Account
- IANA Protocol Number
- IANA Protocol Name

Common Applications

See IP Address (Origin) and Origin Hostname.

Use Case

See IP Address (Origin) and Origin Hostname.

MPE/Data Masking Manipulations

See IP Address (Origin) and Origin Hostname.

Usage Standards

- Use when a log can contain either an IP or a hostname in the same location.
- Must be wrapped in parenthesis to function (<sipn>).
- Do not overload or override.

Examples

- Windows Event Log
    - ```
      <Event xmlns='http://schemas.microsoft.com/win/2004/08/events/
      event'><System><Provider Name='Microsoft-Windows-Time-Service'
      Guid='{06edcfeb-0fd0-4e53-acca-a6f8bbf81bcb}'/><EventID>37</EventID><Version>0</
      Version><Level>Information</Level><Task>None</Task><Opcode>Info</
      Opcode><Keywords></Keywords><TimeCreated
      SystemTime='2016-08-02T19:21:10.521541000Z'/><EventRecordID>5823536</
      EventRecordID><Correlation/><Execution ProcessID='968' ThreadID='6580'/
      ><Channel>System</Channel><Computer> USABLDRRECFLOW01</Computer><Security
      UserID='NT AUTHORITY\LOCAL SERVICE'/></System><EventData
      Name='TMP_EVENT_TIME_SOURCE_REACHABLE'><Data Name='TimeSource'> USABLDRRECFLOW01
       (ntp.d|1.1.1.1:123->1.1.1.1:123)</Data></EventData></Event>
      ```
    - ```
      <Event xmlns='http://schemas.microsoft.com/win/2004/08/events/
      event'><System><Provider Name='Microsoft-Windows-Time-Service'
      Guid='{06edcfeb-0fd0-4e53-acca-a6f8bbf81bcb}'/><EventID>37</EventID><Version>0</
      Version><Level>Information</Level><Task>None</Task><Opcode>Info</
      Opcode><Keywords></Keywords><TimeCreated
      SystemTime='2016-09-10T02:47:47.934071900Z'/><EventRecordID>534913</
      EventRecordID><Correlation/><Execution ProcessID='1008' ThreadID='7908'/
      ><Channel>System</Channel><Computer> USABLDRRECFLOW01</Computer><Security
      UserID='NT AUTHORITY\LOCAL SERVICE'/></System><EventData
      ```

```
Name='TMP_EVENT_TIME_SOURCE_REACHABLE'><Data Name='TimeSource'>1.1.1.1,0x8
(ntp.m|0x8|1.1.1.1:123->1.1.1.1:123)</Data></EventData></Event>
```

TimeSource can either be an IP or a hostname in these examples.

Origin Interface

The network port or interface from which the activity originated (for example, attacker or client).

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Interface (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	Interface (Origin)
Elasticsearch Field Name	originInterface
Rule Builder Column Name	sinterface
Regex Pattern	<sinterface>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- IANA Protocol Number
- IANA Protocol Name

Common Applications

- Switches
- Firewalls
- Network equipment

Use Case

Troubleshooting connectivity.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Origin is Client (In Client-Server Model).
- Origin is Attacker (In Attacker-Target Model).
- If you have more than just a port number (for example, a switch ID), capture full interface name including switch ID.
- A Wireless Access Point can be an interface.

Examples

- FortiGate

```
02 25 2010 13:56:25 1.1.1.1 <LOC5:ALRT> date=2010-02-25 time=13:56:25
devname=FG322222222222222 device_id=FG2222222222 log_id=0419016384 type=ips
subtype=signature pri=alert fwver=040003 severity=critical carrier_ep="N/A"
profile="scan" src=1.1.1.1 dst=1.1.1.1 src_int="port1" dst_int="port2" policyid=48
serial=23455436 status=detected proto=6 service=2612/tcp vd="root" count=1 src_port=80
dst_port=2612 attack_id=107347979 sensor="all_default" ref="http://Host1/ids/
VID107347979" user="N/A" group="N/A" incident_serialno=128862663 msg="http_decoder:
HTTP.Request.Smuggling"
```

> Firewall log showing a signature detection with interface src (origin). In this case, the possible attacker (origin) is represented as source from the Firewall perspective.

- Squid Proxy

```
2014/05/01 10:45:29| Accepting  spoofing HTTP connections at 1.1.1.1:3128, FD 14.
```

Connection origin showing IP and corresponding interface.

- Juniper Firewall

```
08 23 2016 09:56:43 1.1.1.1 <USER:INFO> 1 2016-08-23T14:56:42.429Z USABLDRRECFLOW01
RT_FLOW - RT_FLOW_SESSION_CREATE [junos@1.1.1.1.2.40 source-address="1.1.1.1" source-
port="57101" destination-address="1.1.1.1" destination-port="443" service-name="junos-
https" nat-source-address="1.1.1.1" nat-source-port="57101" nat-destination-
address="1.1.1.1" nat-destination-port="443" src-nat-rule-type="static rule" src-nat-
rule-name="ARUBA_RAP_WLC3600_xlate" dst-nat-rule-type="N/A" dst-nat-rule-name="N/A"
protocol-id="6" policy-name="EMEA_ARUBA_GUEST_ACCESS" source-zone-name="FRONTEND_DMZ"
destination-zone-name="INTERNET" session-id-32="83048" username="N/A" roles="N/A"
packet-incoming-interface="reth5.0" application="UNKNOWN" nested-application="UNKNOWN"
encrypted="UNKNOWN"]
```

Showing inbound interface in flow.

- Cisco Router

```
10 09 2016 01:59:26 1.1.1.1 <LOC7:ERRR> Original Address=1.1.1.1 39296: Oct  9 01:59:48:
%ILPOWER-3-CONTROLLER_PORT_ERR: Controller port error, Interface Gi4/0/38: Power
Controller reports Short detected
```

Parse full interface Gi4/0/38.

Origin MAC Address

The MAC Address from which activity originated.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	MAC Address (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	MAC Address (Origin)
Elasticsearch Field Name	originMac
Rule Builder Column Name	SMAC
Regex Pattern	<smac>
NetMon Name	SrcMAC

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- IANA Protocol Number
- IANA Protocol Name

Common Applications

- Firewall
- IDS/IPS
- Vulnerability scanners

Use Case

- Differentiating hosts and interfaces.
- Detecting MAC ID cloning.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Can be in any format of MAC address:
  - MM:MM:MM:SS:SS:SS
  - MM-MM-MM-SS-SS-SS
  - MMM.MMM.SSS.SSS
  - MM MM MM SS SS SS
- Origin is Client (In Client-Server Model)
- Origin is Attacker (In Attacker-Target Model)

Examples

- FireEye Web MPS

```
02 01 2016 17:13:19 1.1.1.1 <LOC4:WARN> fenotify-609081.warning: CEF:0|FireEye|MPS|
1.1.1.1875|IM|infection-match|1|rt=Feb 01 2016 23:13:10 UTC src=1.1.1.1 cn3Label=cncPort
cn3=80 cn2Label=sid cn2=84575103 shost= https://www.recordflow.biz proto=tcp spt=51997
dst=1.1.1.1 cs5Label=cncHost cs5=1.1.1.1 dvchost= USABLDRRECFLOW01dvc=1.1.1.1
smac=00:00:00:00:00:00 cn1Label=vlan cn1=0 dpt=80 externalId=609081 cs4Label=link
cs4=THING dmac=00:00:00:00:00:00 cs1Label=sname cs1=Exploit.Kit.AnglerDIPv4
```

> Dmac= in this log is the attacker MAC Address (origin).

- Brocade Switch

```
03 01 2017 02:08:41 1.1.1.1 <LOC6:NOTE> Mar  1 02:08:38 USABLDRRECFLOW01dataplane[2287]:
fw rule INTERNAL-IN:10000 block udp(17) src= USABLDRRECFLOW01/0:00:00:00:00:00/
IPV6Address dst=/00:00:00:00/0000::0:0(547) len=159 hoplimit=1 len=119
```

Src= with hostname followed by origin MAC Address. Network traffic shown src->dst will be origin->impacted.

- Windows Event Log – DHCP Ops

```
<Event xmlns='http://Host2/win/2004/08/events/event'><System><Provider Name='Microsoft-
Windows-DHCP-Server' Guid='{6d64f02c-a125-4dac-9a01-f0555b41ca84}'/><EventID>20097</
EventID><Version>0</Version><Level>Information</Level><Task>None</Task><Opcode>Info</
Opcode><Keywords></Keywords><TimeCreated SystemTime='2014-10-07T00:13:02.116745100Z'/
><EventRecordID>445336</EventRecordID><Correlation/><Execution ProcessID='1320'
ThreadID='2952'/><Channel>Microsoft-Windows-Dhcp-Server/FilterNotifications</
Channel><Computer> USABLDRRECFLOW01</Computer><Security UserID='NT AUTHORITY\NETWORK
SERVICE'/></System><EventData>DHCP Services were denied to machine with hardware address
00-00-00-00-00-00, hardware type 1 and FQDN/Hostname USABLDRRECFLOW01because it did not
match any entry in the Allow List.</EventData></Event>
```

Origin MAC Address with dashes instead of colons.

Origin NAT IP

The Network Address Translated IP from which activity originated (for example, attacker or client).

Data Type

IP

Aliases

Use	Alias
Client Console Full Name	NAT IP Address (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	NAT IP Address (Origin)
Elasticsearch Field Name	originNatIp
Rule Builder Column Name	SNATIP
Regex Pattern	<snatip>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Impacted Hostname
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname or IP
- Impacted NAT IP
- Origin Port
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Network equipment

Use Case

Internal host context

MPE/Data Masking Manipulations

Polyfield – Origin Host

Usage Standards

- Do not override/overload, use <snatip> not (?<snatip>.*?).
- NAT Origin is Client (In Client-Server Model).
- NAT Origin is Attacker (In Attacker-Target Model).
- Use when you see an Origin IP address IPv4 or IPv6.

Examples

- Cisco Netflow

```
02 19 2014 06:40:29 NetFlow V9 CONN_ID=- Src=1.1.1.1 SPort=62173 InIfc=4 Dst=1.1.1.1
DPort=8080 OutIfc=3 Prot=6 ICMP_IPV4_TYPE=- ICMP_IPV4_CODE=- XLATE_SRC_ADDR_IPV4=-
XLATE_DST_ADDR_IPV4=- XLATE_SRC_PORT=- XLATE_DST_PORT=- FW_EVENT=- FW_EXT_EVENT=-
EVENT_TIME_MSEC=- IN_PERMANENT_BYTES=- DETAILS=CONN_ID=1632431052 ICMP_IPV4_TYPE=0
ICMP_IPV4_CODE=0 XLATE_SRC_ADDR_IPV4=1.1.1.1 XLATE_DST_ADDR_IPV4=1.1.1.1
 XLATE_SRC_PORT=61695 XLATE_DST_PORT=8080 FW_EVENT=2 FW_EXT_EVENT=2015
EVENT_TIME_MSEC=1392835229440 IN_PERMANENT_BYTES=8807 DefaultDevice TemplateID=263
```

> XLATE-SRC-ADDR indicates an origin IP (source in a network context) utilizing Network Address Translation (NAT). SIP and DIP (Origin and Impacted) are indicated here with src= and dst=.

Serial Number [7.2]

The hardware or software serial number in a log message. Should be a permanent, unique identifier of what it is identifying.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String (128 characters maximum)

Aliases

Use	Alias
Client Console Full Name	Serial Number
Client Console Short Name	Not applicable
Web Console Tab/Name	Serial Number
Elasticsearch Field Name	serialNumber
Rule Builder Column Name	SerialNumber
Regex Pattern	<serialnumber>
NetMon Name	Not applicable

Field Relationships

- This field was previously an overload of object and subject.
- Session is often used for what are called serial numbers, but are closer to session identifiers.

Common Applications

- Palo Alto

- Juniper
- F5
- Asset management systems

Use Case

Uniquely identify systems.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Serial Number is only used for data that uniquely identifies an object, device or application. It is not meant to be used for defining a "session" or "record id."
- Only overload this field with GUID when S/N not present when the GUID is permanent.

Examples

Correct Examples

- Avaya Secure Access Link Remote Access Log

```
Jun 21 16:29:30 Host2ldomain Host1 xgEnterpriseProxy: Device registered with server
https://Host4/eMessage: model: SessionMgr, serial number: (000)222-2222
```

> Serial Number describes the device being registered to the server.

- Bluecat Adonis

```
03 19 2013 14:34:17 1.1.1.1 <LOC1:INFO> Mar 19 14:34:17 USABLDRRECFLOW01named[4476]:
info: zone 10.in-addr.arpa/IN/Internal: transferred serial 324442789: TSIG 'view13530'
```

> Serial used in DNS transaction.

Ambiguous Examples

- FortiGate

```
03 27 2016 12:24:47 1.1.1.1 <LOC5:ALRT> date=2016-03-27 time=12:24:47 devname=SLAVE
devid=FG222222222222222222 logid=0419016384 type=utm subtype=ips eventtype=signature
level=alert vd="Front_End" severity=high srcip=1.1.1.1 dstip=1.1.1.1 srcintf="port14"
dstintf="port13" policyid=1897 sessionid=3487142146 action=detected proto=6
service=HTTPS attack="OpenSSL.ChangeCipherSpec.Injection" srcport=50077 dstport=443
```

```
hostname="recordflow.biz" direction=outgoing attackid=38738 profile="All-All-All" ref="h
ttp://www.fortinet.com/ids/VID38738" incidentserialno=981770026 msg="applications3:
OpenSSL.ChangeCipherSpec.Injection," crscore=30 crlevel=high
```

> Incidentserialno correlates logs describing a single incident, and is closer to a session or record ID than a serial number.

- Cisco Telepresence VCS

```
04 26 2016 18:07:35 1.1.1.1 <USER:NOTE> 2016-04-26T18:07:36-04:00 radvcsx tvcs:
Event="Search Completed" Reason="Not Found" Service="H323" Src-alias-type="H323" Src-
alias="pima_373@Host5" Dst-alias-type="E164" Dst-alias="93516#9#935" Call-serial-
number="e2c39d22-cd9f-222c-a2ea-7b57a39239fc" Tag="f420cf74-2222-45d6-989a-76e32d94525a"
Detail="found:false, searchtype:LRQ" Level="1" UTCTime="2016-04-26 22:07:36,027"
```

> Call-Serial-Number is closer to a session in this context.

SIPv4

Constituent element of <sip> for only IPv4 parsing (not generally used). Completely redundant to SIP.

Data Type

IP

Aliases

Use	Alias
Client Console Full Name	Host (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Origin)
Elasticsearch Field Name	originIpV4
Rule Builder Column Name	SIP
Regex Pattern	<sipv4>
NetMon Name	Not applicable

Field Relationships

- Nested element of <sip> default regex
- Can not be used with <sipv6>

Common Applications

IPv4 only network equipment.

Use Case

For more information, see IP Address (Origin).

MPE/Data Masking Manipulations

Polyfield – Origin Host

Usage Standards

- This field is rarely used because it is redundant to <sip>.
- If you are 100% certain an IPv4 address will always appear.
- Only use if you need an extremely minute performance improvement.

Examples

Not applicable.

SIPv6

Constituent element of <sip> for only IPv6 parsing (not generally used).

Data Type

IP

Aliases

Use	Alias
Client Console Full Name	Host (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Origin)
Elasticsearch Field Name	originIpV6
Rule Builder Column Name	SIP
Regex Pattern	<sipv6>
NetMon Name	Not applicable

Field Relationships

- Nested element of <sip> default regex
- Can not be used with <sipv4>

Common Applications

IPv6 only network equipment.

Use Case

For more information, see IP Address (Origin).

MPE/Data Masking Manipulations

Polyfield – Origin Host

Usage Standards

- This is rarely used.
- Is redundant to <sip>.
- If you are 100% certain an IPv6 address will always appear.
- Use if you need an extremely minute performance improvement.

Examples

- Trend Micro Deep Security

```
11 19 2014 08:21:12 10.100.6.64 <LOC0:INFO> Nov 19 03:25:07 USABLDRRECFLOW01 dsa_mpnp:
REASON=IPv6_Packet HOSTID=230078 ACT=Deny IN=0C:0B:05:07:B0:05 OUT=
MAC=00:00:00:00:00:00:00:BE:00:00:00:0D:00:0d SRC=fe80:0:0:0:0cd0:000f:bd2f:000b
 DST=ff01:0:0:0:0:0:0:1 LEN=86 PROTO=ICMPv6 SPT=0 DPT=0 CNT=1
```

> SRC= shows origin IPv6 Address.

SIPv6E

The IPv4 IP address mapped to IPv6e from which activity originated (for example, attacker or client).

Data Type

IP

Aliases

Use	Alias
Client Console Full Name	Host (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	Host (Origin)
Elasticsearch Field Name	originIpV6
Rule Builder Column Name	SIP
Regex Pattern	<sipv6e>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Impacted Hostname
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname or IP
- Impacted NAT IP
- Origin Port
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Networked equipment.

Use Case

Host context

MPE/Data Masking Manipulations

Polyfield – Origin Host

Usage Standards

- Do not override/overload, use <sipv6e> not (?<sipv6e>.*?).
- Origin is Client (In Client-Server Model).
- Origin is Attacker (In Attacker-Target Model).
- Use when you see an Origin IPv4 address mapped to IPv6.

Examples

- Townsend Alliance LogAgent

```
11 02 2015 22:10:02 1.1.1.1 <ALRT:INFO> Nov  2 22:09:39 USABLDRRECFLOW01QAUDJRN:[PW@0
event="PW-Invalid user or password" event_type="Q-Signon failed profile disabled"
actual_type="PW-Q" user_profile="PSTORE" device="" jrn_seq="6849716"
timestamp="20151102220939315000" job_name="QZSOSIGN" user_name="QUSER"
job_number="535772" eff_user="QUSER" ip_addr="::ffff:1.1.1.1" port="52584"]
```

> ::ffff:1.1.1.1 is an IPv4 IP mapped to IPv6. Traditional <sip> and <dip> IP parsers do not work with this type of IP.

Identity Tab

The identity tab contains metadata fields related to the user associated with the action or object in the log.

The following fields are on the Identity tab:

- Account > User (Impacted)
- Group
- Login > User (Origin)
- Recipient
- Sender

This tab contains one polyfield: User (Impacted).

This tab contains four Identity Analytics fields: User Identity (Origin), User Identity (Impacted), Sender Identity, and Recipient Identity.

Identity Analytics fields aggregate multiple identifiers for a user/email into a single unique ID. Each Identity Field is mapped to the corresponding MPE metadata field described in this section. These fields are not available for parsing. For more information about Identity Analytics fields, see Identity-Derived Data.

Account > User (Impacted)

The user or system account impacted by activity reported in the log.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	User (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	User (Impacted)
Elasticsearch Field Name	account
Rule Builder Column Name	Account
Regex Pattern	<account>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Any applications, systems or devices that utilize accounts.

Use Case

Correlating or monitoring user activity.

MPE/Data Masking Manipulations

Mapped to User Identity (Impacted)

Usage Standards

- Use to indicate the user or account that is being altered or logged off a system by another user or system account.
- Use for User Accounts and System Accounts.

Examples

- Windows Event Log


```
<Event xmlns='http://Host2/win/2004/08/events/event'><System><Provider Name='Microsoft-
Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}'/
><EventID>4738</EventID><Version>0</Version><Level>Information</Level><Task>User Account
Management</Task><Opcode>Info</Opcode><Keywords>Audit Success</Keywords><TimeCreated
SystemTime='2014-02-26T13:18:11.277015700Z'/><EventRecordID>1635656743</
EventRecordID><Correlation/><Execution ProcessID='524' ThreadID='4900'/
><Channel>Security</Channel><Computer> USABLDRRECFLOW01Computer><Security/></
System><EventData>A user account was changed.
```


Subject:

```
Security ID:        safaware\pete.store
Account Name:       pete.store
Account Domain:             safaware
Logon ID:           0x7b1adb067
```

Target Account:

```
Security ID:        S-1-5-21-2222222-2222222222-2222222222-90119
Account Name:       LHR-Reception
Account Domain:             safaware
```

```
Changed Attributes:
        SAM Account Name:   -
        Display Name:       -
        User Principal Name:        -
        Home Directory:             -
        Home Drive:         -
        Script Path:        -
        Profile Path:       -
        User Workstations:  -
        Password Last Set:  -
        Account Expires:            -
        Primary Group ID:   -
        AllowedToDelegateTo:        -
        Old UAC Value:              0x15
        New UAC Value:              0x211
        User Account Control:
                'Password Not Required' - Disabled
                'Don't Expire Password' - Enabled
        User Parameters:    -
        SID History:        -
        Logon Hours:        -
Additional Information:
        Privileges:         -</EventData></Event>
```

Target in Windows indicates Impacted. In this log, the Target Account (Impacted) is being modified by Subject Account (Origin).

Group

The user group or role impacted by activity reported in the log. Do not use for entity group (zone or domain).

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Group
Client Console Short Name	Not applicable
Web Console Tab/Name	Group
Elasticsearch Field Name	group
Rule Builder Column Name	Group
Regex Pattern	<group>
NetMon Name	Not applicable

Field Relationships

- Login
- Account
- Domain
- Session
- SessionType
- Policy

Common Applications

- AD group
- Linux user group

- Security role

Use Case

- Capturing active directory organizational unit.
- Capturing certificate organizational units.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

- Not Zone (internet, network, security).
- Only to capture explicitly called out (user) group, organizational units, and roles.

Examples

- Cylance

```
08 16 2016 22:42:18 1.1.1.1 <USER:NOTE> 250 <44>1 2016-08-17T04:42:20.0816805Z
sysloghost CylancePROTECT - - - Event Type: AuditLog, Event Name: ZoneAddDevice,
Message: Zone: Corporate; Devices: USABLDRRECFLOW01, , User: Dave Foss
(pete.store@recordflow.biz) pete.store@recordflow.biz)
```

> Corporate Zone is parsed here.

- AWS

```
TS=2015-07-03T07:15:21Z ACCT=22222222222 RSRC=sg-22222222222 ARN=
USABLDRRECFLOW01:security-group/sg- USABLDRRECFLOW01CREATETS= STS=ResourceDiscovered
REG=us-west-2 RSRCTYP=AWS::EC2::SecurityGroup DETALS=ownerid=9052222962 groupname=launch
-wizard-1 groupid=gg22222 description=launch-wizard-1 created
2015-07-03T00:07:57.767-07:00 vpcid=vpc-22222226
```

> Groupname= parses into Group. Is explicit as a group.

- Salesforce

```
EVT_TYP=RestApi TS=2015-07-13T22:37:51Z REQ_ID=3z1tWodgfdgdH5TjAgF- ORG_ID=00D0000000000
1 U_N=pete.store@recordflow.biz.isvdev01 RUN_T=77 CPU_T=19 CLNT_IP=1.1.1.1 URI=/
services/data/v33.0/query
```

> Organization ID parsed (specific to LogRhythm in this example).

Login > User (Origin)

The host IP that was affected by the activity (for example, target or server). Destination IP in IPv4 or IPv6 format.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	User (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	User (Origin)
Elasticsearch Field Name	login
Rule Builder Column Name	Login
Regex Pattern	<login>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Impacted Account

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Any applications, systems or devices that utilize accounts.

Use Case

Correlating or monitoring user activity.

MPE/Data Masking Manipulations

Mapped to User Identity (Origin)

Usage Standards

- Use to indicate the user or system account that is performing altering another account or logging in to a system.
- Use for User Accounts and System Accounts.

Examples

- Windows Event Log

```
<Event xmlns='http://Host2/win/2004/08/events/event'><System><Provider Name='Microsoft-
Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}'/
><EventID>4738</EventID><Version>0</Version><Level>Information</Level><Task>User Account
Management</Task><Opcode>Info</Opcode><Keywords>Audit Success</Keywords><TimeCreated
SystemTime='2014-02-26T13:18:11.277015700Z'/><EventRecordID>1635656743</
EventRecordID><Correlation/><Execution ProcessID='524' ThreadID='4900'/
><Channel>Security</Channel><Computer> USABLDRRECFLOW01Computer><Security/></
System><EventData>A user account was changed.

Subject:

       Security ID:        Safaware\pete.store

       Account Name:       pete.store

       Account Domain:            safaware

       Logon ID:           0x7b1adb067

Target Account:

       Security ID:        S-1-5-21-2222222-22222222-22222-90119

       Account Name:       dave.store

       Account Domain:            safaware

Changed Attributes:
```

```
        SAM Account Name:   -
        Display Name:       -
        User Principal Name:       -
        Home Directory:            -
        Home Drive:         -
        Script Path:        -
        Profile Path:       -
        User Workstations:  -
        Password Last Set:  -
        Account Expires:           -
        Primary Group ID:   -
        AllowedToDelegateTo:       -
        Old UAC Value:            0x15
        New UAC Value:            0x211
        User Account Control:
               'Password Not Required' - Disabled
               'Don't Expire Password' - Enabled
        User Parameters:    -
        SID History:        -
        Logon Hours:        -
Additional Information:
        Privileges:         -</EventData></Event>
```

Subject in Windows indicates Origin. In this log, the Subject Account (Origin) is modifying the Target Account (Impacted).

- Cisco Clean Access Appliance

```
03 28 2010 14:55:50 1.1.1.1 <USER:INFO> Perfigo: Authentication:[00:00:00:00:00:00 ##
1.1.1.1] escribne - Successfully logged in, Provider: conncoll, L2 MAC address:
00:00:00:00:00:00, Role: Students, OS: Macintosh OSX
```

User logon event. Listed user is the client (origin) connecting to a server (impacted) (client-server).

Recipient

The recipient of an email or called to number for a VoIP log.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Recipient
Client Console Short Name	Not applicable
Web Console Tab/Name	Recipient
Elasticsearch Field Name	recipient
Rule Builder Column Name	Recipient
Regex Pattern	<recipient>
NetMon Name	Not applicable

Field Relationships

- Sender
- Subject
- Session
- Session Type

Common Applications

- Email logs
- VoIP logs
- Instant messaging services

Use Case

Tracking malware infection vector.

MPE/Data Masking Manipulations

Mapped to Recipient Identity.

Usage Standards

- Recipient shall not be used for identifying the direction of network traffic or network zones.
- Only used for destination email, destination caller, chat, instant messaging, or other communication mediums, such as
  - AOL Instant Messenger
  - IRC
  - Lync
  - Skype
  - Google Hangouts
  - Fax

Examples

- ColdFusion Mailsent Log

```
"Information","scheduler-2","12/28/11","09:14:33",,"Mail: 'Web site submission from Pete
Store' From:'NoReply@recordflow.biz' To:'mdaveman@recordlow.com' was successfully sent
using smtp.recordflow.biz"
```

> To email parsed appropriately.

- Cisco Telepresence Video Communications Server

```
04 26 2016 16:40:14 1.1.1.1 <USER:NOTE> 2016-04-26T16:40:14-04:00 radvcsx tvcs:
Event="Call Attempted" Service="SIP" Src-ip="1.1.1.1" Src-port="1196" Src-alias-
type="SIP" Src-alias="sip:pete.store@Host5" Dst-alias-type="SIP" Dst-
alias="sip:dpackl@Host5" Call-serial-number="d415c222-fd22-47fd-8d0a-222b1a351460"
Tag="02e3b418-f67b-408b-22b2-adafea222e32" Protocol="TLS" Auth="NO" Level="1"
UTCTime="2016-04-26 20:40:14,467"
```

> Dst-Alias in this case a VoIP call destination.

- Cisco Unified Comm Mgr (Call Mgr)

```
05 22 2012 15:05:49 1.1.1.1 <LOC7:WARN> 750: May 22 2012 20:05:49.41 UTC :
%UC_CALLMANAGER-4-MaliciousCall: %[Called Party Number=2755][Called Device Name=
USABLDRRECFLOW01][Called Display Name=Jason Riggins][Calling Party Number=2378][Calling
```

```
Device Name= USABLDRRECFLOW01][Calling Display Name=Dave Store Test][App ID=Cisco
CallManager][Cluster ID=StandAloneCluster][Node ID=KaM-CCM2-SubT]: A malicious call has
been identified
```

Another VoIP call destination.

Sender

The sender of an email or the caller number for a VoIP log. Must relate to a specific user, or unique address in the case of a phone call or email.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Sender
Client Console Short Name	Not applicable
Web Console Tab/Name	Sender
Elasticsearch Field Name	sender
Rule Builder Column Name	Sender
Regex Pattern	<sender>
NetMon Name	Not applicable

Field Relationships

- Recipient
- Subject
- Session
- Session Type

Common Applications

- Email logs
- VoIP logs

Use Case

- Identify spam traffic by looking at top senders of email.
- Track ransomware back to source/spread pattern.

MPE/Data Masking Manipulations

Mapped to Sender Identity.

Usage Standards

- Sender shall not be used for identifying the direction of network traffic or network zones.
- Only used for origin email, origin caller, chat, instant messaging, or other communication mediums, such as
    - AOL Instant Messenger
    - IRC
    - Lync
    - Skype
    - Google Hangouts
    - Fax

Examples

- ColdFusion Mailsent Log

```
"Information","scheduler-2","12/28/11","09:14:33",,"Mail: 'Web site submission from
Donna Hirt' From:'NoReply@recordflow.biz' To:'mcoffman@sagepointadvisor.com' was
successfully sent using mta23.colo.lan"
```

From email parsed appropriately.

- Cisco Telepresence Video Communications Server

```
04 26 2016 16:40:14 1.1.1.1 <USER:NOTE> 2016-04-26T16:40:14-04:00 radvcsx tvcs:
Event="Call Attempted" Service="SIP" Src-ip="1.1.1.1" Src-port="1196" Src-alias-
type="SIP" Src-alias="sip:pete_store@Host5" Dst-alias-type="SIP" Dst-
alias="sip:dpack@Host5" Call-serial-number="d415c736-fd67-47fd-8d0a-892b1a351460"
Tag="02e3b418-f67b-408b-92b2-adafea551e32" Protocol="TLS" Auth="NO" Level="1"
UTCTime="2016-04-26 20:40:14,467"
```

Src-Alias in this case a VoIP call origin.

- Cisco Unified Comm Mgr (Call Mgr)

```
05 22 2012 15:05:49 1.1.1.1 <LOC7:WARN> 750: May 22 2012 20:05:49.41 UTC :
%UC_CALLMANAGER-4-MaliciousCall: %[Called Party Number=2755][Called Device
Name=SEP002414B3815B][Called Display Name=Jason Riggins][Calling Party Number=2378]
```

```
[Calling Device Name=recflow00001][Calling Display Name=recflow Test][App ID=Cisco
CallManager][Cluster ID=StandAloneCluster][Node ID=rec-flow-001]: A malicious call has
been identified
```

Another VoIP call origin.

Location Tab

The Location tab contains fields that attempt to identify a physical or logical location for the object referenced in a log.

The Location fields are either polyfields or enrichment fields added after parsing:

- Entity (Origin)
- Entity (Impacted)
- Zone (Origin)
- Zone (Impacted)
- Location (Origin)
- Location (Impacted)
- Country (Origin)

Log Tab

The Log tab contains metadata that describe the log source rather than the log itself.

Most of these fields are generated by the log source type or agent configuration. They are not parsed into the schema from the raw log:

- Log count
- Log source entity
- Log source type
- Log source host
- Log source
- Log sequence number (from Agent)

Network Tab

The Network tab contains fields that relate to the networks associated with the origin and impacted host.

The following fields are on the Network tab:

- Domain [7.2]
- Impacted NAT Port
- Impacted Port
- Origin NAT Port
- Origin Port

Several of the network fields are polyfields or are auto-calculated based on the entity structure:

- Network (Origin)
- Network (Impacted)
- Protocol

Domain (Origin) [7.4]

The Windows or DNS domain name from which the activity reported in the log originates.

> This field is not available in LogRhythm versions earlier than 7.4.0.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Domain (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	Domain (Origin)
Elasticsearch Field Name	domain
Rule Builder Column Name	Domain
Regex Pattern	<domainorigin>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP

- DIP
- DIPv4
- DIPv6
- DIPv6E
- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP

- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Origin Login
- Impacted Account
- IANA Protocol Number
- IANA Protocol Name

Common Applications

- WebProxy
- Network monitoring
- Active Directory
- SSO

Use Case

Correlating user activity across domains.

MPE/Data Masking Manipulations

Not applicable.

Domain [7.2]

The Windows or DNS domain name referenced or impacted by activity reported in the log.

> This field is not available in LogRhythm versions earlier than 7.2.1.

Data Type

String

Aliases

Use	Alias
Client Console Full Name	Domain (Origin) As of LogRhythm 7.4, the Client Console display reverses the Domain (Origin) and Domain (Impacted) in search results. Grid columns with the Domain (Impacted) heading show Domain (Origin) results and vice versa.
Client Console Short Name	Not applicable
Web Console Tab/Name	Domain (Impacted)
Elasticsearch Field Name	domain
Rule Builder Column Name	Domain
Regex Pattern	<domain> or <domainimpacted>
NetMon Name	Not applicable

Field Relationships

- SIP
- Origin Port

- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Origin Login
- Impacted Account
- IANA Protocol Number
- IANA Protocol Name

Common Applications

- WebpProxy
- Network monitoring
- Active Directory
- SSO

Use Case

Correlating user activity across domains.

MPE/Data Masking Manipulations

Not applicable.

Usage Standards

Used for capturing an Active Directory Domain name.

Examples

- Windows Event Log


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider
Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-
a5ba-3e3b0328c30d}'/><EventID>4742</EventID><Version>0</Version><Level>Information</
Level><Task>Computer Account Management</Task><Opcode>Info</Opcode><Keywords>Audit
Success</Keywords><TimeCreated SystemTime='2016-02-26T03:09:41.988899400Z'/
><EventRecordID>2283625151</EventRecordID><Correlation/><Execution ProcessID='520'
ThreadID='1140'/><Channel>Security</Channel><Computer> USABLDRRECFLOW01</
Computer><Security/></System><EventData><Data Name='ComputerAccountChange'>-</Data><Data
Name='TargetUserName'>USLT0752CROBB$</Data><Data Name='TargetDomainName'>SAFAWARE</
Data><Data Name='TargetSid'>SAFAWARE\ USABLDRRECFLOW01$</Data><Data
Name='SubjectUserSid'>SAFAWARE\pete.store</Data><Data
```

```
Name='SubjectUserName'>pete.store</Data><Data Name='SubjectDomainName'>SAFAWARE</
Data><Data Name='SubjectLogonId'>0x14af66a2b</Data><Data Name='PrivilegeList'>-</
Data><Data Name='SamAccountName'>-</Data><Data Name='DisplayName'>-</Data><Data
Name='UserPrincipalName'>-</Data><Data Name='HomeDirectory'>-</Data><Data
Name='HomePath'>-</Data><Data Name='ScriptPath'>-</Data><Data Name='ProfilePath'>-</
Data><Data Name='UserWorkstations'>-</Data><Data Name='PasswordLastSet'>-</Data><Data
Name='AccountExpires'>-</Data><Data Name='PrimaryGroupId'>-</Data><Data
Name='AllowedToDelegateTo'>-</Data><Data Name='OldUacValue'>0x80</Data><Data
Name='NewUacValue'>0x81</Data><Data Name='UserAccountControl'>

              %%2080</Data><Data Name='UserParameters'>-</Data><Data Name='SidHistory'>-
</Data><Data Name='LogonHours'>-</Data><Data Name='DnsHostName'>-</Data><Data
Name='ServicePrincipalNames'>-</Data></EventData></Event>
```


> TargetDomainName is the Domain of the impacted user in this Account Management event. In Windows Event Logging, Subject refers to Origin and Target refers to Impacted.

Impacted NAT Port

The Network Address Translated (NAT) port to which activity is targeted—for example, server or target port.

Data Type

Integer

Aliases

Use	Alias
Client Console Full Name	TCP/UDP Port (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	TCP/UDP Port (Impacted)
Elasticsearch Field Name	impactedNatPort
Rule Builder Column Name	DNATPort
Regex Pattern	<snatport>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin Port
- Origin NAT Port
- Impacted Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Any network connected application or device.

Use Case

Host and application contexts.

MPE/Data Masking Manipulations

Used to help in determining Application.

Usage Standards

- Use to indicate the Network Address Translated (NAT) impacted port number associated with a server or targeted host.
- Origin is Server (In Client-Server Model).
- Target is Impacted (In Attacker-Target Model).

Examples

- Cisco Netflow

```
02 19 2014 06:40:29 NetFlow V9 CONN_ID=- Src=1.1.1.1 SPort=62173 InIfc=4 Dst=1.1.1.1
DPort=8080 OutIfc=3 Prot=6 ICMP_IPV4_TYPE=- ICMP_IPV4_CODE=- XLATE_SRC_ADDR_IPV4=-
XLATE_DST_ADDR_IPV4=- XLATE_SRC_PORT=- XLATE_DST_PORT=- FW_EVENT=- FW_EXT_EVENT=-
EVENT_TIME_MSEC=- IN_PERMANENT_BYTES=- DETAILS=CONN_ID=1632431052 ICMP_IPV4_TYPE=0
ICMP_IPV4_CODE=0 XLATE_SRC_ADDR_IPV4=1.1.1.1 XLATE_DST_ADDR_IPV4=1.1.1.1
XLATE_SRC_PORT=61695 XLATE_DST_PORT=8080 FW_EVENT=2 FW_EXT_EVENT=2015
EVENT_TIME_MSEC=1392835229440 IN_PERMANENT_BYTES=8807 DefaultDevice TemplateID=263
```

> XLATE_DST_PORT shows the translation IPs destination (impacted) port. In a network flow context, destination and impacted are synonymous.

Impacted Port

The port to which activity is targeted (for example, server or target port).

Data Type

Integer

Aliases

Use	Alias
Client Console Full Name	TCP/UDP Port (Impacted)
Client Console Short Name	Not applicable
Web Console Tab/Name	TCP/UDP Port (Impacted)
Elasticsearch Field Name	impactedPort
Rule Builder Column Name	DPort
Regex Pattern	<dport>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin Port
- Origin NAT Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Any network connected application or device.

Use Case

Host and application contexts.

MPE/Data Masking Manipulations

Used to help in determining Application.

Usage Standards

- Use to indicate the impacted port number associated with a server or targeted host.
- Origin is Server (In Client-Server Model).
- Target is Impacted (In Attacker-Target Model).

Examples

- FireEye Web MPS

```
02 01 2016 17:13:19 1.1.1.1 <LOC4:WARN> fenotify-609081.warning: CEF:0|FireEye|MPS|
1.1.1.1875|IM|infection-match|1|rt=Feb 01 2016 23:13:10 UTC src=1.1.1.1 cn3Label=cncPort
cn3=80 cn2Label=sid cn2=84575103 shost= USABLDRRECFLOW01proto=tcp spt=51997 dst=1.1.1.1
cs5Label=cncHost cs5=1.1.1.1 dvchost= USABLDRRECFLOW01dvc=1.1.1.1 smac=00:00:00:00:00:00
cn1Label=vlan cn1=0 dpt=80 externalId=609081 cs4Label=link cs4URL act=blocked
cs6Label=channel cs6=GET Stuff dmac=00:00:00:00:00:00 cs1Label=sname
cs1=Exploit.Kit.AnglerDIPv4
```

Spt= in this case is the impacted (target) port in the attacker-target paradigm.

- Cisco Next Generation Firewall

```
CISCONGFW EVENT Ev_Id=610 Ev_Severity=6 Ev_TypeId=HTTP_COMPLETE Ev_SrcId=32
Ev_RecvTime=2/24/2013 10:04:34 PM Ev_MetaData=0 Smx_Config_Version=2 Identity_Source=0
Smx_Policy_Id=0 Flow_ConnId=456 Smx_Egress_Interface_Id=0 Smx_Ingress_Interface_Id=0
Avc_App_Id=300003 Ev_GenTime=2/24/2013 10:04:09 PM Flow_Protocol=6 Flow_SrcIp=1.1.1.1
Flow_DstIp=1.1.1.1 Flow_SrcPort=60221 Flow_DstPort=80 Ev_Producer_Id=5
Flow_Transaction_Id=0 Url=recordflow.biz Flow_DstHostName=recordflow.biz Smx_Policy_Id=0
Flow_Bytes_Sent=391 Http_Response_Status=302 Flow_Bytes_Received=647
```

Impacted port (destination in a network traffic flow context).

- Cisco ISE

```
02 10 2014 13:54:24 1.1.1.1 <LOC6:NOTE> Feb 10 13:54:43 USABLDRRECFLOW01
CISE_Failed_Attempts 0000217969 2 0 2014-02-10 13:54:43.264 +02:00 0008145644 5413
NOTICE Failed-Attempt: RADIUS Accounting-Request dropped, ConfigVersionId=143, Device IP
Address=1.1.1.1, Device Port=1646, DestinationIPAddress=1.1.1.1, DestinationPort=1646,
Protocol=Radius, NetworkDeviceName=Switch_ USABLDRRECFLOW01, NAS-IP-Address=1.1.1.1,
NAS-Port=50023, Service-Type=Framed, Acct-Status-Type=Start, Acct-Delay-Time=20, Acct-
Session-Id=000022222, Acct-Authentic=Local, NAS-Port-Type=Ethernet, NAS-Port-
Id=GigabitEthernet0/23, cisco-av-pair=connect-progress=Call Up, AcsSessionID=
USABLDRRECFLOW01/151856948/212124, FailureReason=11038 RADIUS Accounting-Request header
contains invalid Authenticator field, Step=11004, Step=11017, Step=11038, Step=5413,
NetworkDeviceGroups=Device Type#All Device Types#Switch,
NetworkDeviceGroups=Location#All Locations#HQ, NetworkDeviceGroups=Unit#All
Units#Networking, NetworkDeviceGroups=ACS Group#All ACS Groups, ACS Group=ACS Group#All
ACS Groups,
```

Destination Port (Impacted) is the server port being authenticated against (Client-Server relationship).

Origin NAT Port

The Network Address Translated (NAT) port from which activity originated (for example, client or attacker port).

Data Type

Integer

Aliases

Use	Alias
Client Console Full Name	TCP/UDP Port (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	TCP/UDP Port (Origin)
Elasticsearch Field Name	originNatPort
Rule Builder Column Name	SNATPort
Regex Pattern	<snatport>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin Port
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- Impacted Account
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Any network connected application or device.

Use Case

Host and application contexts.

MPE/Data Masking Manipulations

Used to help in determining Application.

Usage Standards

- Use to indicate the Network Address Translated (NAT) origin port number associated with a client or attacker host where Origin is Client (In Client-Server Model).
- Origin is Attacker (In Attacker-Target Model).

Examples

- Cisco Netflow

```
02 19 2014 06:40:29 NetFlow V9 CONN_ID=- Src=1.1.1.1 SPort=62173 InIfc=4 Dst=1.1.1.1
DPort=8080 OutIfc=3 Prot=6 ICMP_IPV4_TYPE=- ICMP_IPV4_CODE=- XLATE_SRC_ADDR_IPV4=-
XLATE_DST_ADDR_IPV4=- XLATE_SRC_PORT=- XLATE_DST_PORT=- FW_EVENT=- FW_EXT_EVENT=-
EVENT_TIME_MSEC=- IN_PERMANENT_BYTES=- DETAILS=CONN_ID=1632431052 ICMP_IPV4_TYPE=0
ICMP_IPV4_CODE=0 XLATE_SRC_ADDR_IPV4=1.1.1.1 XLATE_DST_ADDR_IPV4=1.1.1.1
XLATE_SRC_PORT=61695 XLATE_DST_PORT=8080 FW_EVENT=2 FW_EXT_EVENT=2015
EVENT_TIME_MSEC=1392835229440 IN_PERMANENT_BYTES=8807 DefaultDevice TemplateID=263
```

> XLATE_SRC_PORT shows the translation IP's source (origin) port. In a network flow context, origin and source are synonymous.

Origin Port

The port from which activity originated (for example, client or attacker port).

Data Type

Integer

Aliases

Use	Alias
Client Console Full Name	TCP/UDP Port (Origin)
Client Console Short Name	Not applicable
Web Console Tab/Name	TCP/UDP Port (Origin)
Elasticsearch Field Name	originPort
Rule Builder Column Name	SPort
Regex Pattern	<sport>
NetMon Name	Not applicable

Field Relationships

- SIP
- SIPv4
- SIPv6
- SIPv6E
- Origin Hostname
- Origin Hostname or IP
- Origin NAT IP
- DIP
- DIPv4
- DIPv6
- DIPv6E
- Origin NAT Port
- Impacted Port
- Impacted NAT Port
- Origin MAC Address
- Impacted MAC Address
- Origin Interface
- Impacted Interface
- Origin Domain
- Impacted Domain
- Origin Login
- Impacted Account

- Impacted Hostname
- Impacted Hostname or IP
- Impacted NAT IP
- IANA Protocol Number
- IANA Protocol Name

Common Applications

Any network connected application or device.

Use Case

Host and application contexts.

MPE/Data Masking Manipulations

Used to help in determining Application.

Usage Standards

- Use to indicate the origin port number associated with a client or attacker host.
- Origin is Client (In Client-Server Model).
- Origin is Attacker (In Attacker-Target Model).

Examples

- FireEye Web MPS

```
02 01 2016 17:13:19 1.1.1.1 <LOC4:WARN> fenotify-609081.warning: CEF:0|FireEye|MPS|
1.1.1.1875|IM|infection-match|1|rt=Feb 01 2016 23:13:10 UTC src=1.1.1.1 cn3Label=cncPort
cn3=80 cn2Label=sid cn2=84575103 shost= USABLDRRECFLOW01proto=tcp spt=51997 dst=1.1.1.1
cs5Label=cncHost cs5=1.1.1.1 dvchost= USABLDRRECFLOW01 dvc=1.1.1.1
smac=00:00:00:00:00:00 cn1Label=vlan cn1=0 dpt=80 externalId=609081 cs4Label=link
cs4=STUFF dmac=00:00:00:00:00:00 cs1Label=sname cs1=Exploit.Kit.AnglerDIPv4
```

> Dpt= is Origin in this case as it is the port used by the attacker ip (dst).

- Cisco Next Generation Firewall

```
CISCONGFW EVENT Ev_Id=610 Ev_Severity=6 Ev_TypeId=HTTP_COMPLETE Ev_SrcId=32
Ev_RecvTime=2/24/2013 10:04:34 PM Ev_MetaData=0 Smx_Config_Version=2 Identity_Source=0
Smx_Policy_Id=0 Flow_ConnId=456 Smx_Egress_Interface_Id=0 Smx_Ingress_Interface_Id=0
Avc_App_Id=300003 Ev_GenTime=2/24/2013 10:04:09 PM Flow_Protocol=6 Flow_SrcIp=1.1.1.1
Flow_DstIp=1.1.1.1 Flow_SrcPort=60221 Flow_DstPort=80 Ev_Producer_Id=5
Flow_Transaction_Id=0 Url=recordflow.biz Flow_DstHostName=recordflow.bizSmx_Policy_Id=0
Flow_Bytes_Sent=391 Http_Response_Status=302 Flow_Bytes_Received=647
```

Origin port (source in a network traffic flow context).

- Cisco ISE

```
02 10 2014 13:54:24 1.1.1.1 <LOC6:NOTE> Feb 10 13:54:43 USABLDRRECFLOW01
CISE_Failed_Attempts 0000217969 2 0 2014-02-10 13:54:43.264 +02:00 0008145644 5413
NOTICE Failed-Attempt: RADIUS Accounting-Request dropped, ConfigVersionId=143, Device IP
Address=1.1.1.1, Device Port=1646, DestinationIPAddress=1.1.1.1, DestinationPort=1646,
Protocol=Radius, NetworkDeviceName=Switch_3560-X_2, NAS-IP-Address=1.1.1.1, NAS-
Port=50023, Service-Type=Framed, Acct-Status-Type=Start, Acct-Delay-Time=20, Acct-
Session-Id=00002222, Acct-Authentic=Local, NAS-Port-Type=Ethernet, NAS-Port-
Id=GigabitEthernet0/23, cisco-av-pair=connect-progress=Call Up, AcsSessionID=
USABLDRRECFLOW01/151856948/212124, FailureReason=11038 RADIUS Accounting-Request header
contains invalid Authenticator field, Step=11004, Step=11017, Step=11038, Step=5413,
NetworkDeviceGroups=Device Type#All Device Types#Switch,
NetworkDeviceGroups=Location#All Locations#HQ, NetworkDeviceGroups=Unit#All
Units#Networking, NetworkDeviceGroups=ACS Group#All ACS Groups, ACS Group=ACS Group#All
ACS Groups,
```

Device Port shows the originating RADIUS request Port for the corresponding device IP. Destination (Impacted) is the server being authenticated against (Client-Server relationship).

Other MPE Fields

These fields do not immediately map to a tab in the Web Console:

- Items In
- Items Out
- [Tag1-Tag5]

[Tag1-Tag5]

Used only for subrules, and are invisible to the end user.

Data Type

String

Aliases

Use	**Alias**
Client Console Full Name	Not applicable
Client Console Short Name	Not applicable
Web Console Tab/Name	Not applicable
Elasticsearch Field Name	Not applicable
Rule Builder Column Name	Tag1, Tag2, Tag3, Tag4, Tag5
Regex Pattern	<tag1>, <tag2>, <tag3>, <tag4>, <tag5>
NetMon Name	Not applicable

Field Relationships

Any field you do not use to create subrules—for example, command.

Common Applications

Not applicable.

Use Case

Creating subrules not based on VMID, ThreatID, or Severity.

MPE/Data Masking Manipulations

They are invisible outside of MPE Rule Builder.

Usage Standards

If you want to create a subrule of a value not captured into VMID, ThreatID, or Severity, a tag must be nested within the existing metatag.

Examples

These tags can be used in a wide variety of situations. Because these fields do not appear as parsed fields outside of the rule builder, refer to the usage standards to determine when to use these fields.

Items In

Items—not otherwise defined specifically—received from a device, system, or process.

Data Type

Double

Aliases

Use	Alias
Client Console Full Name	Not applicable
Client Console Short Name	Not applicable
Web Console Tab/Name	Not applicable
Elasticsearch Field Name	itemsPacketsIn
Rule Builder Column Name	ItemsIn
Regex Pattern	<itemsin>
NetMon Name	Not applicable

Field Relationships

- Packets In
- Packets Out
- Bytes In/Bytes Out

Common Applications

Devices that send and receive nonspecific item types.

Use Case

- Inventory control

- Phone logs

MPE/Data Masking Manipulations

- Polyfield
- <items> tag can be used for a single item value.
- In rule builder, parses in itemsin/itemsout.

Usage Standards

- Use when bytes, bits, or packets in field is not appropriate
- Use for the number of objects transferred in.

Examples

- A10 Networks Load Balancer

```
09 08 2011 12:36:04 1.1.1.1 <LOC0:INFO> USABLDRRECFLOW01: [HA]<6> Sent 577, Received
559, Duplicate Id's 0, Wrong Group 0, Missed 0, Inaccurate Time 0
```

> Total received Items.

- SharePoint Audit

```
9/5/2008 5:03:44 PM event=100 account=safaware\pete.store userid=922 machinename=
machineip= doclocation=ncsportal locationtype=0 eventname= eventsource=1 sourcename=
eventdata=<Export><RequestedBy>safaware\pete.store</RequestedBy><Completed /><TotalItems
>5</TotalItems><TotalSizeInBytes>120703</TotalSizeInBytes></Export>
```

> Total Items could be parsed into <items>.

Items Out

Items—not otherwise defined specifically—received from a device, system, or process.

Data Type

Double

Aliases

Use	Alias
Client Console Full Name	Not applicable
Client Console Short Name	Not applicable
Web Console Tab/Name	Not applicable
Elasticsearch Field Name	itemsPacketsOut
Rule Builder Column Name	ItemsOut
Regex Pattern	<itemsout>
NetMon Name	Not applicable

Field Relationships

- Items In
- Packets In
- Packets Out
- Bytes In/Bytes Out

Common Applications

Devices that send and receive nonspecific item types.

Use Case

- Inventory control
- Phone logs

MPE/Data Masking Manipulations

- Polyfield
- <items> tag can be used for a single item value.
- In rule builder, parses into itemsin/itemsout.

Usage Standards

- Use when bytes, bits, or packets out field is not appropriate.
- Number of objects transferred out.

Examples

- A10 Networks Load Balancer

```
09 08 2011 12:36:04 1.1.1.1 <LOC0:INFO> USABLDRRECFLOW01: [HA]<6> Sent 577, Received
559, Duplicate Id's 0, Wrong Group 0, Missed 0, Inaccurate Time 0
```

> Total received Items.

Derived Data

Derived data is not parsed in the schema, but is instead inferred and built from other metadata fields.

Identity-Derived Data

Identity-Derived data is augmented by LogRhythm's Identity feature. If you have configured known identities and have the feature enabled, then these fields are populated with known identity data.

Display Field	Description	Associated Data Sources
User Identity (Origin)	The identity that has the login in the User (Origin) field associated with it.	Only matches login Identifier fields
User Identity (Impacted)	The identity that has the User (Impacted) field associated with it.	Only matches login Identifier fields
Recipient Identity	Identity that has the Recipient's email address associated with it.	Only matches email Identifier fields
Sender Identity	Identity that has the Sender's email address associated with it.	Only matches email Identifier fields

Log-Derived Data

Derived data is generated based on information about the parser (for example, Common Event), on post processing information parsed out of the log (for example, Duration), or contextual information linking the log data to an entity or host (for example, Priority). The following fields are Log-Derived data where the value of the field is not part of the original log.

Display Field	Description	Associated Data Sources
Application Tab		
Application	Application derived by IANA protocol and port number or directly assigned in MPE processing settings.	Protocol Number Protocol Name Origin/Impacted Port
Known Application	Application derived from IANA protocol and port number. If a known application cannot be derived, it is displayed as unknown.	Protocol Name Protocol Number Origin/Impacted Port
Duration	Duration is a polyfield for capturing time derived.	Time Start Time End Days Hours Minutes Seconds Milliseconds Microseconds Nanoseconds
Classification Tab		
Classification	Value is determined based on the MPE Rule's assigned Common Event. Classification choice is a secondary effect of choosing the correct common event for a rule. Each common event has a classification and the classification is automatically associated to the log via the common event selection.	Assigned
Common Event	Value is determined based on the MPE Rule's assigned Common Event.	Assigned
Priority	Value is determined based on the Risk-Based Priority (RBP) calculation.	Risk-Based Priority
Direction	Indicates the directional flow of data between the Origin Host and the Impacted Host — Inbound, Outbound, Internal, External, or Unknown.	Origin/Impacted Host
MPE Rule Name	Name of rule that matched, assigned on rule creation.	Assigned

Display Field	Description	Associated Data Sources
Host Tab		
Host (Origin)	Origin host derived from Origin IP Address, Origin Hostname, or both.	IP Address (Origin) Hostname (Origin)
Host (Impacted)	Impacted host derived from Impacted IP Address, Impacted Hostname, or both.	IP Address (Impacted) Hostname (Impacted)
Known Host (Origin)	A value determined by mapping parsed origin host identifiers, such as IP address or hostname, to a LogRhythm host record.	IP Address (Origin) Hostname (Origin) LogRhythm Host Record
Known Host (Impacted)	A value determined by mapping parsed impacted host identifiers, such as IP address or hostname, to a LogRhythm host record.	IP Address (Impacted) Hostname (Impacted) LogRhythm Host Record
Location Tab		
Entity (Origin)	A value determined based on the origin host's assigned entity.	IP Address (Origin) Hostname (Origin) Entity
Entity (Impacted)	A value determined based on the impacted host's assigned entity.	IP Address (Impacted) Hostname (Impacted) Entity
Zone (Origin)	A value determined based on the zone of the origin host – Internal, External, DMZ, or Unknown.	IP Address (Origin)
Zone (Impacted)	A value determined based on the zone of the impacted host – Internal, External, DMZ, or Unknown.	IP Address (Impacted)
Location (Origin)	A value determined by resolving the parsed origin IP address against a Geo-IP database.	IP Address (Origin)
Location (Impacted)	A value determined by resolving the parsed impacted IP address against a Geo-IP database.	IP Address (Impacted)
Country (Origin)	The country in which the determined origin location exists.	IP Address (Origin)
Country (Impacted)	The country in which the determined impacte location exists.	IP Address (Impacted)
Log Tab		

Display Field	Description	Associated Data Sources
Log Date/Normal Date	Timestamp when the log was generated or received, corrected to UTC.	Agent
Log Count	The number of identical log messages received.	Agent
Log Source Entity	The entity to which the log source belongs.	Agent
Log Source Type	The device or application from which a log was received.	Agent
Log Source Host	The origin host from which the log was received.	Agent
Log Source	The assigned name of a log source.	Agent
Log Sequence Number	The sequence in which a log was collected, generated by the Agent.	Agent
Log Message	The raw log message.	Agent
First Log Date	Timestamp when the first identical log message was received.	Agent
Last Log Date	Timestamp when the last identical log message was received.	Agent
Network Tab		
Network (Origin)	A value determined by mapping the origin IP address to a LogRhythm network record.	IP Address (Origin) LogRhythm Network Record
Network (Impacted)	A value determined by mapping the impacted IP address to a LogRhythm network record.	IP Address (Impacted) LogRhythm Network Record